



Behaviour-Driven Adaptive Presentation in Digital Museums: Modelling Real-Time User Signals to Support Exploratory Engagement (Appendices)

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Final Project Report – Appendices

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*This document is submitted in partial fulfilment of the requirements for the
MSc Data Analytics Degree at Edge Hill University.*

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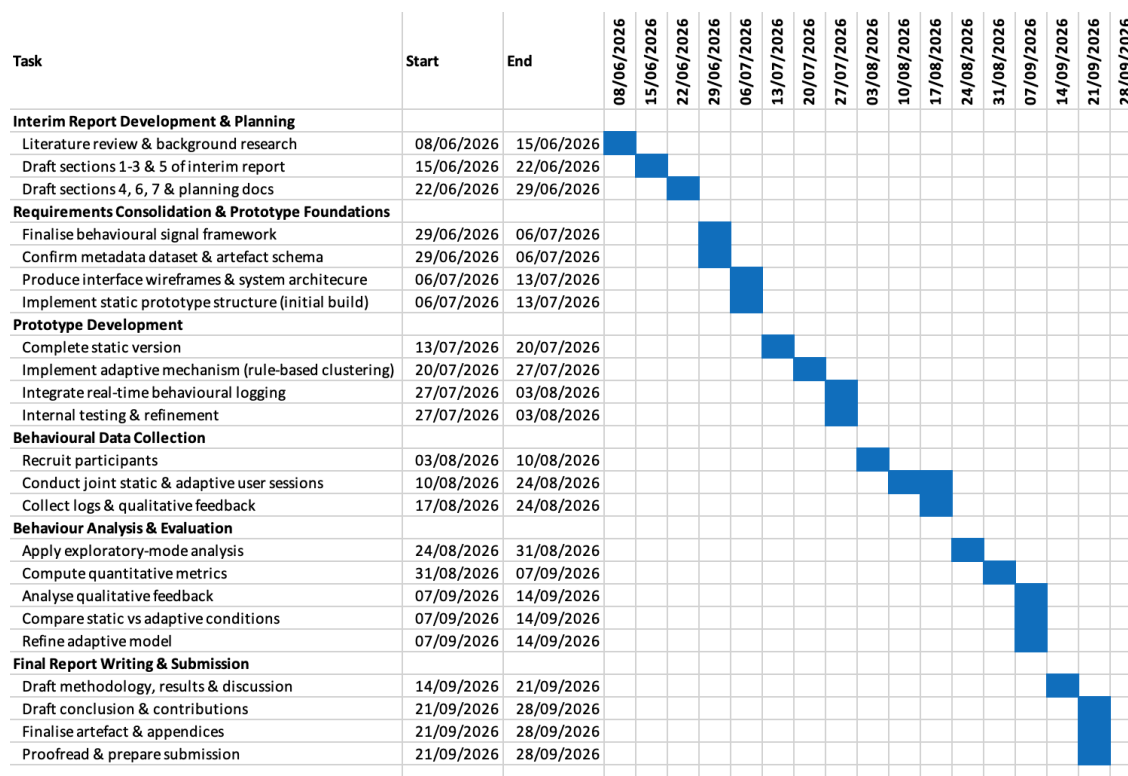
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Appendix A: Project Plan

A.1 Gantt Chart



A.2 Study Tracker

date	time	location	status	timestamp	rank	participant code	version order	consent form	debrief sheet
02/08/2026	10:30:00	Home	Completed	02/08/2026 10:30	1	PILOT1	Static first	Yes	Yes
02/08/2026	11:00:00	Home	Completed	02/08/2026 11:00	2	PILOT2	Adaptive first	Yes	Yes
03/08/2026	10:00:00	Uni	Completed	03/08/2026 10:00	3	ppt1	Static first	Yes	Yes
03/08/2026	11:00:00	Uni	Completed	03/08/2026 11:00	4	ppt2	Adaptive first	Yes	Yes
07/08/2026	11:00:00	Home	Completed	07/08/2026 11:00	8	ppt6	Adaptive first	Yes	Yes
06/08/2026	17:30:00	Home	Completed	06/08/2026 17:30	7	ppt5	Static first	Yes	Yes
10/08/2026	15:00:00	Tilly's	Completed	10/08/2026 15:00	9	ppt7	Static first	Yes	Yes
10/08/2026	15:30:00	Tilly's	Completed	10/08/2026 15:30	10	ppt8	Adaptive first	Yes	Yes
06/08/2026	09:00:00	Home	Completed	06/08/2026 09:00	5	ppt3	Static first	Yes	Yes
06/08/2026	17:00:00	Home	Completed	06/08/2026 17:00	6	ppt4	Adaptive first	Yes	Yes
11/08/2026	15:30:00	Home	Completed	11/08/2026 15:30	11	ppt9	Static first	Yes	Yes
11/08/2026	16:00:00	Home	Completed	11/08/2026 16:00	12	ppt10	Adaptive first	Yes	Yes
18/08/2026	11:30:00	Home	Completed	18/08/2026 11:30	26	ppt24	Adaptive first	Yes	Yes
14/08/2026	19:30:00	Home	Completed	14/08/2026 19:30	14	ppt12	Adaptive first	Yes	Yes
17/08/2026	12:30:00	Home	Completed	17/08/2026 12:30	21	ppt19	Static first	Yes	Yes
14/08/2026	18:00:00	Home	Completed	14/08/2026 18:00	13	ppt11	Static first	Yes	Yes
20/08/2026	14:00:00	Home	Completed	20/08/2026 14:00	31	ppt29	Static first	Yes	Yes
16/08/2026	16:30:00	Dylan's	Completed	16/08/2026 16:30	16	ppt14	Adaptive first	Yes	Yes
16/08/2026	17:00:00	Dylan's	Completed	16/08/2026 17:00	17	ppt15	Static first	Yes	Yes
16/08/2026	17:30:00	Dylan's	Completed	16/08/2026 17:30	18	ppt16	Adaptive first	Yes	Yes
16/08/2026	18:00:00	Dylan's	Completed	16/08/2026 18:00	19	ppt17	Static first	Yes	Yes
15/08/2026	17:00:00	Home	Completed	15/08/2026 17:00	15	ppt13	Static first	Yes	Yes
17/08/2026	18:30:00	Home	Completed	17/08/2026 18:30	25	ppt23	Static first	Yes	Yes
18/08/2026	19:00:00	Home	Completed	18/08/2026 19:00	29	ppt27	Static first	Yes	Yes
17/08/2026	12:00:00	Home	Completed	17/08/2026 12:00	20	ppt18	Adaptive first	Yes	Yes
17/08/2026	14:00:00	Home	Completed	17/08/2026 14:00	22	ppt20	Adaptive first	Yes	Yes
20/08/2026	16:30:00	Home	Completed	20/08/2026 16:30	32	ppt30	Adaptive first	Yes	Yes
17/08/2026	17:30:00	Home	Completed	17/08/2026 17:30	23	ppt21	Static first	Yes	Yes
17/08/2026	18:00:00	Home	Completed	17/08/2026 18:00	24	ppt22	Adaptive first	Yes	Yes
18/08/2026	14:30:00	Home	Completed	18/08/2026 14:30	28	ppt26	Adaptive first	Yes	Yes
18/08/2026	13:30:00	Home	Completed	18/08/2026 13:30	27	ppt25	Static first	Yes	Yes
18/08/2026	19:30:00	Home	Completed	18/08/2026 19:30	30	ppt28	Adaptive first	Yes	Yes

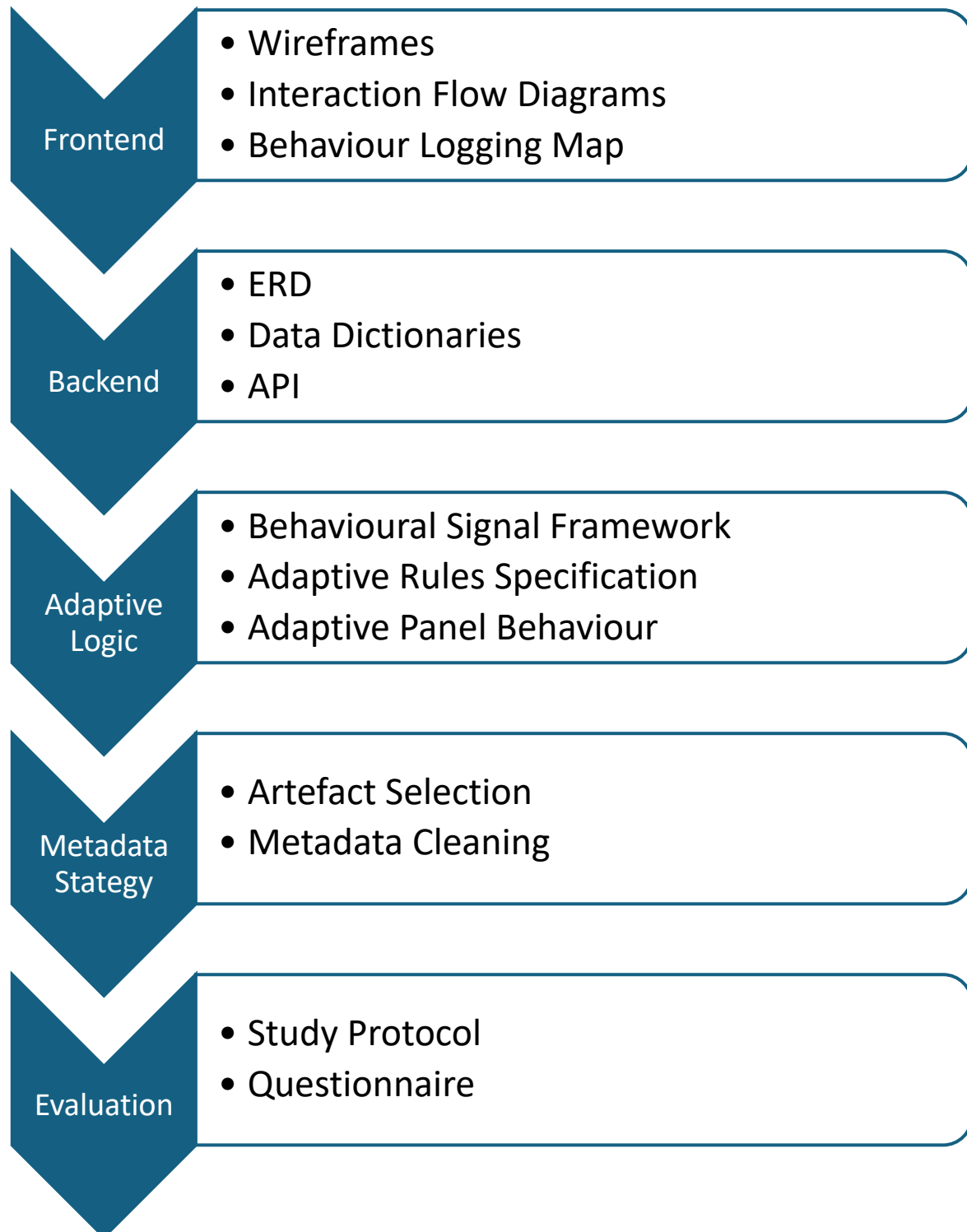
A.3 Literature Review Summary

Author & Year	Topic Category	Relevance
Coburn (2016)	Exploratory Behaviour	Behaviour patterns; need for browsing-first design
Kreiseler et al. (2017)	Exploratory Modes	Clickstream analysis; modelling user pathways
Pousman et al. (2007)	Casual Information Visualisation	Low-effort exploration; design for curiosity
Marchionini (2006)	Exploratory Search	Justifies browsing-based interaction; continuous exploration
Makri & Blandford (2012)	Serendipity Models	Behavioural triggers; unexpected discovery
Pirolli & Card (1999)	Information Foraging	Information scent; adaptive pathways
Chen & Chen (2025)	Museum Personalisation	Behaviour-driven adaptation; real-time cues
Xhembulla (2016)	Contextual Cues	Inferring interest from subtle behaviours
Not & Petrelli (2018)	Adaptive Museum Systems	Personalised narratives; adaptive pathways
Marty (2007)	Digital Museum Users	User motivations; engagement patterns
Stock et al. (2007)	Adaptive Cultural Heritage	Personalised recommendations; user modelling
Khamaj & Ali (2024)	Dynamic Recommendation	Algorithmic selection; artefact surfacing
Amato et al. (2017)	Visual Recommendation	Image-based similarity; visual browsing
Konstantakis et al. (2020)	Semantic Recommendation	Ontology-driven artefact linking
Liu et al. (2018)	User Interest Modelling	Behaviour-based interest prediction
Hyvönen (2020)	Linked Data in Museums	Semantic relationships; thematic clustering
Khoury et al. (2011)	Clickstream Modelling	Path analysis; behavioural segmentation
Bucklin & Sismeiro (2009)	Web Behaviour Analytics	Engagement metrics; dwell time modelling

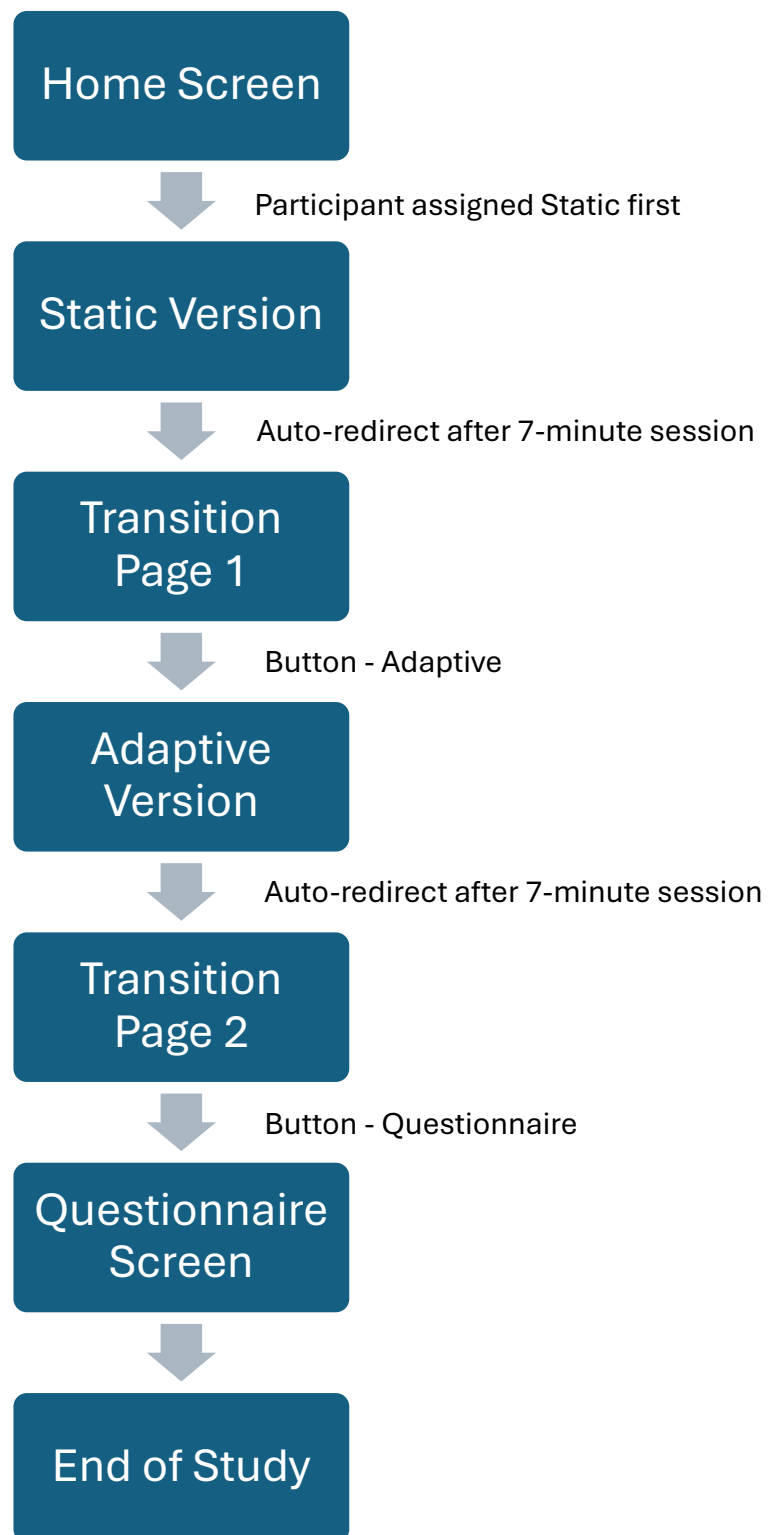
Wang et al. (2022)	Behaviour Prediction	Predictive modelling; next-click inference
Navarrete (2019)	Museum Web Analytics	Visitor pathways; engagement indicators
Cen & Zhao (2024)	Real-Time Personalisation	Adaptive interfaces; live behaviour tracking
Gajos et al. (2010)	Adaptive UI	Interface adaptation; user-dependent layouts
Wang & Li (2025)	Personalised Interfaces	Real-time adaptation logic
Liu & He (2025)	User Engagement	Engagement metrics; acceptance factors
Ivanov & Velkova (2025)	Systematic Review	Landscape mapping; research gaps
Povroznik (2025)	Digital Heritage Review	Trends in digital museums
Almeshari et al. (2019)	Behaviour Typologies	Visitor types; behavioural segmentation
Falk (2016)	Identity-Based Behaviour	Visitor motivations; identity-driven patterns
Bitgood (2016)	Attention & Engagement	Predicting attention; movement patterns
White et al. (2007)	Exploratory Information Behaviour	Browsing strategies; exploratory intent
Makri et al. (2015)	Information Encountering	Serendipity; unplanned discovery

Appendix B: System Design & Diagrams

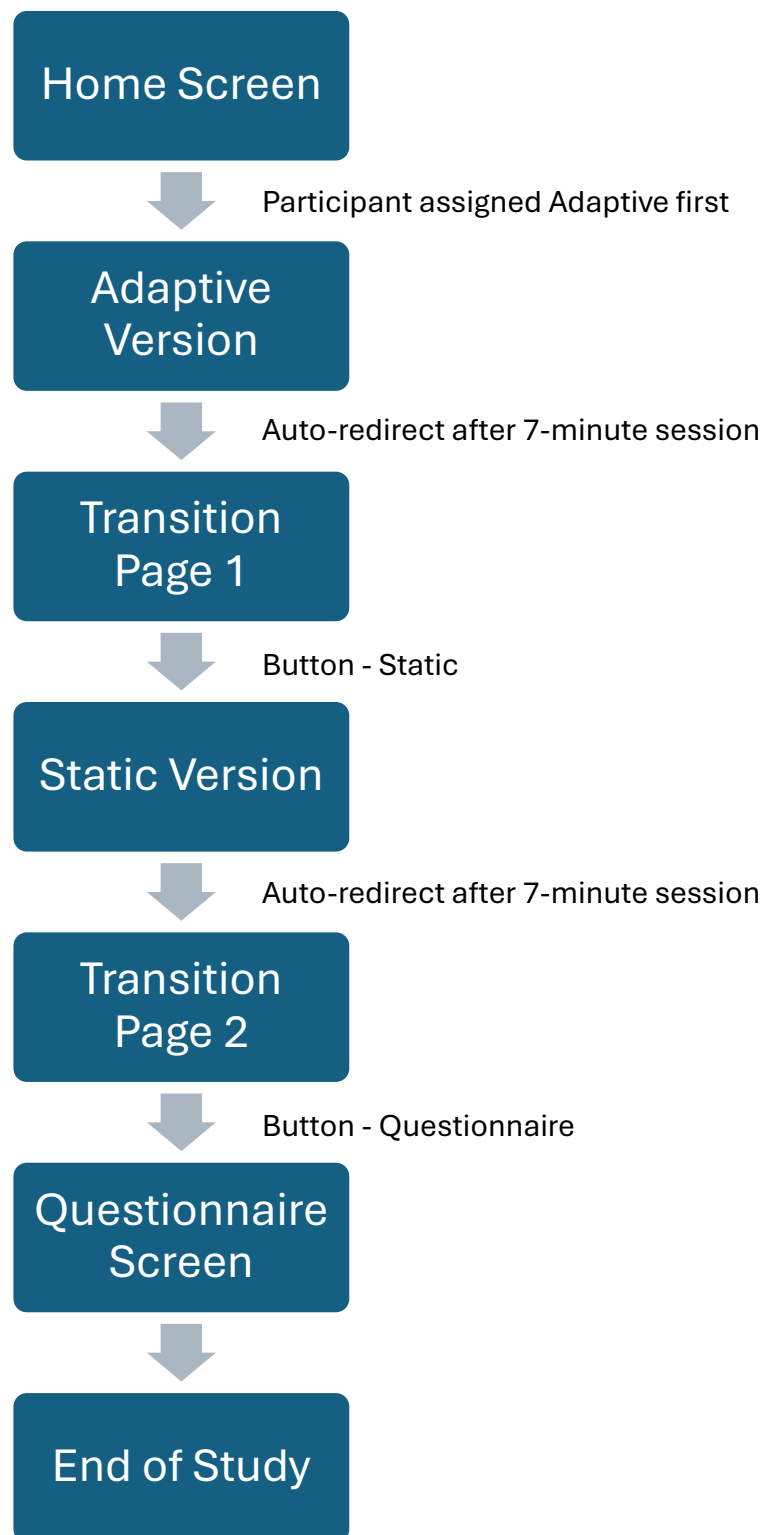
B.1 Design Roadmap/Checklist



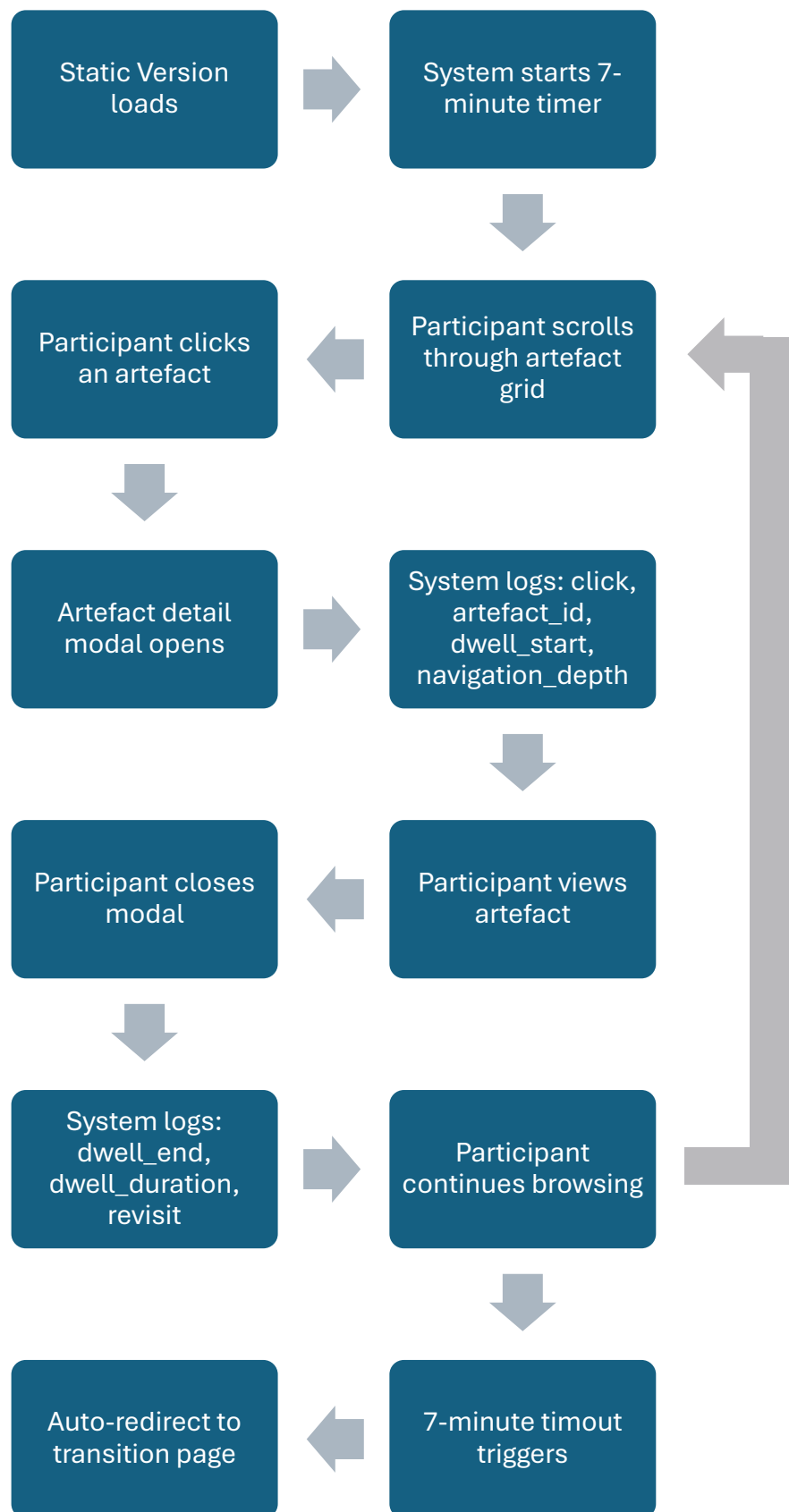
B.2 High-Level Interaction Flow Diagram – Static First



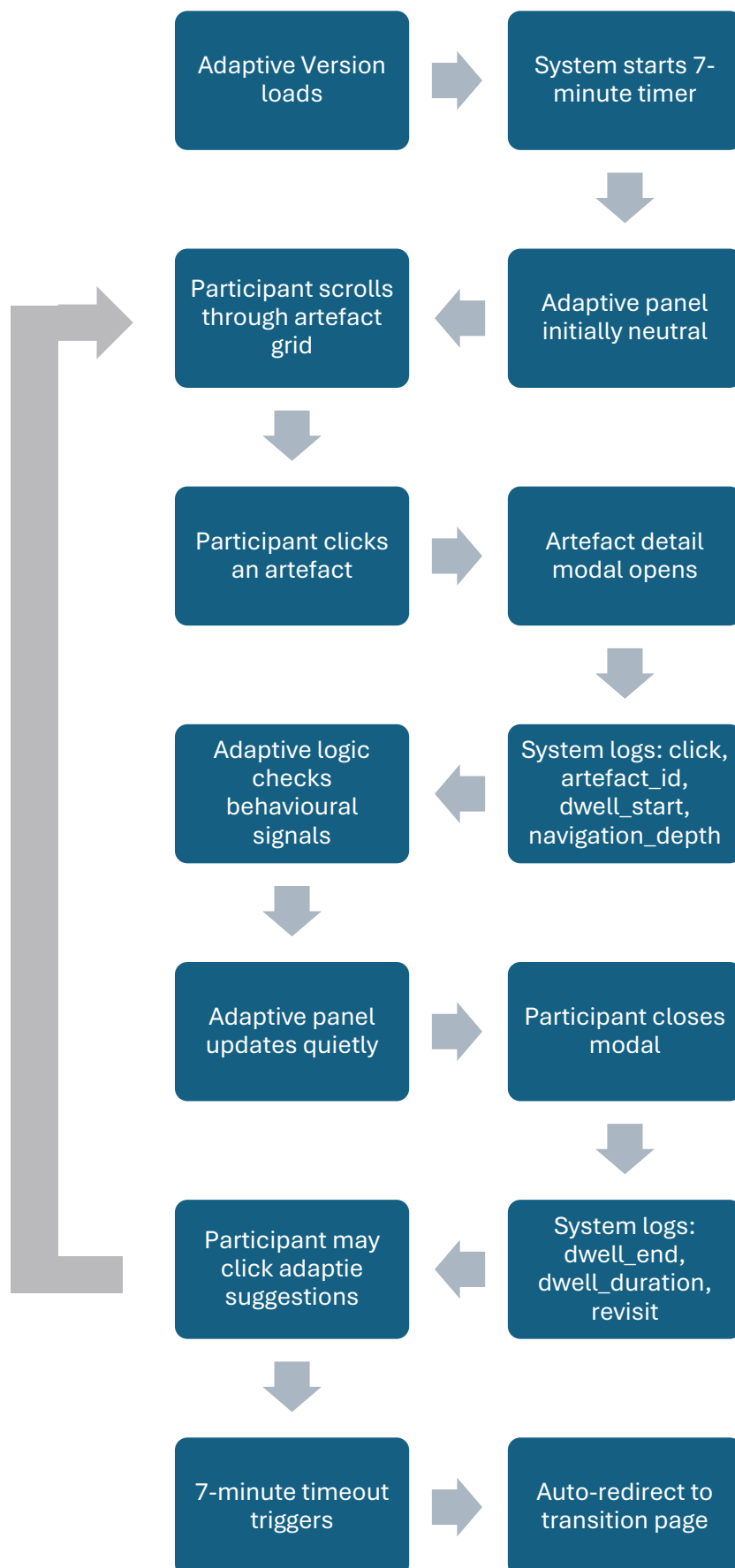
B.3 High-Level Interaction Flow Diagram – Adaptive First



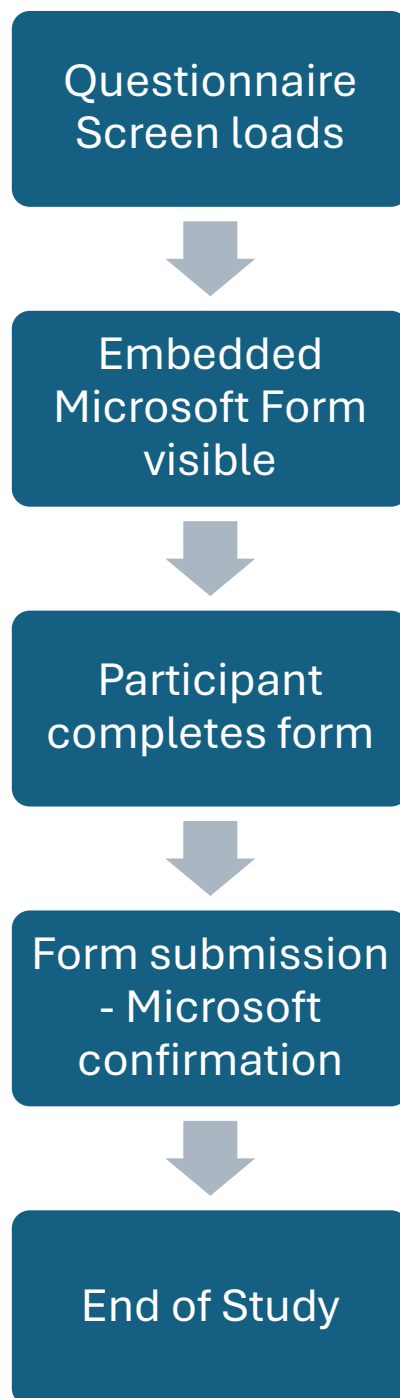
B.4 Detailed Interaction Flow – Static Version

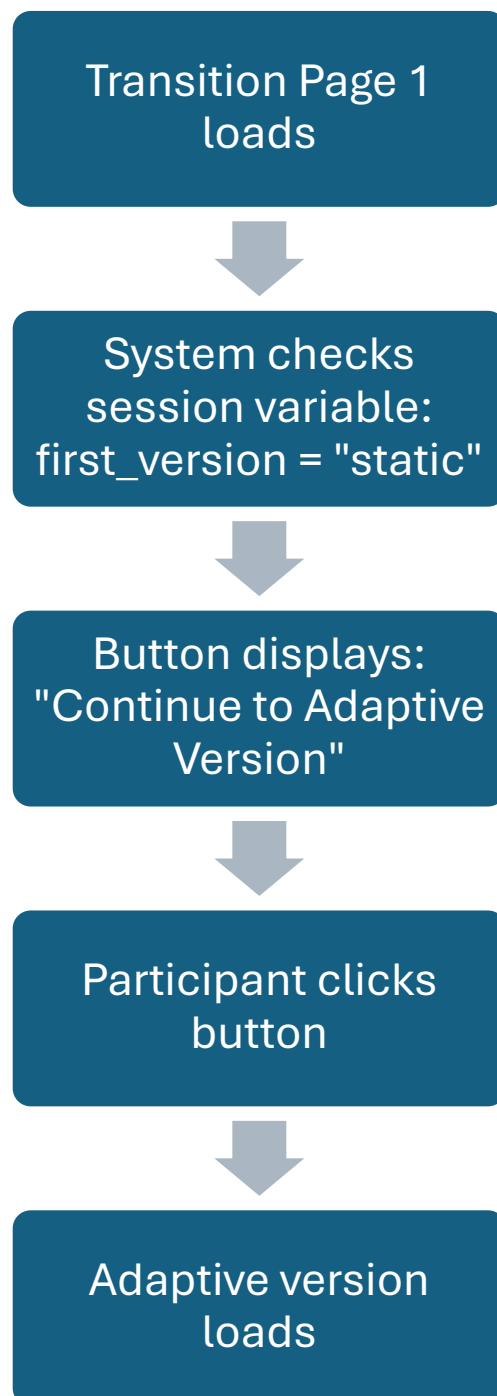


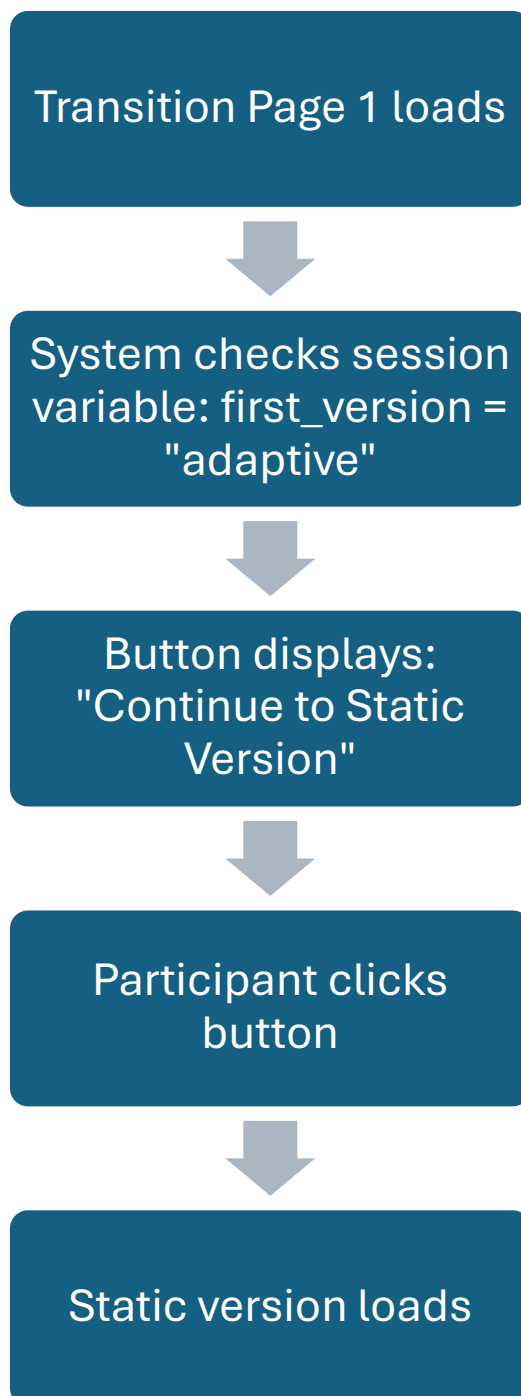
B.5 Detailed Interaction Flow – Adaptive Version



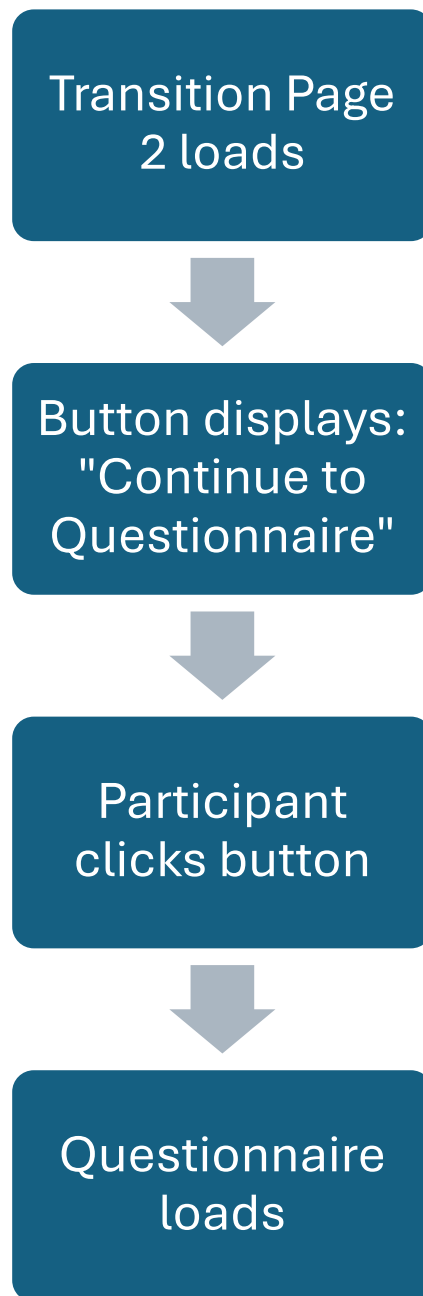
B.6 Detailed Interaction Flow – Questionnaire



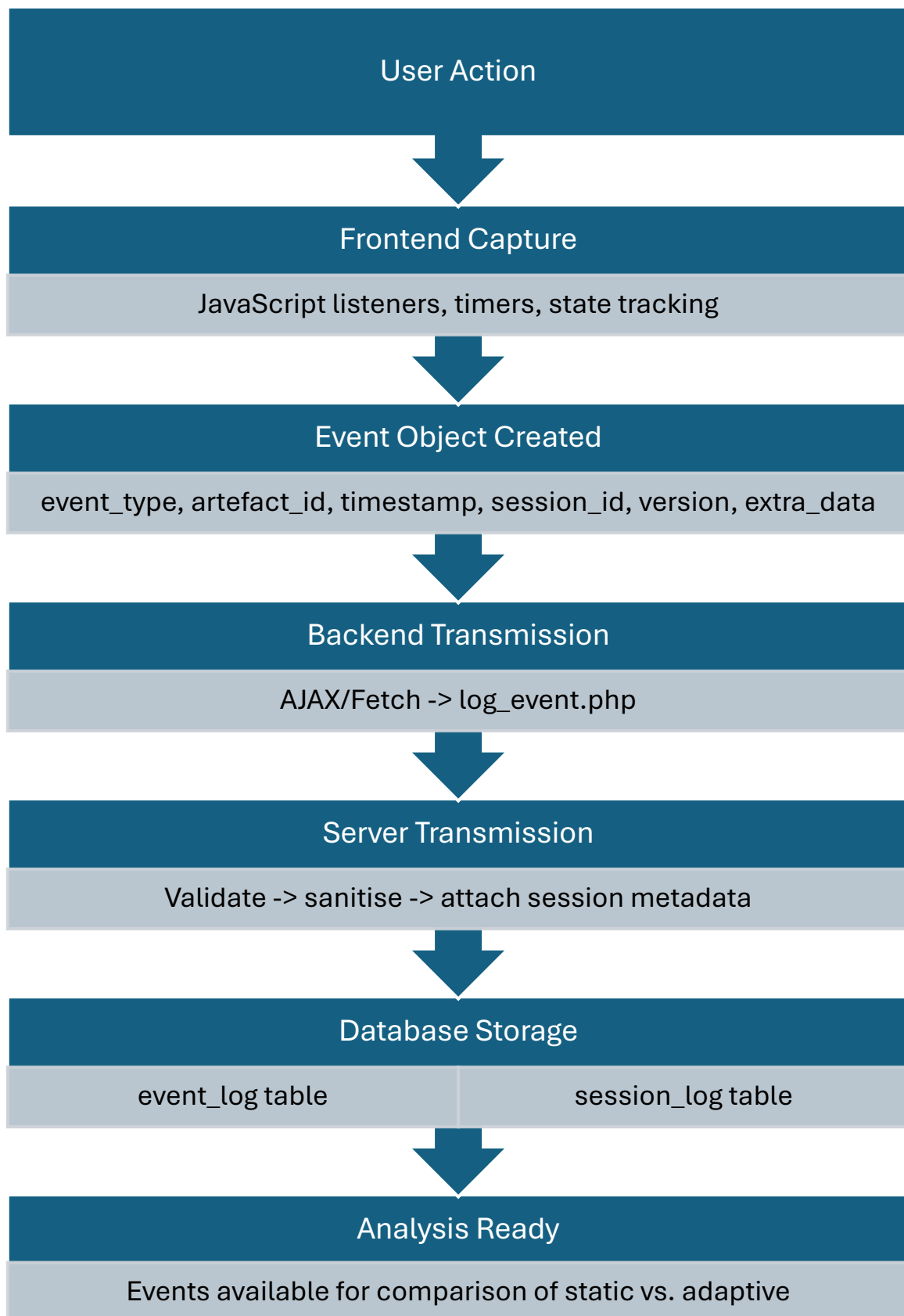
B.7 Detailed Interaction Flow – Transition Page 1, Static First

B.8 Detailed Interaction Flow – Transition Page 1, Adaptive First

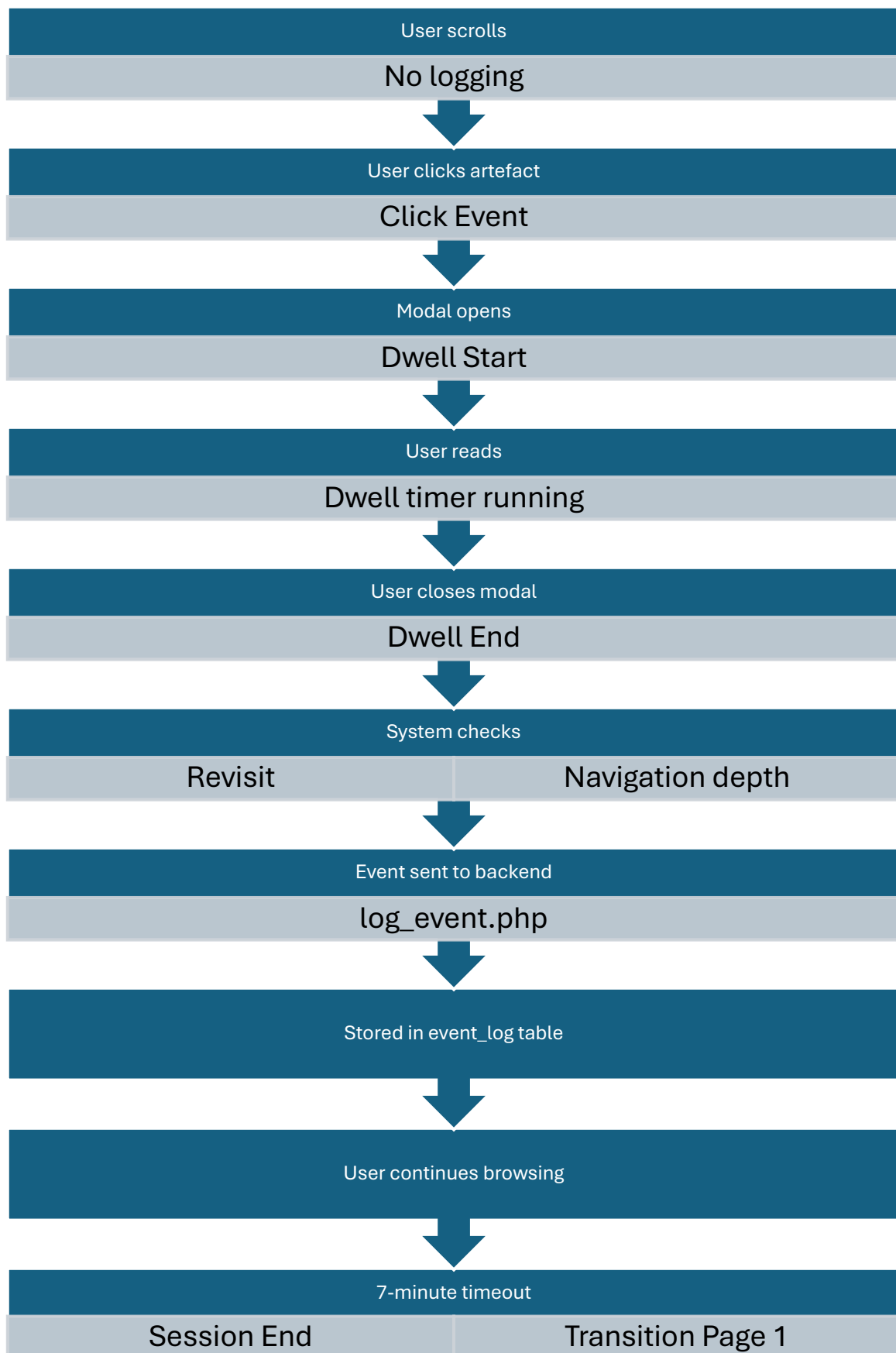
B.9 Detailed Interaction Flow – Transition Page 2



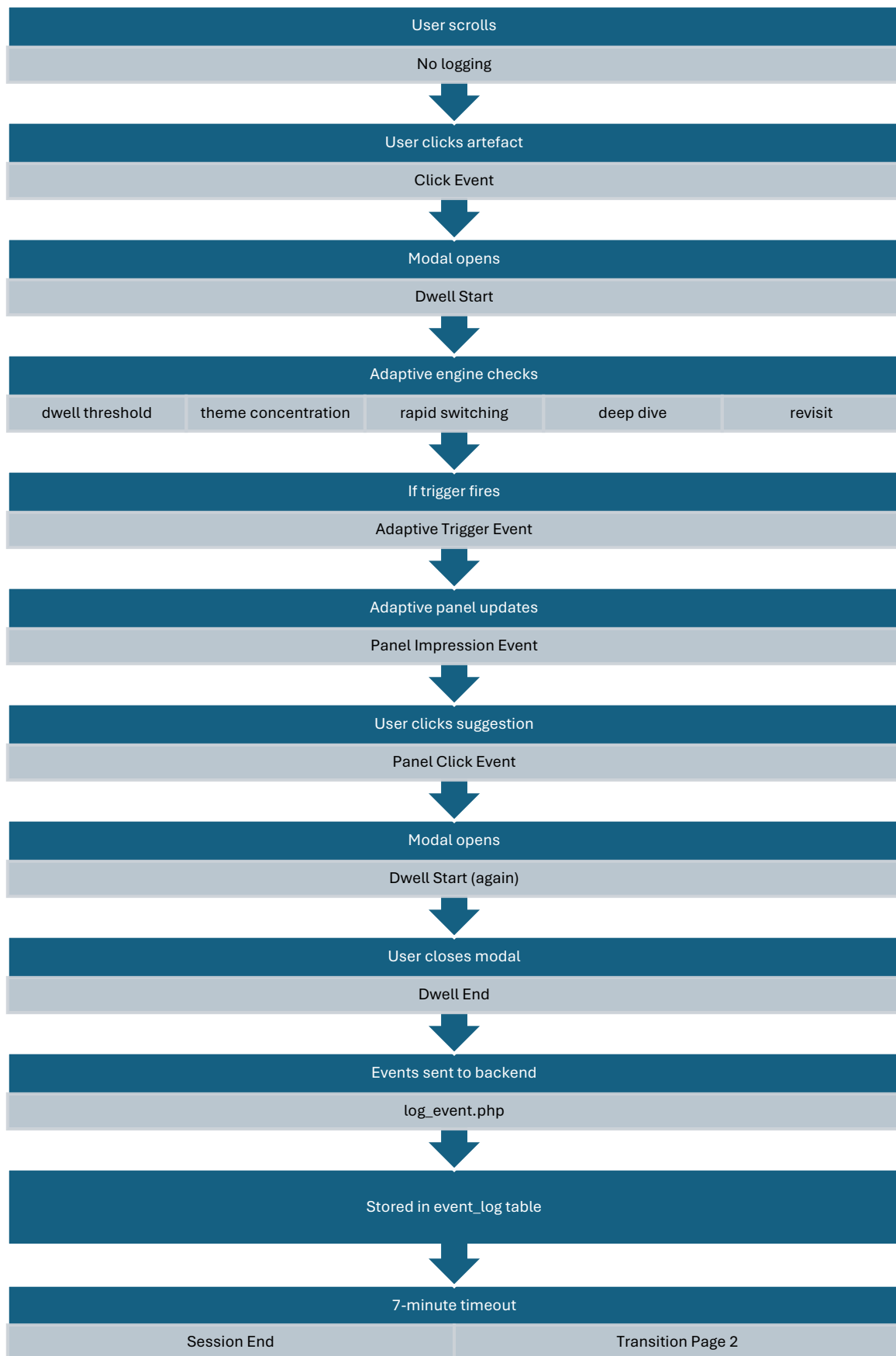
B.10 Behaviour Logging Pipeline



B.11 Behaviour Logging Diagram – Static Version



B.12 Behaviour Logging Diagram – Adaptive Version



B.13 Home Screen Wireframe

Digital Museum Prototype Study

Exploring how people browse and discover artefacts.

This study includes two browsing tasks followed by a short questionnaire.

Static Version

Browse a collection of artefacts
in a non-adaptive interface.

Adaptive Version

Browse a collection where
artefacts update based on your
interactions.

Questionnaire

Share your experience after
completing both versions.

Session Instructions

1. You will complete both versions in the order designated to you.
2. Please explore freely and click on any artefacts that interest you.
3. When finished, complete the questionnaire.

Kate Forshaw | Edge Hill University

This study is ethics approved.

B.14 Static Prototype Wireframe

Static Browsing Interface

Explore the artefacts freely. This version does not adapt to your behaviour.

Click any artefact to view more details.

[Thumbnail
image]
Lion Skull

[Thumbnail
image]
Dress

[Thumbnail
image]
Quill Pen

[Thumbnail
image]
Toy Train

[Thumbnail
image]
Fish Plate

[Thumbnail
image]
Red Fox

(keep scrolling)

Static Version | No adaptive behaviour | [Return to Home](#)

B.15 Adaptive Prototype Wireframe

Adaptive Browsing Interface

This version updates artefacts suggestions based on your browsing behaviour.

Explore freely. Suggestions will appear in the adaptive panel.

[Thumbnail
image]

Lion Skull

[Thumbnail
image]

Dress

[Thumbnail
image]

Quill Pen

**Suggested for
You**

[image]

Ceramic Bowl

(keep scrolling)

[Thumbnail
image]

Toy Train

[Thumbnail
image]

Fish Plate

[Thumbnail
image]

Red Fox

[image]

Typewriter

[image]

Blue Whale

(keep scrolling)

Adaptive Version | Suggestions update based on your behaviour | [Return to Home](#)

B.16 Artefact Detail Modal Wireframe

Close

[Artefact Title]
[Artefact Sub-theme]

[large image]

[Artefact Description]

Theme: _____

Sub-theme: _____

External ID: _____

Source link: _____

Close

B.17 Questionnaire Screen Wireframe

Post-Study Questionnaire

Please complete the short questionnaire about your experience.

Your responses help us understand how people explore digital museum collections.

About the Study Session

Q1) ...

Static Prototype Feedback

Q2) ...

Q3) ...

Q4) ...

Q5) ...

Q6) ...

Adaptive Prototype Feedback

Q7) ...

Q8) ...

Q9) ...

Q10) ...

Q11) ...

Comparative Reflection

Q12) ...

Q13) ...

Final Thoughts

Q14) ...

Instructions

1. Please ensure you have completed both browsing tasks before submitting.
2. Your responses will be anonymous.
3. When finished, you may close the browser window.

Thank you for taking part in this study | [Return to Home](#)

B.18 Transition Page 1 Wireframe

First Browsing Task Complete

You will now continue to the next version of the digital museum prototype.

Thank you for completing the first browsing session.

Click the button below when you are ready to continue.

[Continue to Next
Version](#)

Please explore freely in the next session | Each section lasts approximately 7 minutes.

B.19 Transition Page 2 Wireframe

Browsing Sessions Complete

You will now complete a short questionnaire about your experience.

Thank you for completing both browsing sessions.

Please continue to the questionnaire when you are ready.

[Continue to
Questionnaire](#)

Your responses will help us understand how people explore digital museum collections | Thank you for taking part

B.20 Error/Fallback Screen Wireframe

Something Went Wrong

It looks like this part of the study didn't load correctly.

Don't worry – you can return to the home screen and continue the study.

Your progress so far has been saved.

[Return to Home](#)

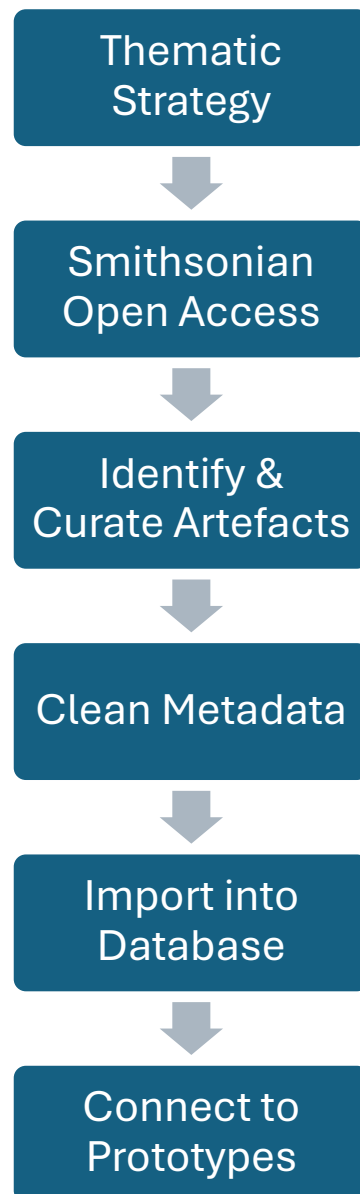
Error code: Session interrupted | If this happens again, please let the researcher know
Digital Museum Prototype Study | Kate Forshaw | Edge Hill University |

B.21 CSS Style Summary Table

Category	Element	Value	Purpose/Notes
Colour	Soft Charcoal	#2E2E2E	Header/title text – museum catalogue feel, high readability
Colour	Warm Off-White	#F7F7F5	Background – neutral, calm, avoids glare
Colour	Muted Sand	#D8D2C4	Section & card backgrounds – subtle gallery tone
Colour	Deep Bronze	#8C6E4E	Accent, buttons, borders – historically resonant without being decorative
Colour	Soft Blue Grey	#6A7A8C	Links – subtle emphasis without drawing focus from artefacts
Colour	Serendipity Accent	#C4D7E0	Adaptive panel borders only – subtle enough not to bias behaviour
Typography	Font family	Inter / Roboto / Helvetica Neue	Clean, modern, consistent with museum digital catalogues
Typography	Font sizes	Header 28–32px, sub-header 20–22px, body 16–18px	Visual hierarchy
Layout	Grid	3–4 column responsive, 20–24px gap	Consistent across static and adaptive versions
Layout	Modal	Max width 700–900px, 24–32px padding	Consistent artefact detail presentation

Layout	Adaptive panel	300–350px, 4 recommendations, no animation/movement/flashing	Keeps adaptive cues subtle and non-intrusive
Accessibility	Contrast ratio	Minimum 4.5:1	WCAG 2.2 alignment
Accessibility	Tap targets	Minimum 44px height	Usability for all participants

B.22 Metadata Process



B.23 Metadata Cleaning Rules

Rule Type	Reasoning	Format	Requirements
Naming conventions	Adaptive rule matching; panel display clarity; database integrity; cross-theme comparison	Proper noun + object type	Title case; no Smithsonian prefixes; no abbreviations; avoid scientific Latin names
Description Style	Short and readable for consistent dwell-time comparison across artefacts, while including one engaging factual detail to encourage genuine reading rather than skimming, supporting more reliable interest and serendipity signals	One sentence identifying the object, with scientific binomial name where applicable, followed by one brief factual or contextual detail (behaviour, function, history, or notable characteristic), occasionally as a short second sentence	Sentence case; one to two sentences; genus and species capitalised correctly and stated once where applicable; common name may be included naturally; avoid subjective or promotional adjectives; avoid long taxonomic chains; no museum-specific jargon
Theme Consistency	2 themes + 12 sub-themes; adaptive rule firing; switching pattern detection, branching factor calculation; panel clustering	Theme: Animals & Everyday Objects; Sub-theme: Mammals, Birds, Reptiles, Insects, Marine Life, Fossils, Tools, Writing, Clothing, Food-related, Household Items, Toys	Exact spelling and capitalisation; no synonyms; no hyphen variation; no new sub-themes introduced
Tag Strategy	Semantic enrichment; serendipity logic; cross-subtheme discovery; adaptive panel diversity	Simple, lowercase keywords, separated by commas	1-3 tags per artefact; lowercase only; no spaces around commas; no duplicates; no overly scientific terms; no subjective tags; visual, material, or functional aspects
Image Quality	Dwell-time accuracy; visual engagement; serendipity triggers; adaptive panel clarity	High-resolution Smithsonian Open Access	Minimum 1200px resolution; JPEG/PNG only; no thumbnails; no watermarked images; no broken links; load within <1s; visually clear; show artefact fully

B.24 API Summary Table

Module	Endpoint	Method	Purpose	Table
Metadata	/artefacts	GET	List artefacts	artefact
Metadata	/artefacts{id}	GET	Retrieve artefact	artefact
Metadata	/themes	GET	List themes	theme
Metadata	/subthemes	GET	List subthemes	subtheme
Metadata	/artefacts/by-subtheme/{subtheme_id}	GET	Returns artefacts filtered by subtheme	artefact
Logging	/event	POST	Log behavioural event	event_log
Logging	/adaptive-event	POST	Log adaptive event	adaptive_event
Logging	/timeout	POST	Log timeout	timeout
Adaptive	/adaptive/suggestions	GET	Get suggestions	adaptive_suggestion
Adaptive	/adaptive/suggestion	POST	Create suggestion	adaptive_suggestion
Adaptive	/adaptive/rules	GET	List rules	adaptive_rule
Adaptive	/adaptive/rules/{rule_id}	GET	Returns one rule	adaptive_rule
Adaptive	/adaptive/trigger	POST	Fire rule	adaptive_event
Session	/session/start	POST	Start session	session_log
Session	/session/end	POST	End session	session_log
Session	/session/{session_id}	GET	Returns session metadata	session_log
Session	/participant/{participant_id}/sessions	GET	Returns all sessions for a participant	session_log
Questionnaire	/questionnaire/items	GET	List items	questionnaire_item
Questionnaire	/questionnaire/response	POST	Submit response	questionnaire_response
Questionnaire	/questionnaire/responses?session_id=	GET	Returns all responses for a session	questionnaire_response

Appendix C: Data Dictionary

artefact				
Attribute	Description	Data type	Field size	Required field
artefact_id (PK)	Unique internal numeric ID for each artefact – auto-increment.	INT	-	Not null
theme_id (FK)	Unique numeric ID linking the artefact to a theme.	INT	-	Not null
subtheme_id (FK)	Unique numeric ID linking the artefact to a sub-theme.	INT	-	Not null
external_id	Smithsonian object ID for reference.	VARCHAR	50	Not null
artefact_title	Official artefact title, in simple terms.	VARCHAR	255	Not null
artefact_description	Short descriptive text summarising artefact, in scientific terms.	TEXT	-	Not null
image_url	Direct link to Smithsonian Open Access image.	VARCHAR	500	Not null
source_url	Smithsonian webpage URL for reference.	VARCHAR	500	Not null
tags	Optional keywords for semantic enrichment.	VARCHAR	255	Null

theme				
Attribute	Description	Data type	Field size	Required field
theme_id (PK)	Unique numeric ID for each theme – auto-increment.	INT	-	Not null
theme_name	Name of the theme category – ‘Animals’, ‘Everyday Objects’.	ENUM	-	Not null
theme_description	Optional descriptive text explaining the theme.	VARCHAR	255	Null

subtheme				
Attribute	Description	Data type	Field size	Required field

subtheme_id (PK)	Unique numeric ID for each theme – auto-increment.	INT	-	Not null
theme_id (FK)	Unique numeric ID linking the subtheme to a theme.	INT	-	Not null
subtheme_name	Name of the subtheme category – ‘Mammals’, ‘Birds’, ‘Fossils’, ‘Reptiles’, ‘Insects’, ‘Marine Life’, ‘Tools’, ‘Writing’, ‘Clothing’, ‘Food-related’, ‘Household Items’, ‘Toys’.	ENUM	-	Not null
subtheme_description	Optional descriptive text explaining the sub-theme.	VARCHAR	255	Null

event_log				
Attribute	Description	Data type	Field size	Required field
event_id (PK)	Unique numeric ID for each logged event – auto-increment.	INT	-	Not null
session_id (FK)	Unique numeric ID linking the event to a participant’s session.	INT	-	Not null
artefact_id (FK)	Unique numeric ID linking the event to an artefact.	INT	-	Not null
suggestion_id (FK)	Unique numeric ID linking the event to an adaptive panel item – null for static version.	INT	-	Null
rule_id	Unique numeric ID linking the event to an adaptive rule/trigger.	INT	-	Null
event_type	Type of behavioural event – ‘click’, ‘dwell_start’, ‘dwell_end’, ‘revisit’, ‘navigation_depth’, ‘adaptive_trigger’, ‘panel_impression’, ‘panel_click’, ‘session_start’, ‘session_end’, ‘timeout_redirect’, ‘error_trigger’.	ENUM	-	Not null
timestamp	Exact date and time the event occurred.	DATETIME	-	Not null

dwellduration	Duration of dwell time in seconds – only populated for dwell_end events.	INT	-	Null
navigation_depth	Depth of exploration, number of artefacts opened sequentially – only populated if a modal is opened.	INT	-	Null

session_log				
Attribute	Description	Data type	Field size	Required field
session_id (PK)	Unique numeric ID for each participant's logged session – auto-increment.	INT	-	Not null
participant_id (FK)	Unique numeric ID linking the session to a participant.	INT	-	Not null
condition_id (FK)	Unique numeric ID linking the session to a study condition.	INT	-	Not null
session_start	Timestamp marking when the participant first loads the home screen.	DATETIME	-	Not null
session_end	Timestamp marking when the participant completes the questionnaire or exits the system.	DATETIME	-	Null
timeout_triggered_static	Flag indicating whether the 7-minute timeout occurred in the static session.	BOOLEAN	-	Not null
timeout_triggered_adaptive	Flag indicating whether the 7-minute timeout occurred in the adaptive session.	BOOLEAN	-	Not null
error_flag	Flag indicating whether the error/fallback screen was triggered at any point.	BOOLEAN	-	Not null
artefact_order	JSON-encoded array of artefact_id in the order the participant saw the artefact grid, seeded by participant_id (Phase 8 > includes/artefact_order.php).	TEXT	-	Null

session_notes	Optional internal notes about session.	TEXT	-	Null
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adaptive_event				
Attribute	Description	Data type	Field size	Required field
adaptive_event_id (PK)	Unique numeric ID for each adaptive event – auto-increment.	INT	-	Not null
event_id (FK)	Unique numeric ID linking the adaptive event to an event.	INT	-	Not null
session_id (FK)	Unique numeric ID linking the adaptive event to a participant's session.	INT	-	Not null
rule_id (FK)	Unique numeric ID linking the adaptive event to an adaptive rule.	INT	-	Not null
suggestion_id (FK)	Unique numeric ID linking the adaptive event to an artefact.	INT	-	Null
trigger_timestamp	Exact timestamp when the adaptive trigger fired.	DATETIME	-	Not null
trigger_confidence	Optional numeric score representing confidence/strength of the trigger.	DECIMAL	5,2	Null

adaptive_rule				
Attribute	Description	Data type	Field size	Required field
rule_id (PK)	Unique numeric ID for each adaptive rule – auto-increment.	INT	-	Not null
rule_name	Short name of the adaptive rule.	VARCHAR	100	Not null
rule_description	Description of what the rule does.	VARCHAR	255	Null
trigger_condition	Optional JSON describing the behavioural threshold.	TEXT	-	Null

active_flag	Flag indicating whether the rule is currently active.	BOOLEAN	-	Not null
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adaptive_suggestions				
Attribute	Description	Data type	Field size	Required field
suggestion_id (PK)	Unique numeric ID for each adaptive suggestion – auto-increment.	INT	-	Not null
rule_id (FK)	Unique numeric ID linking the suggestion to an adaptive rule.	INT	-	Not null
session_id (FK)	Unique numeric ID linking the suggestion to a participant's session.	INT	-	Not null
artefact_id (FK)	Unique numeric ID linking the suggestion to an artefact.	INT	-	Not null
interest_score	Computed interest score behind this suggestion at the time it was generated (Adaptive Logic.docx > Score Normalisation Formula).	DECIMAL(4,2)	-	Null
suggestion_timestamp	Timestamp when the suggestion was generated and shown in the adaptive panel.	DATETIME	-	Not null
panel_position	Position of the suggestion in the panel.	INT	-	Null
clicked_flag	Flag indicating whether the participant clicked the suggestion.	BOOLEAN	-	Not null

participant				
Attribute	Description	Data type	Field size	Required field
participant_id (PK)	Unique numeric ID for each participant – auto-increment.	INT	-	Not null
participant_code	Anonymous participant code.	VARCHAR	20	Not null

participant_notes	Optional notes regarding participant.	TEXT	-	Null
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study_condition				
Attribute	Description	Data type	Field size	Required field
condition_id (PK)	Unique numeric ID for each condition – auto-increment.	INT	-	Not null
condition_name	Name of the counterbalancing condition – ‘static_first’, ‘adaptive_first’.	ENUM	-	Not null
condition_description	Optional description of the condition.	VARCHAR	255	Null

timeout				
Attribute	Description	Data type	Field size	Required field
timeout_id (PK)	Unique numeric ID for each timeout event – auto-increment.	INT	-	Not null
session_id (FK)	Unique numeric ID linking the suggestion to a participant’s session.	INT	-	Not null
version_name	Which version timed out – ‘static’, ‘adaptive’	ENUM	-	Not null
timeout_timestamp	Exact timestamp when the 7-minute timeout occurred.	DATETIME	-	Not null
auto_redirect_flag	Flag indicating whether the system successfully redirected to the transition page.	BOOLEAN	-	Not null
timeout_notes	Optional notes regarding the timeout.	TEXT	-	Null

questionnaire_item				
Attribute	Description	Data type	Field size	Required field

item_id (PK)	Unique numeric ID for each questionnaire item – auto-increment.	INT	-	Not null
item_text	Full text of the questionnaire question.	VARCHAR	500	Not null
item_type	Type of question – ‘multiple_choice’, ‘likert’, ‘free_text’	ENUM	-	Not null
item_order	The order the question appears in the questionnaire.	INT	-	Not null
scale_min_label	For likert items – null for non-likert items.	VARCHAR	100	Null
scale_max_label	For likert items – null for non-likert items.	VARCHAR	100	Null
item_notes	Optional notes for questionnaire items.	TEXT	-	Null

questionnaire_response				
Attribute	Description	Data type	Field size	Required field
response_id (PK)	Unique numeric ID for each questionnaire response – auto-increment.	INT	-	Not null
session_id (FK)	Unique numeric ID linking the response to a session.	INT	-	Not null
participant_id (FK)	Unique numeric ID linking the response to a participant.	INT	-	Not null
item_id (FK)	Unique numeric ID linking the response to a questionnaire item.	INT	-	Not null
response_value	The participant’s response.	VARCHAR	500	Not null
response_timestamp	Timestamp when the participant submitted the questionnaire.	DATETIME	-	Not null
response_notes	Optional notes for questionnaire responses.	TEXT	-	Null

Appendix D: Algorithms & Rule Specifications

D.1 Behavioural Signal Framework

Dwell Time Thresholds			
Category	Threshold	Interpretation	Adaptive Meaning
Micro-dwell	0-3 seconds	Glance, low interest	Ignore; user is scanning
Light dwell	3-8 seconds	Mild curiosity	Weak interest signal
Engaged dwell	8-15 seconds	Reading metadata	Moderate interest signal
Deep dwell	15-30 seconds	Strong interest	Trigger theme reinforcement
Immersive dwell	>30 seconds	High engagement	Trigger serendipity or deep-dive suggestions

Revisit Rules			
Category	Threshold	Interpretation	Adaptive Meaning
Single revisit	2 views of the same artefact	Confirming interest	Reinforce theme/subtheme
Multiple revisits	3+ views	Strong attraction	Show closely related artefacts
Rapid revisit	Revisit within <20 seconds	Curiosity spike	Trigger serendipity suggestion
Delayed revisit	Revisit after >1 minute	Renewed interest	Suggest deeper thematic items

Navigation Depths			
Category	Threshold	Interpretation	Adaptive Meaning
Shallow	1-3 artefacts	Early exploration	No adaptation yet
Moderate	4-7 artefacts	Browsing mode	Suggest related items
Deep	8-12 artefacts	Strong exploration	Trigger deep-dive suggestions
Very deep	>12 artefacts	High engagement	Trigger serendipity + thematic clusters

Theme Switching Patterns			
Category	Threshold	Interpretation	Adaptive Meaning
No switching	Stays within 1 theme	Focused interest	Reinforce current theme
Single switch	Switches once (A -> B or B -> A)	Exploratory drift	Suggest cross-theme artefacts
Oscillation	A -> B -> A or B -> A -> B	Uncertain interest	Suggest bridging artefacts
Rapid switching	3 or more switches	Serendipitous browsing	Trigger serendipity suggestions

Subtheme Switching Patterns			
Category	Threshold	Interpretation	Adaptive Meaning
Low switching	1-2 subthemes	Focused exploration	Reinforce subtheme
Moderate switching	3-4 subthemes	Exploratory drift	Suggest related subthemes
High switching	5-7 subthemes	Broad exploration	Suggest thematic clusters
Rapid switching	8 or more subthemes	Serendipitous mode	Trigger unexpected artefacts

Branching Factors			
Category	Threshold	Interpretation	Adaptive Meaning
Low branching	1-2 subthemes	Narrow interest	Reinforce subtheme
Moderate branching	3-5 subthemes	Balanced exploration	Suggest deeper related artefacts
High branching	6-9 subthemes	Broad exploration	Suggest cross-subtheme clusters
Very high branching	10 or more subthemes	Serendipitous browsing	Trigger serendipitous suggestions

D.2 Adaptive Rule Specification

Dwell-Based Adaptive Rules			
IF	THEN	Priority	Effect
dwell_time is light dwell (3-8s)	Add weak interest weight to artefact's subtheme	Low	No panel update yet; contributes to cumulative interest score
dwell_time is engaged dwell (8-15s)	Add moderate interest weight to artefact's subtheme	Medium	Panel may update if cumulative interest score passes threshold
dwell_time is deep dwell (15-30s)	Trigger theme reinforcement	High	Panel updates with 1-2 artefacts from same subtheme
dwell_time is immersive dwell (>30s)	Trigger deep-dive or serendipity mode	Very high	Panel shows deeper subtheme items + 1 serendipity artefact

Revisit-Based Adaptive Rules			
IF	THEN	Priority	Effect
revisit_count = 2	Reinforce subtheme	Medium	Panel shows 1 related artefact
revisit_count >= 3	Show closely related artefacts	High	Panel shows 2-3 artefacts from same subtheme
Revisit occurs within < 20s	Trigger serendipity suggestion	Very high	Panel shows 1 unexpected artefact from a different subtheme
Revisit occurs after >1 min	Suggest deeper thematic items	Medium	Panel shows deeper items within same subtheme

Navigation Depth Adaptive Rules			
IF	THEN	Priority	Effect
navigation_depth = 4-7	Suggest related items	Medium	Panel shows 1 artefact from recently explored subthemes
navigation_depth = 8-12	Trigger deep-dive suggestions	High	Panel shows deeper items from the dominant subtheme
navigation_depth > 12	Trigger serendipity + thematic clusters	Very high	Panel shows 2 related + 1 serendipity artefact

Theme Switching Adaptive Rules			
IF	THEN	Priority	Effect
theme_switches = 0	Reinforce current theme	Medium	Panel shows items from same theme
theme_switches = 1	Suggest cross-theme artefacts	Medium	Panel shows 1 artefact from the other theme
theme_switches = 2	Suggest bridging artefacts	High	Panel shows artefacts linking both themes
theme_switches >= 3	Trigger serendipity suggestions	Very high	Panel shows unexpected cross-theme artefacts

Subtheme Switching Adaptive Rules			
IF	THEN	Priority	Effect
subtheme_switches = 1-2	Reinforce subtheme	Medium	Panel shows items from same subtheme
subtheme_switches = 3-4	Suggest related subthemes	Medium	Panel shows items from adjacent subthemes
subtheme_switches = 5-7	Suggest thematic clusters	High	Panel shows items from multiple subthemes
subtheme_switches ≥ 8	Trigger unexpected artefacts	Very high	Panel shows 1-2 serendipity artefacts

Branching Factor Adaptive Rules			
IF	THEN	Priority	Effect
branching_factor = 1-2	Reinforce subtheme	Medium	Panel shows deeper items from same subtheme
branching_factor = 3-5	Suggest deeper related artefacts	Medium	Panel shows items from subthemes explored most
branching_factor = 6-9	Suggest cross- subtheme clusters	High	Panel shows items from multiple subthemes
branching_factor ≥ 10	Trigger serendipity suggestions	Very high	Panel shows unexpected artefacts

Serendipity Adaptive Rules			
IF	THEN	Priority	Effect
immersive_dwell (>30s) AND artefact_popularity = low	Show rare/unique artefacts	Very high	Panel displays 1-2 rare or unusual artefacts from low- frequency subthemes
theme_switches >= 2 AND subtheme_switches >= 5	Trigger serendipity mode	Very high	Panel mixes 1 unexpected artefact with 1 unrelated artefact
navigation_depth >= 8 AND revisit_count >= 2	Show deeper thematic items	High	Panel displays deeper items from the dominant subtheme
subtheme_switches >= 8 AND revisit_count >= 1	Show unexpected artefacts	Very high	Panel displays 1-2 artefacts from distant subthemes

Priority Levels		
Level	Description	Colour
Very High	Serendipity rules, immersive dwell, rapid switching	Red
High	Deep exploration, multiple revisits	Orange
Medium	Moderate dwell, exploratory drift	Yellow
Low	Light dwell, shallow navigation	Grey

D.3 Panel Composition Rules

Panel Composition Rules	
Mode	Composition
Reinforcement mode	3 related + 1 neutral
Deep-dive mode	2 deeper items + 1 related + 1 neutral
Exploratory drift	2 related + 2 cross-subtheme
Serendipity mode	2 serendipities + 1 related + 1 neutral
Balanced browsing	1 related + 1 deeper + 1 cross-subtheme + 1 serendipity

D.4 Rule Weighting Table

Priority	Multiplier	Colour
Very High	*2.0	Red
High	*1.5	Orange
Medium	*1.2	Yellow
Low	*1.0	Grey

D.5 Panel Selection Logic

Interest	Score Range	Panel Composition
Weak interest	0.0 – 0.5	Neutral items only
Moderate interest	0.6 – 1.0	2 related + 2 neutral
Strong interest	1.1 – 1.5	2 related + 1 deeper + 1 neutral
Very strong interest	1.6 – 2.0	2 deeper + 1 related + 1 serendipity
Serendipity trigger	>2.0	2 serendipities + 1 related + 1 deeper

D.6 Signal Weight Tables

Dwell Time Weights		
Category	Weight	Meaning
Micro-dwell (0-3s)	0.0	Ignore
Light dwell (3-8s)	0.4	Weak interest
Engaged dwell (8-15s)	0.8	Moderate interest
Deep dwell (15-30s)	1.2	Strong interest
Immersive dwell (>30s)	1.8	Very strong interest

Revisit Weights		
Category	Weight	Meaning
Single revisit	0.7	Confirmed interest
Multiple revisits (3+)	1.3	Strong attraction
Rapid revisit (<20s)	1.8	Curiosity spike
Delayed revisit (>1min)	1.0	Renewed interest

Navigation Depth Weights		
Category	Weight	Meaning
Shallow (1-3)	0.2	Early exploration
Moderate (4-7)	0.7	Browsing mode
Deep (8-12)	1.3	Strong exploration
Very deep (>12))	1.9	High engagement

Theme Switching Weights		
Category	Weight	Meaning
No switching	0.4	Focused interest
Single switch	0.7	Exploratory drift
Oscillation	1.2	Uncertain interest
Rapid switching (>=3)	1.8	Serendipitous browsing

Subtheme Switching Weights		
Category	Weight	Meaning
Low (1-2)	0.4	Focused exploration
Moderate (3-4)	0.8	Exploratory drift
High (5-7)	1.3	Broad exploration
Rapid (>=8)	1.9	Serendipity mode

Branching Factor Weights		
Category	Weight	Meaning
Low (1-2)	0.4	Narrow interest
Moderate (3-5)	0.9	Balanced exploration
High (6-9)	1.4	Broad exploration
Very high (≥ 10)	1.9	Serendipitous browsing

Serendipity Weights		
Category	Weight	Meaning
Immersive + rare (>30 s dwell, low popularity)	1.9	Rare, high-engagement artefact
Theme + subtheme drift (≥ 2 theme, ≥ 5 subtheme switches)	1.9	Broad, serendipitous browsing
Deep navigation + revisit (≥ 8 depth, ≥ 2 revisits)	1.4	Deep, repeated engagement
Subtheme drift + revisit (≥ 8 subtheme switches, ≥ 1 revisit)	1.9	Distant, exploratory browsing

D.7 Artefact Order Design Principles

Design Decision	Constraint/Rationale	Rule Applied	Verified By
Seeded by participant_id, not per-page load	A participant must see the same artefact order in both their static and adaptive sessions, so the shuffle can't depend on random state that changes each page load	Order is a deterministic function of participant_id, generated by a self-contained seeded LCG (SeededRandom) rather than PHP's global rand() or mt_rand(), so it's unaffected by – and can't leak into – any other randomness in the same request	Testing plans, AO2 “Same participant, same order twice”
Round-robin interleave across subthemes, not a pure shuffle	A pure random shuffle can still cluster several same-subtheme cards together by chance; literature cautions against pure randomness in serendipity-focused interfaces	All 12 subthemes are interleaved round-robin so adjacent cards are rarely from the shuffled subtheme visiting order and per-subtheme shuffle still keep every participant's grid different	Testing plans, AO1 “Different participants, different order” / AO3 “Subthemes interleaved, not blocked”

Appendix E: Test Cases & Results

E.1 Testing Plan & Test Case Tables

Phase 1 Testing Plan	
Section	Description
Goal	Ensure the database schema is correctly implemented, artefact metadata loads cleanly, and all relationships function as expected.
Components Under Test	Database scheme, tables, foreign keys, artefact metadata, theme/subtheme mapping, indexing, study_condition, questionnaire_item, and adaptive_rule reference data.
Test Types	Structural testing, CRUD testing, referential integrity testing, basic performance checks.
Entry Criteria	All tables created: metadata imported: DB connection established: reference data seeded (study_condition, questionnaire_item, adaptive_rule).
Exit Criteria	All tables validated: artefacts retrievable; foreign keys functioning; reference data (study_condition, questionnaire_item, adaptive_rule) verified; no broken records.
Tools Used	phpMyAdmin/MySQL Workbench, SQL queries, browser test page.
Test Evidence	Screenshots of queries, logs of successful inserts, artefact fetch results.

Phase 1 Test Cases				
ID	Description	Steps	Expected Result	Pass
DF1	Verify artefact table loads correctly	Run SELECT * FROM artefact	All 60 artefacts appear with correct fields	Yes
DF2	Verify theme/subtheme mapping	Run join query linking artefact -> subtheme -> theme	Correct theme/subtheme returned for each artefact	Yes
DF3	Test foreign key constraints	Attempt to insert artefact with invalid subtheme_id	Insert rejected with FK error	Yes

DF4	Test session insertion	Insert dummy session into session_log	Session appears with correct timestamp	Yes
DF5	Test event insertion	Insert dummy event into event_log	Event stored with correct session_id + artefact_id	Yes
DF6	Verify study_condition table populated correctly	Run SELECT * FROM study_condition	Both static_first and adaptive_first rows appear with descriptions	Yes
DF7	Verify questionnaire_item table populated correctly	Run SELECT * FROM questionnaire_item ORDER BY item_order	All 14 items appear in order 1-14 with correct item_type values	Yes
DF8	Verify adaptive_rule table populated correctly	Run SELECT * FROM adaptive_rule	All 27 rules appear with active_flag = TRUE and valid trigger_condition JSON	Yes
DF9	Test session_log foreign key to study_condition	Attempt to insert session_log row with invalid condition_id	Insert rejected with FK error	Yes

Phase 2 Testing Plan	
Section	Description
Goal	Validate that all behavioural events are captured, transmitted, and stored correctly.
Components Under Test	/event, /adaptive-event, /timeout, /session/start, /session/end endpoints; backend validation; event_log table.
Test Types	API testing, backend validation, timestamp accuracy, data integrity.
Entry Criteria	Database stable; endpoints implemented; text page available.
Exit Criteria	All event types log correctly; timestamps accurate; no missing fields.
Tools Used	Browser console, network inspector, SQL queries, test page.
Test Evidence	Logged events in DB, console logs, network request screenshots.

Phase 2 Test Cases				
ID	Description	Steps	Expected Result	Pass
BL1	Test click event logging	Click test button	event_log stores click event	Yes
BL2	Test dwell_start + dwell_end	Open modal -> close modal	Two events logged with correct dwell duration	Yes
BL3	Test navigation depth	Open 3 artefacts sequentially	navigation_depth increments correctly	Yes
BL4	Test revisit detection	Open same artefact twice	revisit flag logged	Yes
BL5	Test session_start	Load home page	session_start event logged	Yes
BL6	Test timeout event	Trigger timeout manually	timeout event logged with correct version	Yes
BL7	Test session_end	Submit questionnaire / end session	session_end event logged	Yes
BL8	Test adaptive event logging	Simulate adaptive_trigger, panel_impression, panel_click	Adaptive events logged correctly to event_log and adaptive_event	Yes

Phase 3 Testing Plan	
Section	Description
Goal	Ensure static browsing works flawlessly, logs behaviour correctly, and transitions correctly.
Components Under Test	/index.php, static.php, transition page 1, modal, timeout, logging.
Test Types	Functional testing, UI testing, behavioural logging validation, flow testing.
Entry Criteria	Behaviour logging engine stable; home page implemented.
Exit Criteria	Static version fully functional; all events logged; transitions correct.
Tools Used	Browser, console, SQL queries.
Test Evidence	Screenshots, event logs, session logs.

Phase 3 Test Cases				
ID	Description	Steps	Expected Result	Pass

SP1	Test home page loads	Open index.php	Home page displays correctly	Yes
SP2	Test static grid loads	Click “Static Version”	Grid loads with all artefacts	Yes
SP3	Test modal behaviour	Click artefact -> close modal	Modal opens/closes; dwell logged	Yes
SP4	Test timeout redirect	Wait 7 minutes	Redirect to Transition Page 1	Yes
SP5	Test transition logic	Check button text	Correct next version displayed	Yes
SP6	Test revisit detection	Open same artefact twice	revisit logged	Yes
SP7	Test researcher session setup	Submit participant code + condition on index.php	session_log row + session_start event created; condition stored correctly	Yes
SP8	Test navigation guards	Access static.php / transition1.php directly with no active session (or task already completed)	Safely redirects to index.php - no page content, no logging	Yes
SP9	Test emergency end-session control	With a session active, click “End session (researcher only)” on index.php and confirm the dialog	session_log.session_end set; session_end event logged; page returns to researcher setup form with no active session	Yes

Phase 4 Testing Plan	
Section	Description
Goal	Validate backend adaptive rule evaluation and suggestion generation.
Components Under Test	Rule evaluation functions, scoring model, suggestion selection, backend test harness.
Test Types	Unit testing, rule testing, scoring validation, output correctness.
Entry Criteria	Static version stable; logging engine stable.
Exit Criteria	Engine returns correct suggestions; rule priority works; serendipity triggers correctly.
Tools Used	Backend test harness, SQL queries, console logs.
Test Evidence	Suggestion lists, rule triggers, scoring outputs.

Phase 4 Test Cases				
ID	Description	Steps	Expected Result	Pass
AE1	Test dwell threshold rule	Simulate 20s dwell	Deep-dive rule fires	Yes
AE2	Test revisit rule	Simulate 3 revisits	Multiple-revisit rule fires	Yes
AE3	Test switching rule	Simulate rapid switching	Serendipity rule fires	Yes
AE4	Test scoring model	Feed known weights	Correct interest score returned	Yes
AE5	Test suggestion diversity	Simulate broad browsing	4 unique suggestions returned	Yes
AE6	Test rule priority	Trigger multiple rules	Serendipity overrides others	Yes
AE7	Test navigation depth rule	Simulate navigation_depth reaching 8 (deep)	navdepth_deep rule fires; deep-dive suggestion signal generated	Yes
AE8	Test branching factor rule	Simulate branching_factor reaching 6 (high)	branching_high rule fires; cross-subtheme cluster signal generated	Yes

Phase 5 Testing Plan	
Section	Description
Goal	Validate adaptive UI behaviour, panel updates, and logging.
Components Under Test	adaptive.php, adaptive panel, heartbeat updates, event-driven updates, transition page 2.
Test Types	UI testing, behavioural testing, adaptive logic integration, flow testing.
Entry Criteria	Adaptive engine stable; static version stable.
Exit Criteria	Panel updates correctly; no layout shifts; adaptive events logged.
Tools Used	Browser, console, SQL queries.
Test Evidence	Panel update logs, adaptive_event entries, screenshots.

Phase 5 Test Cases				
ID	Description	Steps	Expected Result	Pass
AP1	Test panel loads	Open adaptive.php	Panel shows "Suggested for You"	Yes
AP2	Test heartbeat update	Wait 20s	Panel updates quietly	Yes
AP3	Test dwell-trigger update	Dwell > 30s	Panel updates with deep dive + serendipity	Yes
AP4	Test panel click logging	Click suggestion	panel_click logged	Yes
AP5	Test freeze-before-timeout	Wait final 30s	Panel stops updating	Yes
AP6	Test transition page 2	Wait 7 minutes	Redirect to questionnaire	Yes

Phase 6 Testing Plan	
Section	Description
Goal	Ensure questionnaire loads, submits, and logs session_end correctly.
Components Under Test	questionnaire.php, Microsoft Form embed, session_end logging.
Test Types	UI testing, form submission testing, logging validation.
Entry Criteria	Transition page 2 functional.
Exit Criteria	Questionnaire submits; session_end logged; flow complete.
Tools Used	Browser, SQL queries.
Test Evidence	Form submission confirmation, session_end logs.

Phase 6 Test Cases				
ID	Description	Steps	Expected Result	Pass
Q11	Test questionnaire loads	Open questionnaire.php	Form visible	Yes
Q12	Test submission	Submit form	Microsoft confirmation appears	Yes
Q13	Test session_end logging	Submit form	session_end logged	Yes

Phase 7 Testing Plan	
Section	Description

Goal	Ensure system recovers gracefully from errors and preserves study flow.
Components Under Test	fallback.php, error_trigger logging, session reset logic.
Test Types	Error simulation, flow testing, recovery testing.
Entry Criteria	All main pages implemented.
Exit Criteria	Fallback always returns to home; no broken flow.
Tools Used	Browser, console, SQL queries.
Test Evidence	Error logs, fallback behaviour screenshots.

Phase 7 Test Cases				
ID	Description	Steps	Expected Result	Pass
FB1	Test missing session variable	Load adaptive.php directly	Fallback screen appears	Yes
FB2	Test broken DB connection	Disable DB temporarily	Fallback screen appears	Yes
FB3	Test incorrect navigation	Jump from static -> questionnaire	Fallback screen appears	Yes
FB4	Test reset logic	Click "Return to Home"	Home loads cleanly	Yes

Phase 8 Testing Plan	
Section	Description
Goal	Randomise initial artefact display order to reduce position/subtheme clustering, without breaking comparability between the static and adaptive versions.
Components Under Test	includes/artefact_order.php, static.php grid query, adaptive.php grid query, session_log.artefact_order.
Test Types	Randomisation testing, consistency testing, regression testing.
Entry Criteria	Phase 7 complete; static.php and adaptive.php functional.
Exit Criteria	Order is randomised per participant, stable within a session, and identical between static and adaptive for the same participant.
Tools Used	Browser, SQL queries, PHP test scripts.
Test Evidence	Rendered grid screenshots, stored artefact_order values.

Phase 8 Test Cases				
ID	Description	Steps	Expected Result	Pass
AO1	Test order randomisation	Load static.php as two different participants	Different artefact orders appear	Yes
AO2	Test order consistency within participant	Load static.php then adaptive.php as the same participant	Same artefact order appears in both	Yes
AO3	Test subtheme interleaving	Inspect the first 12 artefacts shown	No two consecutive artefacts share the same subtheme	Yes
AO4	Test order stability on reload	Reload static.php mid-session	Artefact order remains unchanged	Yes

Phase 9 Testing Plan	
Section	Description
Goal	Validate full participant flow end-to-end.
Components Under Test	Entire system.
Test Types	End-to-end testing, flow testing, data integrity testing.
Entry Criteria	All components implemented.
Exit Criteria	Full session runs without errors; logs complete.
Tools Used	Browser, SQL queries, log exports.
Test Evidence	Full session logs, screenshots, exported CSV.

Phase 9 Test Cases				
ID	Description	Steps	Expected Result	Pass
FS1	Full static-first flow	Static -> Transition -> Adaptive -> Transition -> Questionnaire	Entire flow works	Yes
FS2	Full adaptive-first flow	Adaptive -> Transition -> Static -> Transition -> Questionnaire	Entire flow works	Yes
FS3	Data integrity check	Export logs	All events present	Yes

Phase 10 Testing Plan	
Section	Description
Goal	Validate system with real participants before full evaluation.
Components Under Test	Full system + UX.

Test Types	Usability testing, behavioural testing, logging validation.
Entry Criteria	Full simulation successful.
Exit Criteria	No major issues; ready for evaluation.
Tools Used	Participant sessions, SQL queries, observation notes.
Test Evidence	Pilot logs, participant feedback.

Phase 10 Test Cases				
ID	Description	Steps	Expected Result	Pass
PT1	Run pilot session	Participant completes full study	No errors	Yes
PT2	Check adaptive behaviour	Observe panel updates	Behaviour matches rules	No
PT3	Check logs	Export logs	All events present	Yes
PT4	Check serendipity frequency control	Trigger serendipity twice within 20s	Second trigger blocked; third (after gap) allowed	Yes

E.2 Phase 1 Test Case Results

DF1 - Verify artefact table loads correctly

Rows returned: 60 (expected 60)

RESULT: PASS

All 60 artefacts were returned, each with every required field populated.

DF2 - Verify theme/subtheme mapping

Joined rows returned: 60 (expected 60)

RESULT: PASS

All 60 artefacts joined correctly to a subtheme and theme, and each artefact's theme_id matches its subtheme's parent theme.

DF3 - Test foreign key constraints

Attempted: INSERT INTO artefact with subtheme_id = 999999 (does not exist)
Artefact row count before: 60, after: 60

RESULT: PASS

Insert was correctly rejected with a foreign key constraint error, and no row was left behind:

- SQLSTATE[23000]: Integrity constraint violation: 1452 Cannot add or update a child row: a foreign key constraint fails (`digitalmuseum`.`artefact`, CONSTRAINT `fk\_artefact\_subtheme` FOREIGN KEY (`subtheme\_id`) REFERENCES `subtheme` (`subtheme\_id`) ON DELETE RESTRICT ON UPDATE CASCADE)

DF4 - Test session insertion

Inserted dummy session_id: 1 with session_start = 2026-07-18 11:24:59
(Test data cleaned up after verification.)

RESULT: PASS

Session was inserted and retrieved with the correct participant_id, condition_id, and timestamp.

DF5 - Test event insertion

Inserted dummy event_id: 1 for session_id=2, artefact_id=1
(Test data cleaned up after verification.)

RESULT: PASS

Event was inserted and retrieved with the correct session_id, artefact_id, event_type, and timestamp.

DF6 - Verify study_condition table populated correctly

Rows returned: 2 (expected 2)
condition_name values found: static_first, adaptive_first

RESULT: PASS

Both 'static_first' and 'adaptive_first' rows are present, each with a description.

DF7 - Verify questionnaire_item table populated correctly

Rows returned: 14 (expected 14)

item_order sequence: 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14

RESULT: PASS

All 14 items are present in order 1-14, each with a valid item_type matching the questionnaire design, and likert items carry both scale labels.

DF8 - Verify adaptive_rule table populated correctly

Rows returned: 27 (expected 27)

RESULT: PASS

All 27 rules are present, active, with unique names and well-formed trigger_condition JSON (signal, priority, effect all present).

DF9 - Test session_log foreign key to study_condition

Attempted: INSERT INTO session_log with condition_id = 999999 (does not exist)

session_log row count before: 0, after: 0

RESULT: PASS

Insert was correctly rejected with a foreign key constraint error, and no row was left behind:

- SQLSTATE[23000]: Integrity constraint violation: 1452 Cannot add or update a child row: a foreign key constraint fails (`digitalmuseum`.`session\_log`, CONSTRAINT `fk\_session\_condition` FOREIGN KEY (`condition\_id`) REFERENCES `study\_condition` (`condition\_id`) ON DELETE RESTRICT ON UPDATE CASCADE)

E.3 Phase 2 Test Case Results

BL5 - Test session_start

```
-----
Endpoint tested: http://localhost:8888/digitalmuseum/api/session_start.php
Session(s) created during test: 4, 5
(Test data cleaned up after verification.)
```

RESULT: PASS

session_start creates a session_log row and a matching session_start event_log row, reuses existing participants, and rejects invalid requests correctly.

BL7 - Test session_end

```
-----
Endpoint tested: http://localhost:8888/digitalmuseum/api/session_end.php
Session used for test: 9
(Test data cleaned up after verification.)
```

RESULT: PASS

session_end closes the session, logs exactly one matching session_end event, stays idempotent on repeat calls, and rejects invalid requests correctly.

BL1-BL4 - Test click, dwell_start/dwell_end, navigation_depth, revisit

```
-----
Endpoint tested: http://localhost:8888/digitalmuseum/api/log_event.php
Session used for test: 7
(Test data cleaned up after verification.)
```

RESULT: PASS

click, dwell_start/dwell_end (with correct dwell_duration), navigation_depth (incrementing correctly), and revisit events are all logged correctly, and invalid requests are rejected.

BL8 - Test adaptive event logging (adaptive_trigger, panel_impression, panel_click)

```
-----
Endpoint tested: http://localhost:8888/digitalmuseum/api/log_adaptive_event.php
Session used for test: 10
(Test data cleaned up after verification.)
```

RESULT: PASS

adaptive_trigger, panel_impression, and panel_click events all log correctly to both event_log and adaptive_event, and invalid requests are rejected.

BL6 - Test timeout event

```
-----
Endpoint tested: http://localhost:8888/digitalmuseum/api/log_timeout.php
Session used for test: 11
(Test data cleaned up after verification.)
```

RESULT: PASS

Static and adaptive timeouts both log a timeout row and a matching timeout_redirect event, flip the correct session_log flag, and invalid requests are rejected.

Phase 2 - Behaviour Logging Test Page

Drives js/logging.js and js/timeout.js through real browser interaction (BL1-BL6). Open your browser console and/or network inspector alongside the log panel below.

Session

session_id: 12 | navigation_depth: 4

Artefacts

Artefact 1

Artefact 2

Artefact 3

Open artefact 1 twice (in separate modal opens) to trigger a revisit (BL4). Open 1 -> 2 -> 3 in sequence to watch navigation depth increment (BL3).

Timeout (manual trigger, does not redirect)

Console output for debugging

```
[11:02:59] INFO Test page ready. Enter a participant code and click "Start Session" to begin.
[11:03:54] DEBUG [logging] /digitalmuseum/api/session_start.php
{"participant_code":"TEST_PAGE_779443","condition_name":"adaptive_first"} ->
{"success":true,"session_id":12,"participant_id":11,"condition_id":2,"session_start":07-19 09:03:54"}
[11:03:58] DEBUG [logging] /digitalmuseum/api/log_event.php
{"session_id":12,"artefact_id":1,"event_type":"click"} ->
{"success":true,"event_id":22,"session_id":12,"artefact_id":1,"event_type":"click","07-19 09:03:58","dwell_duration":null,"navigation_depth":null}
[11:04:00] DEBUG [logging] /digitalmuseum/api/log_event.php
{"session_id":12,"artefact_id":1,"event_type":"navigation_depth","navigation_depth":1} ->
{"success":true,"event_id":23,"session_id":12,"artefact_id":1,"event_type":"navigation_depth","07-19 09:04:00","dwell_duration":null,"navigation_depth":1}
[11:04:00] DEBUG [logging] /digitalmuseum/api/log_event.php
{"session_id":12,"artefact_id":1,"event_type":"dwell_start"} ->
```

E.4 Phase 3 Test Case Results

RESEARCHER SETUP

Participant code

Condition **Select...**

Start Session

Digital Museum Prototype Study

Exploring how people browse and discover artefacts.

This study includes two browsing tasks followed by a short questionnaire.

Static Version
Browse a collection of artefacts in a non-adaptive interface.

Adaptive Version
Browse a collection where artefacts update based on your interactions.

Post-Study Questionnaire
Share your experience after completing both versions.

You will complete both versions in the order designated to you.

Please explore freely and click on any artefacts that interest you.

When finished, complete the questionnaire.

Study ID: DC_25259628

Researcher: Kate Forshaw

Edge Hill University

Ethics approval reference: [to be confirmed]

Session active - Participant: 13 (session #14)

End session (researcher only)

Digital Museum Prototype Study

Exploring how people browse and discover artefacts.

This study includes two browsing tasks followed by a short questionnaire.

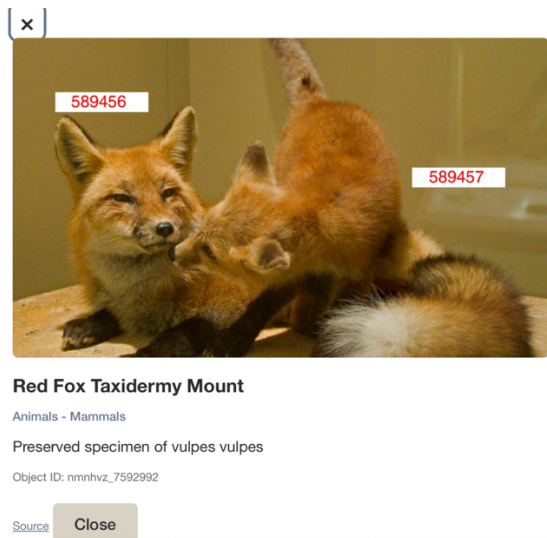
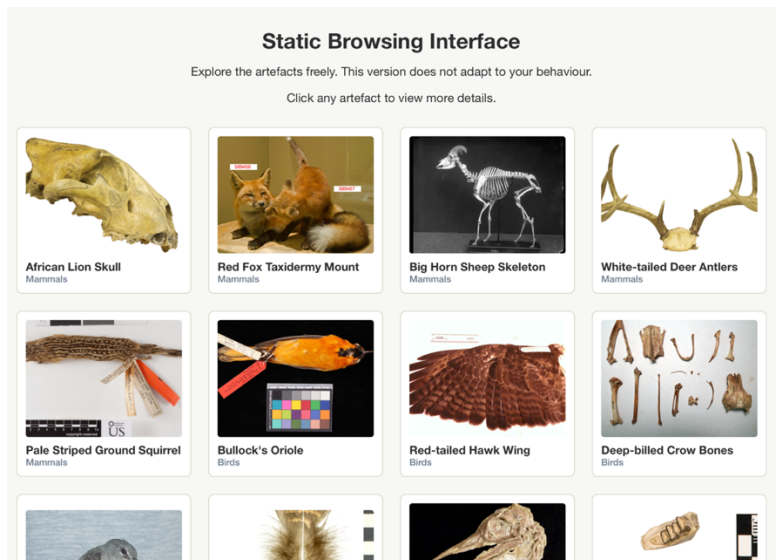
Static Version
Browse a collection of artefacts in a non-adaptive interface.

Adaptive Version
Browse a collection where artefacts update based on your interactions.

Post-Study Questionnaire
Share your experience after completing both versions.

You will complete both versions in the order designated to you.

Please explore freely and click on any artefacts that interest you.



49

14

2

NULL

NULL revisit

2026-07-21 07:28:59

First Browsing Task Complete

You will now continue to the next version of the digital museum prototype.

Thank you for completing the first browsing session.

Click the button below when you are ready to continue.

[Continue to Adaptive Version](#)

End this session now?

This is for the researcher only - not part of the participant task. It will immediately close session #14.

[Cancel](#) [OK](#)

E.5 Phase 4 Test Case Results

```

AE1, AE2, AE3, AE7, AE8 - Adaptive rule firing
-----
Active rules loaded from adaptive_rule: 27

RESULT: PASS
Each simulated signal fired exactly the rule Testing Plans.docx expects.

```

```

AE6 - Test rule priority
-----
Active rules loaded from adaptive_rule: 27

RESULT: PASS
Serendipity rules override same- and higher-priority non-serendipity rules, plain
priority rank decides when no serendipity rule fires, and no rules firing resolves to
null rather than an error.

```

```

AE4 - Test scoring model
-----
No database connection used - pure arithmetic over hand-fed weights.

RESULT: PASS
calculateInterestScore()/categoriseInterestScore()/meetsSerendipityThreshold() all
return the correct values for known weight inputs, including the >2.0 boundary.

```

```

AE1, AE2, AE3, AE5, AE7, AE8 - Adaptive rule firing + suggestion diversity
-----
Active rules loaded from adaptive_rule: 27

RESULT: PASS
Each simulated signal fired exactly the rule Testing Plans.docx expects, and every panel
mode - plus the full engine pipeline - returned 4 unique artefacts for a simulated
broad-browsing session.

```

E.6 Phase 5 Test Case Results

AP1 - Test panel loads

 Pages tested: http://localhost:8888/digitalmuseum/index.php,
 http://localhost:8888/digitalmuseum/adaptive.php
 (Test data cleaned up after verification.)

RESULT: PASS

adaptive.php redirects away with no active session, and loads correctly with one - showing "Suggested for You", the neutral empty-state message, and 4 empty panel slots ready for the engine to fill.

AP2 - Test heartbeat update

 Endpoint tested: http://localhost:8888/digitalmuseum/api/trigger_rule.php
 (Test data cleaned up after verification.)

RESULT: PASS

A heartbeat-style call (session_id only, no artefact_id) stays silent on an idle session, and, once navigation_depth crosses the threshold, quietly produces and persists a real 4-suggestion update.

AP3 - Test dwell-trigger update

 Endpoint tested: http://localhost:8888/digitalmuseum/api/trigger_rule.php
 (Test data cleaned up after verification.)

RESULT: PASS

A >30s dwell on a rarely-viewed artefact resolves to serendipity mode (2 serendipity + 1 related + 1 neutral) via the correctly-attributed serendipity_immersive_rare rule, and is fully persisted.

AP4 - Test panel click logging

 Endpoint tested: http://localhost:8888/digitalmuseum/api/log_adaptive_event.php
 (Test data cleaned up after verification.)

RESULT: PASS

Clicking a suggestion logs a panel_click event correctly and sets adaptive_suggestion.clicked_flag, scoped to exactly that suggestion, and repeat clicks keep logging without disturbing the flag.

AP5 - Test freeze-before-timeout

 Endpoint tested: http://localhost:8888/digitalmuseum/api/trigger_rule.php
 (Test data cleaned up after verification.)

RESULT: PASS

The same qualifying browsing signals update the panel normally right up to the 390s mark, and are correctly suppressed (with nothing persisted) from that point through the end of the session.

AP6 - Test transition page 2

 Pages/endpoints tested: http://localhost:8888/digitalmuseum/index.php,
 http://localhost:8888/digitalmuseum/transition2.php,
 http://localhost:8888/digitalmuseum/api/log_timeout.php
 (Test data cleaned up after verification.)

RESULT: PASS

transition2.php stays hidden with no session, and after only one task times out, and shows its content with the Continue to Questionnaire link only once both browsing tasks have timed out.

E.7 Phase 6 Test Case Results

QI1 - Test questionnaire loads

Pages/endpoints tested: http://localhost:8888/digitalmuseum/index.php,
http://localhost:8888/digitalmuseum/questionnaire.php,
http://localhost:8888/digitalmuseum/api/log_timeout.php
(Test data cleaned up after verification.)

RESULT: PASS

questionnaire.php stays hidden with no session and before both tasks time out, then renders the full 14-item native form (not an iframe) once both browsing tasks are complete.

QI2 - Test submission

Pages/endpoints tested: http://localhost:8888/digitalmuseum/index.php,
http://localhost:8888/digitalmuseum/questionnaire.php,
http://localhost:8888/digitalmuseum/api/log_timeout.php
(Test data cleaned up after verification.)

RESULT: PASS

An incomplete submission is rejected and redisplayes the form; a complete submission saves all 14 responses, closes the session, and shows the completion screen; resubmitting afterwards changes nothing.

QI3 - Test session_end logging

Endpoint tested: http://localhost:8888/digitalmuseum/api/post_questionnaire_response.php
Session used for test: 28
(Test data cleaned up after verification.)

RESULT: PASS

A valid submission closes the session and logs exactly one matching session_end event; resubmitting a closed session is rejected (not silently accepted); and invalid requests (unknown session, mismatched participant, missing responses, wrong method) are all rejected correctly.

E.8 Phase 7 Test Case Results

FB1 - Test missing session variable

 Pages tested: adaptive.php (per FB1's steps), plus static.php, transition1.php, transition2.php, questionnaire.php (same guard, same page, extra coverage).
 (No database records were created for this test.)

RESULT: PASS

Every participant-facing page redirects to fallback.php?reason=missing_session when there is no active session, and fallback.php renders the correct calm screen with a working Return to Home button.

FB2 - Test broken DB connection

 Automated: fallback.php's broken_db message, and safe_db_connect()'s resilience against an unreachable DB (simulated via a closed port, not the real MySQL instance).

AUTOMATED RESULT: PASS

fallback.php shows the correct message for reason=broken_db, and the connection helper every guarded page relies on fails gracefully (returns null, doesn't hang) when the database is unreachable.

MANUAL STEP STILL NEEDED TO FULLY CLOSE OUT FB2:

=====

This script cannot safely stop your live MySQL instance mid-test, so the literal FB2 step ("Disable DB temporarily") needs one manual check:

1. In MAMP, stop the MySQL server (leave Apache running).
2. In a browser, load static.php or adaptive.php directly.
3. Confirm the fallback screen appears with: "We're having trouble reaching the study database right now."
4. Restart MySQL in MAMP before continuing any other testing.

Something Went Wrong

We're having trouble reaching the study database right now.

No progress has been lost. Please return to the home screen to continue.

[Return to Home](#)

FB3 - Test incorrect navigation

 Pages/endpoints tested: http://localhost:8888/digitalmuseum/index.php, http://localhost:8888/digitalmuseum/static.php, http://localhost:8888/digitalmuseum/transition1.php, http://localhost:8888/digitalmuseum/transition2.php, http://localhost:8888/digitalmuseum/questionnaire.php, http://localhost:8888/digitalmuseum/api/log_timeout.php
 (Test data cleaned up after verification.)

RESULT: PASS

Every out-of-order navigation attempt (skipping ahead to questionnaire.php or transition2.php before both tasks are done, reaching transition1.php before the first task is done, or re-entering a finished static.php) correctly shows the fallback screen instead of the requested page.

FB4 - Test reset logic

Pages tested: <http://localhost:8888/digitalmuseum/index.php>,
<http://localhost:8888/digitalmuseum/fallback.php>,
<http://localhost:8888/digitalmuseum/static.php>
(Test data cleaned up after verification.)

RESULT: PASS

Clicking Return to Home on the fallback screen closes out the open DB session, fully clears the local PHP session, and lands cleanly back on index.php's researcher-setup screen - with no way for the same browser session to keep riding on the old, now-closed session.

E.9 Phase 8 Test Case Results

A01 - Test order randomisation

 Pages tested: <http://localhost:8888/digitalmuseum/index.php>,
<http://localhost:8888/digitalmuseum/static.php>
 (Test data cleaned up after verification.)

RESULT: PASS

Two different participants loading static.php were shown different, but each fully complete, artefact grid orders.

A02 - Test order consistency within participant

 Pages tested: <http://localhost:8888/digitalmuseum/index.php>,
<http://localhost:8888/digitalmuseum/static.php>,
<http://localhost:8888/digitalmuseum/adaptive.php>
 (Test data cleaned up after verification.)

RESULT: PASS

The same participant was shown the exact same, fully complete, artefact grid order on both static.php and adaptive.php.

A03 - Test subtheme interleaving

 Pages tested: <http://localhost:8888/digitalmuseum/index.php>,
<http://localhost:8888/digitalmuseum/static.php>
 (Test data cleaned up after verification.)

RESULT: PASS

No two consecutive artefacts share the same subtheme, in the first 12 cards shown or across the full rendered grid.

A04 - Test order stability on reload

 Pages tested: <http://localhost:8888/digitalmuseum/index.php>,
<http://localhost:8888/digitalmuseum/static.php>
 (Test data cleaned up after verification.)

RESULT: PASS

Reloading static.php mid-session showed the exact same, fully complete, artefact grid order every time.

E.10 Phase 9 Tests Case Results

Session active - Participant: 49 (session #50) End session (researcher only)

Digital Museum Prototype Study

Exploring how people browse and discover artefacts.
This study includes two browsing tasks followed by a short questionnaire.

Static Version
Browse a collection of artefacts in a non-adaptive interface.

Adaptive Version
Browse a collection where artefacts update based on your interactions.

Post-Study Questionnaire
Share your experience after completing both versions.

You will complete both versions in the order designated to you.
Please explore freely and click on any artefacts that interest you.
When finished, complete the questionnaire.

Study ID: DC_25259628
Researcher: Kate Forshaw
Edge Hill University
Ethics approval: Confirmed

Static Browsing Interface

Explore the artefacts freely. This version does not adapt to your behaviour.
Click any artefact to view more details.



Agamidae Lizard
Reptiles



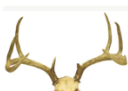
Tongue Plane
Tools



Shoe Buckles
Clothing



Bed Frame
Household Items



White-tailed Deer Antlers
Mammals



California Spiny Lobster
Marine Life



Soup Ladle
Food-related



Toy Bicycle Rider
Toys









First Browsing Task Complete

You will now continue to the next version of the digital museum prototype.

Thank you for completing the first browsing session.
Click the button below when you are ready to continue.

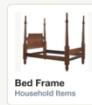
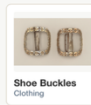
[Continue to Adaptive Version](#)

Please explore freely in the next session.
Each browsing session lasts approximately 7 minutes.

Adaptive Browsing Interface

This version updates artefact suggestions based on your browsing behaviour.

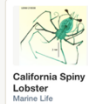
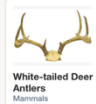
Explore freely. Suggestions will appear in the adaptive panel.



Suggested for You

Based on your browsing

Suggestions will appear here as you explore.



Browsing Sessions Complete

You will now complete a short questionnaire about your experience.

Thank you for completing both browsing sessions.

Please continue to the questionnaire when you are ready.

[Continue to Questionnaire](#)

Your responses will help us understand how people explore digital museum collections.

Thank you for taking part.

Post-Study Questionnaire

Please complete the short questionnaire about your experience.

Your responses help us understand how people explore digital museum collections.

About the Study Session

Which version did you complete first?

- ☐ Static version
☐ Adaptive version

Static Prototype Feedback

The artefacts shown in static version felt relevant to my interests.

Strongly disagree ☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 Strongly agree

I encountered artefacts in the static version that were unexpected or surprising.

Strongly disagree ☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 Strongly agree

The static version allowed me to explore freely without feeling restricted.

Strongly disagree ☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 Strongly agree

Overall, my experience using the static version was positive.

Strongly disagree ☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 Strongly agree

Post-Study Questionnaire

Please complete the short questionnaire about your experience.
Your responses help us understand how people explore digital museum collections.

Thank you for completing the questionnaire.
Your responses have been recorded. You may now close this window, or
return to the home screen.

[Return to Home](#)

Thank you for taking part in this study.

[Return to Home](#)

Session active - Participant: 50 (session #51)

End session (researcher only)

Digital Museum Prototype Study

Exploring how people browse and discover artefacts.

This study includes two browsing tasks followed by a short questionnaire.

Static Version

Browse a collection of artefacts in a non-adaptive interface.

Adaptive Version

Browse a collection where artefacts update based on your interactions.

Post-Study Questionnaire

Share your experience after completing both versions.

You will complete both versions in the order designated to you.

Please explore freely and click on any artefacts that interest you.

When finished, complete the questionnaire.

Study ID: DC_25259628
Researcher: Kate Forshaw
Edge Hill University
Ethics approval: Confirmed

Adaptive Browsing Interface

This version updates artefact suggestions based on your browsing behaviour.

Explore freely. Suggestions will appear in the adaptive panel.



Teddy Bear
Toys



Stegosaurus Dinosaur Skeleton
Reptiles



Scroll-Saw Blade
Tools



Ancient Ostrich Relative Fossil
Fossils



Knife
Food-related



Pale Striped Ground Squirrel
Mammals



Desk Clock
Household Items



Deep-billed Crow Bones
Birds



Staghorn Coral Fragment
Marine Life



Queen Honeybee Cage
Insects



Ink Bottle
Writing



Wool Embroidered Cap

Suggested for You
Based on your browsing
Suggestions will appear here as you explore.

First Browsing Task Complete

You will now continue to the next version of the digital museum prototype.

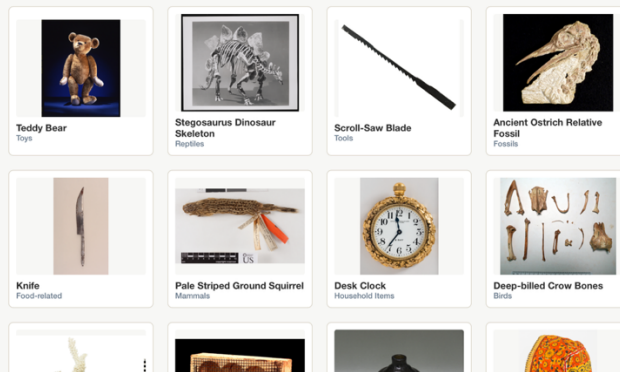
Thank you for completing the first browsing session.
Click the button below when you are ready to continue.

Continue to Static Version

Please explore freely in the next session.
Each browsing session lasts approximately 7 minutes.

Static Browsing Interface

Explore the artefacts freely. This version does not adapt to your behaviour.
Click any artefact to view more details.



Browsing Sessions Complete

You will now complete a short questionnaire about your experience.

Thank you for completing both browsing sessions.
Please continue to the questionnaire when you are ready.

Continue to Questionnaire

Your responses will help us understand how people explore digital museum collections.
Thank you for taking part.

Post-Study Questionnaire

Please complete the short questionnaire about your experience.
Your responses help us understand how people explore digital museum collections.

About the Study Session
Which version did you complete first?

☐ Static version
☐ Adaptive version

Static Prototype Feedback
The artefacts shown in static version felt relevant to my interests.

Strongly disagree ☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 Strongly agree

I encountered artefacts in the static version that were unexpected or surprising.

Strongly disagree ☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 Strongly agree

The static version allowed me to explore freely without feeling restricted.

Strongly disagree ☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 Strongly agree

Overall, my experience using the static version was positive.

Strongly disagree ☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 Strongly agree

Post-Study Questionnaire

Please complete the short questionnaire about your experience.
Your responses help us understand how people explore digital museum collections.

Thank you for completing the questionnaire.
Your responses have been recorded. You may now close this window, or return to the home screen.

[Return to Home](#)

Thank you for taking part in this study.

[Return to Home](#)

```

FS3 - Data integrity check (export + completeness)
-----
admin_test_1 (session 38):
  click: 41
  navigation_depth: 41
  dwell_start: 41
  dwell_end: 41
  revisit: 4
  adaptive_trigger: 32
  panel_impression: 128
  panel_click: 15
  questionnaire_response: 14 of 10 required answered
  Exported: fs3_exports/admin_test_1_event_log.csv, admin_test_1_timeout.csv,
admin_test_1_questionnaire_response.csv

admin_test_2 (session 39):
  click: 45
  navigation_depth: 45
  dwell_start: 45
  dwell_end: 45
  revisit: 5
  adaptive_trigger: 39
  panel_impression: 156
  panel_click: 17
  questionnaire_response: 10 of 10 required answered
  Exported: fs3_exports/admin_test_2_event_log.csv, admin_test_2_timeout.csv,
admin_test_2_questionnaire_response.csv

RESULT: PASS
Every requested session has exactly the events a complete run guarantees
(session_start/session_end, both timeouts, both timeout_redirects, no error_trigger,
every required questionnaire item answered), and full CSV exports have been written to
tests/phase9/fs3_exports/.

```

E.11 Phase 10 Test Case Results

participant_id	participant_code	participant_notes
37	admin_test_1	NULL
38	admin_test_2	NULL
39	admin_test_3	NULL
40	admin_test_4	NULL
41	admin_test_5	NULL
42	admin_test_6	NULL
43	admin_test_7	NULL
44	admin_test_8	NULL
45	admin_test_9	NULL
46	PILOT1	NULL
47	PILOT2	NULL
48	admin_test_10	NULL
49	test_FS1	NULL
50	test_FS2	NULL
52	ppt1	NULL
53	ppt2	NULL

```
PT2 - Check adaptive behaviour
-----
Sessions checked: 2 (participant_code PILOT1/PILOT2)
Adaptive triggers checked: 72

RESULT: FAIL
4 issue(s) found:
- PILOT1 / session 47 / rule 'serendipity_theme_subtheme_switch' @ 2026-08-02 09:42:18:
expected 4 adaptive_suggestion rows, found 8.
- PILOT1 / session 47 / rule 'serendipity_theme_subtheme_switch' @ 2026-08-02 09:42:18:
expected 4 adaptive_suggestion rows, found 8.
- PILOT2 / session 48 / rule 'serendipity_theme_subtheme_switch' @ 2026-08-02 10:03:34:
expected 4 adaptive_suggestion rows, found 8.
- PILOT2 / session 48 / rule 'serendipity_theme_subtheme_switch' @ 2026-08-02 10:03:34:
expected 4 adaptive_suggestion rows, found 8.
```

```
PT4 - Check serendipity frequency control
-----
Endpoint tested: http://localhost:8888/digitalmuseum/api/trigger_rule.php
(Test data cleaned up after verification.)

RESULT: PASS
A serendipity-qualifying click sequence updates the panel normally, an immediate second
call on the same still-qualifying signals is correctly rate-limited with nothing
persisted, and a third call made once the gap has passed (>=20s) updates the panel
again.
```

```
PT3 - Check logs (export + completeness)
-----
PILOT1 (session 47):
click: 34
navigation_depth: 34
dwell_start: 34
dwell_end: 34
revisit: 4
adaptive_trigger: 32
panel_impression: 128
panel_click: 5
questionnaire_response: 13 of 10 required answered
Exported: pt3_exports/PILOT1_event_log.csv, PILOT1_timeout.csv,
PILOT1_questionnaire_response.csv

PILOT2 (session 48):
click: 39
navigation_depth: 39
dwell_start: 39
dwell_end: 37
revisit: 5
adaptive_trigger: 40
panel_impression: 160
panel_click: 16
questionnaire_response: 10 of 10 required answered
Exported: pt3_exports/PILOT2_event_log.csv, PILOT2_timeout.csv,
PILOT2_questionnaire_response.csv

RESULT: PASS
Every pilot session has exactly the events a complete run guarantees
(session_start/session_end, both timeouts, both timeout_redirects, no error_trigger,
every required questionnaire item answered), and full CSV exports have been written to
tests/phase10/pt3_exports/.
```

Appendix F: Experimental Datasets

F.1 Artefact Metadata (First 30 rows)

artefact_id	external_id	title	description
1	nmnheducation_10026876	African Lion Skull	Skull of <i>Panthera leo</i> , the African lion, whose wide jaw and long canines reflect its role as an apex predator.
2	nmnhvz_7592992	Red Fox Taxidermy Mount	Preserved specimen of <i>Vulpes vulpes</i> . Red foxes are adaptable omnivores found across forests, farmland, and cities throughout the northern hemisphere.
3	siris_arc_387812	Big Horn Sheep Skeleton	Full skeletal mount of <i>Ovis canadensis</i> . Bighorn sheep use their curved horns in dramatic head-butting contests to establish dominance.
4	nmnheducation_10841923	White-tailed Deer Antlers	Naturally shed antler pair from <i>Odocoileus virginianus</i> . Male deer regrow their antlers each year, shedding them after the breeding season.
5	nmnhvz_7227114	Pale Striped Ground Squirrel	Preserved specimen of <i>Ictidomys tridecemlineatus pallidus</i> , a burrowing ground squirrel known for the pale stripes running along its back.
6	nmnhvz_4301642	Bullock's Oriole	Preserved specimen of <i>Icterus galbula bullockii</i> . Males display bright orange and black plumage used to attract mates each breeding season.
7	nmnhvz_4439094	Red-tailed Hawk Wing	Spread wing of a <i>Buteo jamaicensis</i> . Red-tailed hawks are common North American raptors, often seen soaring on broad, rounded wings.
8	nmnhpaleobiology_3354311	Deep-billed Crow Bones	Partial skeletal remains of <i>Corvus impluviatus</i> , an extinct crow species known only from fossils found in Hawaiian lava tube caves.
9	nmnhvz_15224411	Spectacled Flowerpecker	Preserved specimen of <i>Dicaeum dayakorum</i> , a small songbird first formally described by scientists as recently as 2019.
10	nmnheducation_11174829	Wild Turkey Feather	Down feather from a <i>Meleagris gallopavo</i> , the wild turkey, native to North America and long domesticated by Indigenous peoples.
11	nmnhpaleobiology_3368443	Ancient Ostrich Relative Fossil	Skull and axial remains of <i>Pseudocrypturus cercanaxius</i> , an early bird relative of modern ostriches that lived millions of years ago.
12	nmnheducation_100204	Fossil Rabbit	Fossilized mandible with teeth from <i>Megalagus</i> , an early rabbit relative that lived in North America roughly 30 million years ago.
13	nmnheducation_10020826	Fossil Rhinoceros	Fossilized partial mandible with teeth from <i>Subhyracodon</i> , an early rhinoceros ancestor that roamed North America over 30 million years ago.
14	nmnheducation_10021002	Fossil Crocodile	Fossilized tooth from <i>Thecachampsa</i> , an extinct crocodilian that lived in what is now North America during the Miocene epoch.
15	nmnheducation_10019905	Fossil Sperm Whale	Fossilized molar tooth from <i>Scaldicetus</i> , an extinct sperm whale ancestor that hunted prey in ancient oceans millions of years ago.
16	nmnhvz_6146546	Crocodile Skull	Skull specimen of <i>Caiman yacare</i> , a South American caiman species found across wetlands, rivers, and floodplains of the Pantanal region.
17	nmnhvz_6042435	Agamidae Lizard	Specimen of <i>Agama rueppelli occidentalis</i> , an agamid lizard native to arid regions of northeastern Africa, known for its rapid movement.
18	nmnhvz_6283987	Egyptian Tortoise	Specimen of <i>Testudo kleinmanni</i> , one of the smallest tortoise species in the world and now critically endangered in the wild.
19	nmnhvz_6277586	African Chameleon	Specimen of <i>Chamaeleo africanus</i> , a chameleon species known for its colour-changing skin, used for camouflage and communicating mood.
20	siris_arc_402282	Stegosaurus Dinosaur Skeleton	Full skeletal mount of <i>Stegosaurus</i> , a plant-eating dinosaur recognised by the row of bony plates running along its spine.
21	nmnhentomology_16355587	Malaria Mosquito Microscope Slide	Prepared slide of <i>Anopheles quadrimaculatus</i> , a mosquito species historically responsible for spreading malaria across parts of North America.
22	nmnheducation_10004232	Butterfly Specimen	Mounted specimen of <i>Panacea divalis</i> , a brightly coloured butterfly native to the tropical forests of Central and South America.
23	nmah_1815861	Insect Wing	Preserved section of an insect wing, showing the delicate vein structure that provides both strength and flexibility for flight.
24	npm_1991.0414.1	Queen Honeybee Cage	Wooden box used for shipping a queen honeybee, allowing beekeepers to safely transport and introduce new queens to a hive.
25	nmnheducation_10866557	Giant Stag Beetle	Specimen of <i>Dorcus titanus</i> , a giant stag beetle whose males use oversized jaw-like mandibles to battle rivals for mates.
26	nmnheducation_10008363	Pelagic Octopus Shell	Shell specimen of the winged argonaut, <i>Argonauta hians</i> , a pelagic octopus that secretes a thin shell to protect its eggs.
27	siris_arc_402549	Blue Whale	Life-sized model of <i>Balaenoptera musculus</i> , the blue whale, the largest animal known to have ever lived on Earth.
28	nmnhinvertebratezoology_164793	California Spiny Lobster	Specimen of <i>Panulirus interruptus</i> , a spiny lobster found along the Pacific coast, identified by its long antennae and spiny shell.
29	nmnheducation_10008991	Common Starfish	Specimen of <i>Asterias rubens</i> , the common starfish, which can regenerate lost arms and moves using tiny tube feet.
30	nmnheducation_10009334	Staghorn Coral Fragment	Branching coral sample from the staghorn coral family, reef-building corals that grow in fast branching shapes resembling deer antlers.

theme	sub_theme	image_url	source_url	tags
Animals	Mammals	https://ids.si.edu/ids/download?id=NMNH-EO_067340_SKULL_Lion_Pan	https://www.si.edu/object/lion:nmnheducation_10026876	skull,mammal,bone
Animals	Mammals	https://ids.si.edu/ids/download?id=NMNH-Vulpesvulpes1.jpg	https://www.si.edu/object/vulpes-vulpes:nmnhvz_7592992	mammal,fur,specimen
Animals	Mammals	https://ids.si.edu/ids/download?id=SIA-MNH-2320.jpg	https://www.si.edu/object/mounted-skeleton-bighorn-sheep:siris_arc_38	skeleton,mammal,bone
Animals	Mammals	https://ids.si.edu/ids/download?id=NMNH-EO_400085_White_Tailed_Deer	https://www.si.edu/object/white-tailed-deer:nmnheducation_10841923	antler,mammal,bone
Animals	Mammals	https://ids.si.edu/ids/download?id=NMNH-Spermophilustridecemlineatus	https://www.si.edu/object/ictidomys-tridecemlineatus-pallidus:nmnhvz_	mammal,fur,specimen
Animals	Birds	https://ids.si.edu/ids/download?id=NMNH-USNM186125IcteridaeIcterus	https://www.si.edu/object/icterus-galbula-bullockii:nmnhvz_4301642	bird,feather,specimen
Animals	Birds	https://ids.si.edu/ids/download?id=NMNH-IMAG0135.jpg	https://www.si.edu/object/buteo-jamaicensis:nmnhvz_4439094	wing,bird,feather
Animals	Birds	https://ids.si.edu/ids/download?id=NMNH-USNM_PAL_386427_Corvus	https://www.si.edu/object/corvus-impluviatus-james-olson-1991:nmnhp	bird,bone,fossil
Animals	Birds	https://ids.si.edu/ids/download?id=NMNH-USNM_663246Dicaeum_day	https://www.si.edu/object/dicaeum-dayakorum-saucier-et-al-2019:nmnh	bird,feather,specimen
Animals	Birds	https://ids.si.edu/ids/download?id=NMNH-EO_401247_Wild_Turkey_Me	https://www.si.edu/object/wild-turkey:nmnheducation_11174829	feather,bird,specimen
Animals	Fossils	https://ids.si.edu/ids/download?id=NMNH-USNM336103.jpg	https://www.si.edu/object/pseudocrypturus-cercanaxius-houde-1988:nm	fossil,bone,bird
Animals	Fossils	https://ids.si.edu/ids/download?id=NMNH-EO_050878_Fossil_Rabbit_M	https://www.si.edu/object/fossil-rabbit:nmnheducation_10020422	fossil,bone,mammal
Animals	Fossils	https://ids.si.edu/ids/download?id=NMNH-EO_051283_Running_Rhinoc	https://www.si.edu/object/fossil-rhinoceros:nmnheducation_10020826	fossil,bone,mammal
Animals	Fossils	https://ids.si.edu/ids/download?id=NMNH-EO_051460_Fossil_Crocodile	https://www.si.edu/object/fossil-crocodile-crocodile:nmnheducation_10	fossil,bone,reptile
Animals	Fossils	https://ids.si.edu/ids/download?id=NMNH-EO_050360_Fossil_Sperm_W	https://www.si.edu/object/fossil-sperm-whale:nmnheducation_10019905	fossil,bone,marine
Animals	Reptiles	https://ids.si.edu/ids/download?id=NMNH-281839_skull_dorsal.jpg	https://www.si.edu/object/caiman-yacare:nmnhvz_6146546	skull,reptile,bone
Animals	Reptiles	https://ids.si.edu/ids/download?id=NMNH-USNM_66927_dorsal.jpg	https://www.si.edu/object/agama-rueppelli-occidentalis:nmnhvz_604245	reptile,scale,specimen
Animals	Reptiles	https://ids.si.edu/ids/download?id=NMNH-USNM_139092_dorsal.jpg	https://www.si.edu/object/testudo-kleinmanni:nmnhvz_6283987	reptile,shell,specimen
Animals	Reptiles	https://ids.si.edu/ids/download?id=NMNH-USNM_132157_lateral.jpg	https://www.si.edu/object/chamaeleo-africanus:nmnhvz_6277586	reptile,scale,specimen
Animals	Reptiles	https://ids.si.edu/ids/download?id=SIA-SIA_000095_B44A_F14_018.jpg	https://www.si.edu/object/stegosaurus-skeleton:siris_arc_402282	fossil,bone,reptile
Animals	Insects	https://ids.si.edu/ids/download?id=NMNH-USNMENT00863207_Anophe	https://www.si.edu/object/anopheles-anopheles-quadrinaculatus:nmnh	insect,wing,slide
Animals	Insects	https://ids.si.edu/ids/download?id=NMNH-EO_023404_Butterfly_Panac	https://www.si.edu/object/butterfly:nmnheducation_10004232	insect,wing,specimen
Animals	Insects	https://ids.si.edu/ids/download?id=NMAH-ET2017-12948-000001.jpg	https://www.si.edu/object/insect-wing:nmah_1815861	wing,insect,feather
Animals	Insects	https://ids.si.edu/ids/download?id=NPM-1991_0414_1az.jpg	https://www.si.edu/object/queen-honeybee-cage:npm_1991.0414.1	insect,wood,tool
Animals	Insects	https://ids.si.edu/ids/download?id=NMNH-EO_400264_Giant_Stag_Bee	https://www.si.edu/object/giant-stag-beetle:nmnheducation_10866557	insect,specimen,mandible
Animals	Marine Life	https://ids.si.edu/ids/download?id=NMNH-EO_031602_Brown_Paper_N	https://www.si.edu/object/winged-argonaut:nmnheducation_10008363	shell,marine,specimen
Animals	Marine Life	https://ids.si.edu/ids/download?id=SIA-SIA_000095_B44_F21_007.jpg	https://www.si.edu/object/hall-life-sea-museum-natural-history-blue-wh	marine,model,whale
Animals	Marine Life	https://ids.si.edu/ids/download?id=NMNH-USNM_210038_Panulirus_int	https://www.si.edu/object/panulirus-interruptus:nmnhinvertebratezoolog	marine,shell,invertebrate
Animals	Marine Life	https://ids.si.edu/ids/download?id=NMNH-EO_032529_Northern_Sea_S	https://www.si.edu/object/common-sea-star-starfish-starfish:nmnheduc	marine,starfish,specimen
Animals	Marine Life	https://ids.si.edu/ids/download?id=NMNH-EO_033255_Stony_Coral_Ac	https://www.si.edu/object/hard-coral-staghorn-coral-stony-coral-stagho	coral,marine,specimen

F.2 Behavioural Log Data (First 30 rows)

session_id	participant_id	condition_id	session_start	session_end	timeout_triggered_static	timeout_triggered_adaptive	error_flag	artefact_order
53	52	1	03/08/2026 09:01	03/08/2026 09:22	1	1	1	0 [13,54,9,46,2,59,23,32,17,26,38
54	53	2	03/08/2026 09:53	03/08/2026 10:21	1	1	1	0 [43,32,5,40,20,21,9,50,52,27,56
58	57	1	06/08/2026 08:10	06/08/2026 08:35	1	1	1	0 [22,49,13,8,39,51,44,2,34,27,60
59	58	2	06/08/2026 15:49	06/08/2026 16:13	1	1	1	0 [5,53,7,15,32,45,16,38,22,28,58
60	59	1	06/08/2026 16:14	06/08/2026 16:35	1	1	1	0 [43,38,32,52,13,5,7,47,57,27,22
61	60	2	07/08/2026 09:45	07/08/2026 10:05	1	1	1	0 [37,41,32,20,3,6,54,21,15,27,60
62	61	1	10/08/2026 15:07	10/08/2026 15:27	1	1	1	0 [47,6,42,54,32,14,40,18,4,28,57
63	62	2	10/08/2026 15:29	10/08/2026 15:49	1	1	1	0 [50,15,17,32,37,23,1,7,43,28,57
65	64	1	11/08/2026 15:02	11/08/2026 15:19	1	1	1	0 [48,42,24,6,11,2,19,38,34,28,57
66	65	2	11/08/2026 15:20	11/08/2026 15:44	1	1	1	0 [43,18,11,50,40,10,1,34,60,51,2
67	66	1	14/08/2026 17:18	14/08/2026 17:41	1	1	1	0 [37,57,8,11,33,45,50,25,20,27,2
68	67	2	14/08/2026 18:28	14/08/2026 18:51	1	1	1	0 [2,37,10,48,33,22,45,17,12,55,6
69	68	1	15/08/2026 16:14	15/08/2026 16:55	1	1	1	0 [39,41,22,17,33,48,55,13,7,27,3
70	69	2	16/08/2026 15:35	16/08/2026 15:54	1	1	1	0 [38,13,53,24,20,5,46,10,42,60,3
71	70	1	16/08/2026 15:55	16/08/2026 16:16	1	1	1	0 [24,46,16,4,55,34,15,41,36,27,8
72	71	2	16/08/2026 16:17	16/08/2026 16:37	1	1	1	0 [46,10,12,36,24,18,4,41,53,56,3
73	72	1	16/08/2026 16:38	16/08/2026 16:59	1	1	1	0 [16,4,40,53,12,33,25,42,49,29,7
74	73	2	17/08/2026 11:10	17/08/2026 11:31	1	1	1	0 [60,24,51,6,50,3,13,43,18,28,36
75	74	1	17/08/2026 11:45	17/08/2026 12:10	1	1	1	0 [59,21,46,38,19,55,32,45,14,30,]
76	75	2	17/08/2026 13:08	17/08/2026 13:30	1	1	1	0 [3,19,60,48,55,11,23,42,10,26,3
78	77	2	17/08/2026 16:59	17/08/2026 17:20	1	1	1	0 [6,15,4,25,18,26,54,48,36,57,45
79	78	1	17/08/2026 17:22	17/08/2026 17:43	1	1	1	0 [20,41,51,36,7,58,25,30,47,34,1
80	79	1	17/08/2026 17:49	17/08/2026 18:12	1	1	1	0 [51,17,14,4,60,47,8,25,30,34,44
81	80	2	18/08/2026 10:34	18/08/2026 10:58	1	1	1	0 [40,1,22,57,27,13,48,42,52,35,2
82	81	1	18/08/2026 12:41	18/08/2026 13:04	1	1	1	0 [26,45,17,58,2,52,22,8,12,33,50
83	82	2	18/08/2026 13:26	18/08/2026 13:49	1	1	1	0 [40,25,48,15,55,27,43,4,57,35,2
84	83	1	18/08/2026 17:55	18/08/2026 18:19	1	1	1	0 [25,18,30,12,10,45,53,59,3,35,4
85	84	2	18/08/2026 18:19	18/08/2026 18:40	1	1	1	0 [43,60,52,48,5,12,16,29,36,31,2
86	85	1	20/08/2026 13:07	20/08/2026 13:37	1	1	1	0 [14,60,3,39,28,20,7,25,53,31,48
87	86	2	20/08/2026 17:34	20/08/2026 18:00	1	1	1	0 [44,36,56,4,6,51,49,20,30,31,21

event_id	session_id	artefact_id	suggestion_id	rule_id	event_type	timestamp	dwell_duration	navigation_depth
2636	53				session_start	03/08/2026 09:01		
2637	53	9			click	03/08/2026 09:05		
2638	53	9			dwell_start	03/08/2026 09:05		
2639	53	9			navigation_depth	03/08/2026 09:05		1
2640	53	9			dwell_end	03/08/2026 09:06	27	
2641	53	55			click	03/08/2026 09:06		
2642	53	55			navigation_depth	03/08/2026 09:06		2
2643	53	55			dwell_start	03/08/2026 09:06		
2644	53	55			dwell_end	03/08/2026 09:06	2	
2645	53	33			click	03/08/2026 09:07		
2646	53	33			navigation_depth	03/08/2026 09:07		3
2647	53	33			dwell_start	03/08/2026 09:07		
2648	53	33			dwell_end	03/08/2026 09:07	14	
2649	53	19			click	03/08/2026 09:07		
2650	53	19			dwell_start	03/08/2026 09:07		
2651	53	19			navigation_depth	03/08/2026 09:07		4
2652	53	19			dwell_end	03/08/2026 09:08	11	
2653	53	53			click	03/08/2026 09:08		
2654	53	53			navigation_depth	03/08/2026 09:08		5
2655	53	53			dwell_start	03/08/2026 09:08		
2656	53	53			dwell_end	03/08/2026 09:08	8	
2657	53	18			click	03/08/2026 09:08		
2658	53	18			dwell_start	03/08/2026 09:08		
2659	53	18			navigation_depth	03/08/2026 09:08		6
2660	53	18			dwell_end	03/08/2026 09:08	11	
2661	53	11			click	03/08/2026 09:09		
2662	53	11			navigation_depth	03/08/2026 09:09		7
2663	53	11			dwell_start	03/08/2026 09:09		
2664	53	11			dwell_end	03/08/2026 09:09	10	

timeout_id	session_id	version_name	timeout_timestamp	auto_redirect_flag
32	53	static	03/08/2026 09:12	1
33	53	adaptive	03/08/2026 09:19	1
34	54	adaptive	03/08/2026 10:04	1
35	54	static	03/08/2026 10:11	1
36	58	static	06/08/2026 08:24	1
37	58	adaptive	06/08/2026 08:31	1
38	59	adaptive	06/08/2026 16:00	1
39	59	static	06/08/2026 16:07	1
40	60	static	06/08/2026 16:25	1
41	60	adaptive	06/08/2026 16:32	1
42	61	adaptive	07/08/2026 09:54	1
43	61	static	07/08/2026 10:01	1
44	62	static	10/08/2026 15:18	1
45	62	adaptive	10/08/2026 15:25	1
46	63	adaptive	10/08/2026 15:39	1
47	63	static	10/08/2026 15:46	1
49	65	static	11/08/2026 15:09	1
50	65	adaptive	11/08/2026 15:16	1
51	66	adaptive	11/08/2026 15:31	1
52	66	static	11/08/2026 15:38	1
53	67	static	14/08/2026 17:31	1
54	67	adaptive	14/08/2026 17:38	1
55	68	adaptive	14/08/2026 18:40	1
56	68	static	14/08/2026 18:47	1
57	69	static	15/08/2026 16:44	1
58	69	adaptive	15/08/2026 16:51	1
59	70	adaptive	16/08/2026 15:45	1
60	70	static	16/08/2026 15:52	1
61	71	static	16/08/2026 16:08	1

adaptive_event_id	event_id	session_id	rule_id	suggestion_id	trigger_timestamp	trigger_confidence
1546	2701	53	2		03/08/2026 09:13	0.64
1547	2702	53	2	1173	03/08/2026 09:13	0.64
1548	2703	53	2	1174	03/08/2026 09:13	0.64
1549	2704	53	2	1176	03/08/2026 09:13	0.64
1550	2705	53	2	1175	03/08/2026 09:13	0.64
1551	2711	53	2	1174	03/08/2026 09:13	
1552	2715	53	2		03/08/2026 09:13	0.6
1553	2716	53	2	1177	03/08/2026 09:13	0.6
1554	2717	53	2	1178	03/08/2026 09:13	0.6
1555	2718	53	2	1179	03/08/2026 09:13	0.6
1556	2719	53	2	1180	03/08/2026 09:13	0.6
1557	2721	53	2	1179	03/08/2026 09:13	
1558	2724	53	9		03/08/2026 09:13	0.72
1559	2727	53	9	1182	03/08/2026 09:14	0.72
1560	2726	53	9	1183	03/08/2026 09:14	0.72
1561	2725	53	9	1181	03/08/2026 09:14	0.72
1562	2728	53	9	1184	03/08/2026 09:14	0.72
1563	2730	53	2		03/08/2026 09:14	0.77
1564	2731	53	2	1186	03/08/2026 09:14	0.77
1565	2732	53	2	1185	03/08/2026 09:14	0.77
1566	2733	53	2	1188	03/08/2026 09:14	0.77
1567	2734	53	2	1187	03/08/2026 09:14	0.77
1568	2738	53	9		03/08/2026 09:14	0.84
1569	2740	53	9	1190	03/08/2026 09:14	0.84
1570	2739	53	9	1189	03/08/2026 09:14	0.84
1571	2741	53	9	1191	03/08/2026 09:14	0.84
1572	2742	53	9	1192	03/08/2026 09:14	0.84
1573	2744	53	2		03/08/2026 09:14	0.86
1574	2745	53	2	1194	03/08/2026 09:14	0.86

suggestion_id ▼	rule_id ▼	session_id ▼	artefact_id ▼	interest_score ▼	suggestion_timestamp ▼	panel_position ▼	clicked_flag ▼
1173	2	53	19	0.64	03/08/2026 09:13	0	0
1174	2	53	16	0.64	03/08/2026 09:13	1	1
1175	2	53	22	0.64	03/08/2026 09:13	2	0
1176	2	53	14	0.64	03/08/2026 09:13	3	0
1177	2	53	19	0.6	03/08/2026 09:13	0	0
1178	2	53	22	0.6	03/08/2026 09:13	1	0
1179	2	53	14	0.6	03/08/2026 09:13	2	1
1180	2	53	21	0.6	03/08/2026 09:13	3	0
1181	9	53	20	0.72	03/08/2026 09:13	0	0
1182	9	53	14	0.72	03/08/2026 09:13	1	0
1183	9	53	50	0.72	03/08/2026 09:13	2	0
1184	9	53	18	0.72	03/08/2026 09:13	3	0
1185	2	53	16	0.77	03/08/2026 09:14	0	0
1186	2	53	14	0.77	03/08/2026 09:14	1	0
1187	2	53	50	0.77	03/08/2026 09:14	2	0
1188	2	53	2	0.77	03/08/2026 09:14	3	0
1189	9	53	18	0.84	03/08/2026 09:14	0	0
1190	9	53	8	0.84	03/08/2026 09:14	1	0
1191	9	53	5	0.84	03/08/2026 09:14	2	0
1192	9	53	49	0.84	03/08/2026 09:14	3	0
1193	2	53	17	0.86	03/08/2026 09:14	0	0
1194	2	53	23	0.86	03/08/2026 09:14	1	1
1195	2	53	5	0.86	03/08/2026 09:14	2	0
1196	2	53	55	0.86	03/08/2026 09:14	3	0
1197	9	53	18	0.84	03/08/2026 09:14	0	0
1198	9	53	29	0.84	03/08/2026 09:14	1	0
1199	9	53	49	0.84	03/08/2026 09:14	2	0
1200	9	53	1	0.84	03/08/2026 09:14	3	0
1201	2	53	18	0.86	03/08/2026 09:14	0	0

F.3 Post-Study Questionnaire Data (First 30 rows)

response_id	session_id	participant_id	item_id	response_value	response_timestamp
64	53	52	1	Static version	03/08/2026 09:22
65	53	52	2	4	03/08/2026 09:22
66	53	52	3	2	03/08/2026 09:22
67	53	52	4	4	03/08/2026 09:22
68	53	52	5	3	03/08/2026 09:22
69	53	52	7	5	03/08/2026 09:22
70	53	52	8	4	03/08/2026 09:22
71	53	52	9	5	03/08/2026 09:22
72	53	52	10	5	03/08/2026 09:22
73	53	52	11	Interesting options shown for next click. Made view simpler and aligned to what I wanted to view.	03/08/2026 09:22
74	53	52	12	Adaptive version	03/08/2026 09:22
75	53	52	13	Quicker navigation. Ease of search. Offered different ways of using the site.	03/08/2026 09:22
76	54	53	1	Adaptive version	03/08/2026 10:21
77	54	53	2	4	03/08/2026 10:21
78	54	53	3	4	03/08/2026 10:21
79	54	53	4	5	03/08/2026 10:21
80	54	53	5	4	03/08/2026 10:21
81	54	53	6	The static version seems to be well suited to someone who is interested in most/all artefacts as the	03/08/2026 10:21
82	54	53	7	5	03/08/2026 10:21
83	54	53	8	5	03/08/2026 10:21
84	54	53	9	4	03/08/2026 10:21
85	54	53	10	5	03/08/2026 10:21
86	54	53	11	The adaptive version is excellent if you have a specific interest and displays all relevant suggestions	03/08/2026 10:21
87	54	53	12	Adaptive version	03/08/2026 10:21
88	54	53	13	I thought the adaptive version was more suited to me as I have a narrow range of particular interest:	03/08/2026 10:21
89	54	53	14	Both experiences have benefits and drawbacks, I think it depends on the individuals interests in his	03/08/2026 10:21
90	58	57	1	Adaptive version	06/08/2026 08:35
91	58	57	2	3	06/08/2026 08:35
92	58	57	3	4	06/08/2026 08:35

item_id	item_text	item_type	item_order	scale_min_label	scale_max_label
1	Which version did you complete first?	multiple_choice	1		
2	The artefacts shown in static version felt relevant to my interests.	likert	2	Strongly disagree	Strongly agree
3	I encountered artefacts in the static version that were unexpected or surprising.	likert	3	Strongly disagree	Strongly agree
4	The static version allowed me to explore freely without feeling restricted.	likert	4	Strongly disagree	Strongly agree
5	Overall, my experience using the static version was positive.	likert	5	Strongly disagree	Strongly agree
6	Is there anything you would like to share about your experience with the static version?	free_text	6		
7	The artefacts shown in adaptive version felt relevant to my interests.	likert	7	Strongly disagree	Strongly agree
8	I encountered artefacts in the adaptive version that were unexpected or surprising.	likert	8	Strongly disagree	Strongly agree
9	The adaptive version allowed me to explore freely without feeling restricted.	likert	9	Strongly disagree	Strongly agree
10	Overall, my experience using the adaptive version was positive.	likert	10	Strongly disagree	Strongly agree
11	Is there anything you would like to share about your experience with the adaptive version?	free_text	11		
12	Which version did you prefer overall?	multiple_choice	12		
13	Why did you prefer this version?	free_text	13		
14	Any final comments about your experience today?	free_text	14		

Appendix G: Analysis Notebook & Outputs

G.1 Analysis Notebook

#Step 1 – Setup

```
!pip install pinguin -q #provides proper effect sizes

import re
from pathlib import Path

import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import pinguin as pg
from scipy import stats

pd.set_option("display.width", 140)
pd.set_option("display.max_columns", 30)
sns.set_theme(style="whitegrid")
VERSION_ORDER = ["static", "adaptive"] #keep this order for consistency
```

#Step 2 – Get data

```
USE_DRIVE = True #False = upload-a-zip option instead

if USE_DRIVE:
    from google.colab import drive
    drive.mount('/content/drive')
    DATA_DIR = Path('/content/drive/MyDrive/DigitalMuseum/csv')
else:
    from google.colab import files
    import zipfile
    print("select the zipped csv export folder (e.g. full_study_csv.zip)")
    uploaded = files.upload()
    zip_name = next(iter(uploaded))
    with zipfile.ZipFile(zip_name) as z:
        z.extractall('/content/full_study_csv')
    DATA_DIR = Path('/content/full_study_csv')

print("looking in:", DATA_DIR)
print(sorted(p.name for p in DATA_DIR.glob('*.*csv')))
```

#Step 3 – Load all tables

```

session_log = pd.read_csv(DATA_DIR / "session_log.csv", parse_dates=["session_start", "session_end"])
participant = pd.read_csv(DATA_DIR / "participant.csv")
study_condition = pd.read_csv(DATA_DIR / "study_condition.csv")
theme = pd.read_csv(DATA_DIR / "theme.csv")
subtheme = pd.read_csv(DATA_DIR / "subtheme.csv")
artefact = pd.read_csv(DATA_DIR / "artefact.csv")
event_log = pd.read_csv(DATA_DIR / "event_log.csv", parse_dates=["timestamp"])
timeout = pd.read_csv(DATA_DIR / "timeout.csv", parse_dates=["timeout_timestamp"])
adaptive_rule = pd.read_csv(DATA_DIR / "adaptive_rule.csv")
adaptive_event = pd.read_csv(DATA_DIR / "adaptive_event.csv", parse_dates=["trigger_timestamp"])
adaptive_suggestion = pd.read_csv(DATA_DIR / "adaptive_suggestion.csv", parse_dates=["suggestion_timestamp"])
questionnaire_item = pd.read_csv(DATA_DIR / "questionnaire_item.csv")
questionnaire_response = pd.read_csv(DATA_DIR / "questionnaire_response.csv", parse_dates=["response_timestamp"])
completeness = pd.read_csv(DATA_DIR / "session_completeness_summary.csv", parse_dates=["session_start", "session_end"])

print("Rows loaded:")
for name, df in [("session_log", session_log), ("event_log", event_log), ("timeout", timeout),
                ("adaptive_event", adaptive_event), ("adaptive_suggestion", adaptive_suggestion),
                ("questionnaire_response", questionnaire_response)]:
    print(f"{name}: {len(df)}")

```

#Step 4 – Validate & clean raw data

#full-row and PK duplicates

#nulls in columns that should never be empty

#implausible values

#referential integrity

def validate_data():

issues = {}

#duplicates

```

dup_rows = {name: df.duplicated().sum() for name, df in [
    ("session_log", session_log), ("event_log", event_log), ("timeout", timeout),
    ("adaptive_event", adaptive_event), ("adaptive_suggestion", adaptive_suggestion),
    ("questionnaire_response", questionnaire_response)]
} if df.duplicated().sum() > 0:

```

issues["duplicate full rows"] = dup_rows

dup_pks = {}

if event_log["event_id"].duplicated().sum():

dup_pks["event_log.event_id"] = event_log["event_id"].duplicated().sum()

if questionnaire_response["response_id"].duplicated().sum():

dup_pks["questionnaire_response.response_id"] = questionnaire_response["response_id"].duplicated().sum()

dup_item = questionnaire_response.duplicated(subset=["session_id", "item_id"]).sum()

if dup_item:

dup_pks["questionnaire_response (session_id.item_id)"] = dup_item

issues["duplicate keys"] = dup_pks

#nulls in columns that should always be populated

null_counts = event_log[["timestamp", "event_type", "session_id"]].isna().sum()

issues["event_log nulls"] = null_counts[null_counts > 0].to_dict()

#implausible values

implausible = {}

bad_session_span = (session_log["session_end"] <= session_log["session_start"]).sum()

if bad_session_span:

implausible["sessions with end <= start"] = bad_session_span

neg_dwell = (event_log["dwell_duration"] < 0).sum()

if neg_dwell:

implausible["negative dwell duration"] = neg_dwell

zero_dwell = (event_log["dwell_duration"] == 0).sum()

if zero_dwell:

implausible["dwell duration == 0 (kept, not dropped)"] = zero_dwell

likert_ids = questionnaire_item.loc[questionnaire_item["item_type"] == "likert", "item_id"]

lik = questionnaire_response[questionnaire_response["item_id"].isin(likert_ids)].copy()

lik["response_value"] = pd.to_numeric(lik["response_value"], errors="coerce")

out_of_range = ((lik["response_value"] < 1) | (lik["response_value"] > 5) | lik["response_value"].isna()).sum()

if out_of_range:

implausible["likert responses outside 1-5 (or non-numeric)"] = out_of_range

bad_codes = participant[~participant["participant_code"].str.match(r"^(p|12)[0-9]{1,2}\$")]

if len(bad_codes):

implausible["participant_code not matching pptN (1-30)"] = bad_codes["participant_code"].tolist()

```

issues["implausible values"] = implausible

#referential integrity
ref = {}
bad_artefacts = set(event_log["artefact_id"].dropna().unique()) - set(artefact["artefact_id"].unique())
if bad_artefacts:
    ref["event_log.artefact_id not in artefact.csv"] = bad_artefacts
bad_sessions = set(event_log["session_id"].unique()) - set(session_log["session_id"].unique())
if bad_sessions:
    ref["event_log.session_id not in session_log.csv"] = bad_sessions
issues["referential integrity"] = ref

return issues

validation_issues = validate_data()
for check, found in validation_issues.items():
    status = "OK - none found" if not found else found
    print(f"{check}: {status}")

```

```

#Step 5 - Build readable sessions table, apply completeness filter
#merges participant_code & condition_name

sessions = session_log.merge(participant[["participant_id", "participant_code"]], on="participant_id")
sessions = sessions.merge(study_condition[["condition_id", "condition_name"]], on="condition_id")

included_ids = completeness.loc[completeness["looks_complete"] == "yes", "session_id"]
n_before = len(sessions)
sessions = sessions[sessions["session_id"].isin(included_ids)].reset_index(drop=True)
print(f"Sessions: {n_before} total -> {len(sessions)} pass the completeness filter")

#readable artefact lookup (theme & subtheme names attached)
#artefact-level breakdowns
artefact_lookup = (
    artefact.merge(subtheme[["subtheme_id", "subtheme_name"]], on="subtheme_id")
    .merge(theme[["theme_id", "theme_name"]], on="theme_id")
)
sessions.head()

```

```

#Step 6 - Split each session into static & adaptive halves
#each version has fixed 7-minute time limit
#phase_start is exactly 420s before the timeout

def build_phase_windows(sessions, timeout):
    """Returns one row per (session, version) with the phase's start/end timestamps."""

    PHASE_DURATION_S = 420

    to = timeout[timeout["session_id"].isin(sessions["session_id"])].copy()

    #handle duplicate timeout rows
    dupe_counts = to.groupby(["session_id", "version_name"]).size()
    dupes = dupe_counts[dupe_counts > 1]
    if len(dupes):
        print(f"[data quality] {len(dupes)} session/version pairs logged duplicate timeout rows - keeping the latest:")
        to = to.loc[to.groupby(["session_id", "version_name"])["timeout_timestamp"].idxmax()]

    #build output
    out = (
        to[["session_id", "version_name", "timeout_timestamp"]]
        .rename(columns={"version_name": "version", "timeout_timestamp": "phase_end"})
        .reset_index(drop=True)
    )
    out["phase_start"] = out["phase_end"] - pd.Timedelta(seconds=PHASE_DURATION_S)
    out["duration_s"] = (out["phase_end"] - out["phase_start"]).dt.total_seconds()

    #sort by completion order (earliest phase_start first)
    out = out.sort_values(["session_id", "phase_start"])
    out = out[["session_id", "version", "phase_start", "phase_end", "duration_s"]]
    return out

phase_windows = build_phase_windows(sessions, timeout)
assert phase_windows.groupby("session_id").size().eq(2).all(), "expected exactly 2 phases per session"
phase_windows.head()

```

```

def tag_events_with_version(event_log, phase_windows):
    """Adds a 'version' column to event_log by checking which phase each event's timestamp falls into.
    Events outside any window (e.g. the final session_end, logged after both phases finish) are left as <NA>."""
    ev = event_log.copy()
    ev["version"] = pd.NA
    for sid, windows in phase_windows.groupby("session_id"):
        mask_sess = ev["session_id"] == sid
        for _, w in windows.iterrows():
            lower = ev["timestamp"] >= w["phase_start"]
            upper = ev["timestamp"] <= w["phase_end"]
            ev.loc[mask_sess & lower & upper, "version"] = w["version"]
    return ev

event_log_tagged = tag_events_with_version(
    event_log[event_log["session_id"].isin(sessions["session_id"])], phase_windows
)

print("Event counts by type * version:")
display(event_log_tagged.groupby(["event_type", "version"],
    dropna=False).size().unstack(fill_value=0))

#Sanity check: panel_impression / panel_click / adaptive_trigger only ever happen during the adaptive task,
#so if the tagging logic is right, none of them should ever be tagged 'static'.
#This is a free correctness check on the phase-splitting logic above.
adaptive_only_types = {"panel_impression", "panel_click", "adaptive_trigger"}
mistagged = event_log_tagged[
    event_log_tagged["event_type"].isin(adaptive_only_types) & (event_log_tagged["version"] != "adaptive")
]
print(f"\nadaptive-only events mistagged as non-adaptive: {len(mistagged)} (should be 0)")

#regression check
#every timeout_redirect tagged with its version
untagged_timeouts = event_log_tagged[
    (event_log_tagged["event_type"] == "timeout_redirect") & event_log_tagged["version"].isna()
]
print(f"timeout_redirect events left untagged: {len(untagged_timeouts)} (should be 0)")

```

```

#Step 7 - Behavioural metrics per participant * version
#builds one row per participant per version
#with core behavioural measures
#useful for statistical test & most plots

def summarise_behaviour(event_log_tagged, sessions):
    df = event_log_tagged.merge(sessions[["session_id", "participant_code", "condition_name"]], on="session_id")
    df = df.dropna(subset=["version"])

    clicks = df[df.event_type == "click"].groupby(["participant_code", "version"]).size().rename("n_clicks")
    dwell = (
        df[df.event_type == "dwell_end"]
        .groupby(["participant_code", "version"])["dwell_duration"]
        .agg(mean_dwell_s="mean", total_dwell_s="sum", n_dwells="count")
    )
    navdepth = (
        df[df.event_type == "navigation_depth"]
        .groupby(["participant_code", "version"])["navigation_depth"].max()
        .rename("max_navigation_depth")
    )
    revisits = df[df.event_type == "revisit"].groupby(["participant_code", "version"]).size().rename("n_revisits")
    unique_artefacts = (
        df[df.event_type == "click"]
        .groupby(["participant_code", "version"])["artefact_id"].nunique()
        .rename("n_unique_artefacts_viewed")
    )

```

```

out = pd.concat([clicks, dwell, navdepth, revisits, unique_artefacts], axis=1).reset_index()
out["n_revisits"] = out["n_revisits"].fillna(0)
return out

behaviour_summary = summarise_behaviour(event_log_tagged, sessions)

pw = phase_windows.merge(sessions[["session_id", "participant_code"]], on="session_id")
behaviour_summary = behaviour_summary.merge(
    pw[["participant_code", "version", "duration_s"]].rename(columns={"duration_s": "time_on_task_s"}),
    on=["participant_code", "version"], how="left",
)

assert (behaviour_summary["time_on_task_s"] == 420).all(), "expected every phase to be exactly the fixed 420s time limit"
print("time_on_task_s is fixed at 420s for every participant * version by design - kept as sanity check")

print("behaviour_summary:", behaviour_summary.shape, "(expect 60 rows = 30 participants * 2 versions)")
behaviour_summary.head()

```

```

#Step 8 - Adaptive-only engagement metrics
#how often adaptive logic fired & how confident it was
#how many artefacts the panel suggested
#what fraction the participant actually clicked
#the model's average interest score

ae = adaptive_event.merge(sessions[["session_id", "participant_code"]], on="session_id")
mean_conf = ae.groupby("participant_code")["trigger_confidence"].mean().rename("mean_trigger_confidence")
n_triggers = ae.groupby("participant_code").size().rename("n_adaptive_triggers")

sugg = adaptive_suggestion.merge(sessions[["session_id", "participant_code"]], on="session_id")
suggestion_click_rate = sugg.groupby("participant_code")["clicked_flag"].mean().rename("suggestion_click_rate")
mean_interest_score = sugg.groupby("participant_code")["interest_score"].mean().rename("mean_interest_score")
n_suggestions = sugg.groupby("participant_code").size().rename("n_suggestions_shown")

panel_ev = event_log_tagged[event_log_tagged["event_type"].isin(["panel_impression", "panel_click"]).merge(
    sessions[["session_id", "participant_code"]], on="session_id"
)
panel_impressions = panel_ev[panel_ev.event_type == "panel_impression"].groupby("participant_code").size().rename("n_panel_impressions")
panel_clicks = panel_ev[panel_ev.event_type == "panel_click"].groupby("participant_code").size().rename("panel_clicks")

adaptive_summary = pd.concat(
    [n_triggers, mean_conf, n_suggestions, suggestion_click_rate, mean_interest_score, panel_impressions, panel_clicks],
    axis=1,
).reset_index().fillna(0)

adaptive_summary.head()

```

```

#Step 9 - Questionnaire responses -> wide format
#pulls out each item's formal name from notes
#responses split 3 ways by item_type

def extract_formal_name(notes):
    m = re.search(r"Formal name:\s*([A-Za-z0-9_]+)", str(notes))
    return m.group(1) if m else None

questionnaire_item["formal_name"] = questionnaire_item["item_notes"].apply(extract_formal_name)
display(questionnaire_item[["item_id", "item_type", "formal_name"]])

qr = questionnaire_response.merge(sessions[["session_id", "participant_code"]], on="session_id")
qr = qr.merge(questionnaire_item[["item_id", "formal_name", "item_type"]], on="item_id")

qr_likert = qr[qr["item_type"] == "likert"].copy()
qr_likert["response_value"] = pd.to_numeric(qr_likert["response_value"], errors="raise")
wide_likert = qr_likert.pivot(index="participant_code", columns="formal_name", values="response_value")

wide_cat = qr[qr["item_type"] == "multiple_choice"].pivot(
    index="participant_code", columns="formal_name", values="response_value"
)

qualitative = qr[qr["item_type"] == "free_text"][["participant_code", "formal_name", "response_value"]].rename(
    columns={"response_value": "response_text"}
)

DIMENSIONS = ["relevance", "surprise", "freedom", "experience"] #4 paired Likert dimensions
wide_likert.head()

```

```

#Step 10 - Descriptive statistics
#quick means/SDs before running any tests
#final small validity check
#catches self-report slip ups for questionnaire item 1

print("=== Questionnaire dimensions (1-5 Likert) ===")
desc_rows = []
for d in DIMENSIONS:
    for v in VERSION_ORDER:
        col = wide_likert[f"{v}_{d}"]
        desc_rows.append({"dimension": d, "version": v, "mean": col.mean(), "sd": col.std(), "n": col.count()})
display(pd.DataFrame(desc_rows).round(2))

print("\n=== Behavioural metrics ===")
display(behaviour_summary.drop(columns="participant_code").groupby("version").agg(["mean", "std"]).round(2))

print("\n=== Preference (item 12) ===")
display(wide_cat["preferred_version"].value_counts())

print("\n=== Order completed first (item 1) vs stated condition ===")
display(wide_cat["order_completed_first"].value_counts())

print("\n=== Actual (logged) vs self-reported order completed first ===")
order_check = sessions[["participant_code", "condition_name"]].copy()
order_check["actual_first"] = order_check["condition_name"].str.replace("_first", "", regex=False)
order_check = order_check.merge(wide_cat[["order_completed_first"]].reset_index(), on="participant_code")
order_check["stated_first"] = order_check["order_completed_first"].str.replace(" version", "", regex=False).str.lower()
order_check["match"] = order_check["actual_first"] == order_check["stated_first"]
display(order_check[["participant_code", "actual_first", "order_completed_first", "match"]])

n_mismatch = (~order_check["match"]).sum()
mismatched_ppts = order_check.loc[~order_check["match"], "participant_code"].tolist()
print(f"{n_mismatch} mismatch(es) out of {len(order_check)}: {mismatched_ppts}")

```

```

#Step 11 - Paired tests: questionnaire dimensions
#static vs adaptive
#Wilcoxon signed rank safer
#normality checked to decide which test

def paired_compare(a, b, label):
    diff = (a - b).dropna()
    shapiro_p = stats.shapiro(diff).pvalue if len(diff) >= 3 else np.nan
    if shapiro_p >= 0.05:
        r = pg.ttest(a, b, paired=True)
        return {"measure": label, "n": len(diff), "mean_static": a.mean(), "mean_adaptive": b.mean(),
                "test": "paired t-test", "statistic": r["T"].iloc[0], "p": r["p_val"].iloc[0],
                "effect_size": r["cohen_d"].iloc[0], "effect_label": "Cohen's d", "shapiro_p_on_diffs": shapiro_p}
    else:
        r = pg.wilcoxon(a, b)
        return {"measure": label, "n": len(diff), "mean_static": a.mean(), "mean_adaptive": b.mean(),
                "test": "Wilcoxon signed-rank", "statistic": r["W_val"].iloc[0], "p": r["p_val"].iloc[0],
                "effect_size": r["RBC"].iloc[0], "effect_label": "rank-biserial r", "shapiro_p_on_diffs": shapiro_p}

questionnaire_results = pd.DataFrame([
    paired_compare(wide_likert[f"static_{d}"], wide_likert[f"adaptive_{d}"], d) for d in DIMENSIONS
])
questionnaire_results.round(4)

```

```

#Step 12 - Paired tests: behavioural metrics
#static vs adaptive
#same paired comparison logic

beh_wide = behaviour_summary.pivot(index="participant_code", columns="version")
BEH_METRICS = ["n_clicks", "mean_dwell_s", "total_dwell_s", "max_navigation_depth",
               "n_revisits", "n_unique_artefacts_viewed"]

behaviour_results = pd.DataFrame([
    paired_compare(beh_wide[(m, "static")], beh_wide[(m, "adaptive")], m) for m in BEH_METRICS
])
behaviour_results.round(4)

```

```
#Step 13 - Visualise: questionnaire dimensions
#paired boxplots for each dimension

fig, axes = plt.subplots(1, 4, figsize=(16, 4), sharey=True)
for ax, d in zip(axes, DIMENSIONS):
    plot_df = wide_likert[[f"static_{d}", f"adaptive_{d}"]].reset_index().melt(
        id_vars="participant_code", var_name="version", value_name="score"
    )
    plot_df["version"] = plot_df["version"].str.replace(f"_{d}", "", regex=False)
    sns.boxplot(data=plot_df, x="version", y="score", order=VERSION_ORDER, ax=ax, width=0.5)
    sns.stripplot(data=plot_df, x="version", y="score", order=VERSION_ORDER, ax=ax, color="black", alpha=0.4, jitter=0.08)
    ax.set_title(d.capitalize())
    ax.set_xlabel("")
    ax.set_ylim(0.5, 5.5)
fig.suptitle("Static vs Adaptive - questionnaire dimensions (1-5 Likert)")
fig.tight_layout()
plt.savefig("questionnaire_dimensions.png", dpi=150, bbox_inches="tight")
plt.show()
```

```
#Step 14 - Visualise: within-subject change per participant
#slope plot drawing a line per participant
#looking for a shift across participants

fig, axes = plt.subplots(1, 4, figsize=(16, 4), sharey=True)
for ax, d in zip(axes, DIMENSIONS):
    for _, row in wide_likert.iterrows():
        ax.plot([0, 1], [row[f"static_{d}"], row[f"adaptive_{d}"]], color="grey", alpha=0.4, marker="o", markersize=4)
    ax.set_xticks([0, 1])
    ax.set_xticklabels(VERSION_ORDER)
    ax.set_title(d.capitalize())
    ax.set_ylim(0.5, 5.5)
fig.suptitle("Per-participant change, static -> adaptive")
fig.tight_layout()
plt.savefig("questionnaire_slopes.png", dpi=150, bbox_inches="tight")
plt.show()
```

```
#Step 15 - Visualise: behavioural metrics
#bar chart of the mean for each behavioural metric by version
#with standard error bars

fig, axes = plt.subplots(2, 3, figsize=(15, 8))
for ax, m in zip(axes.flat, BEH_METRICS):
    means = behaviour_summary.groupby("version")[m].mean().reindex(VERSION_ORDER)
    sems = behaviour_summary.groupby("version")[m].sem().reindex(VERSION_ORDER)
    ax.bar(means.index, means.values, yerr=sems.values, capsize=5, color=["#4C72B0", "#DD8452"])
    ax.set_title(m)
axes.flat[-1].axis("off")
fig.suptitle("Behavioural metrics: static vs adaptive (mean +/- SE)")
fig.tight_layout()
plt.savefig("behavioural_metrics.png", dpi=150, bbox_inches="tight")
plt.show()
```

```

#Step 16 - Visualise: preference & order effects
#bar chart of stated overall preference
#cross-tab of first version & preferred version

fig, ax = plt.subplots(figsize=(5, 4))
order = ["Static version", "Adaptive version", "No preference"]
counts = wide_cat["preferred_version"].value_counts().reindex(order).fillna(0)
ax.bar(counts.index, counts.values, color=["#4C72B0", "#DD8452", "#999999"])
ax.set_title("Overall preference")
ax.set_ylabel("n participants")
plt.xticks(rotation=15)
plt.savefig("preference.png", dpi=150, bbox_inches="tight")
plt.show()

#system-logged 'did first' not self-reported
#corrects the known ppt3 mismatch
actual_first_labelled = (
    order_check.set_index("participant_code")["actual_first"].str.capitalize() + " version"
).reindex(wide_cat.index)

order_effect = pd.crosstab(actual_first_labelled, wide_cat["preferred_version"])
print("Rows = which version was actually completed first")
print("columns = which version preferred:")
display(order_effect)

```

```

#Step 17 - Visualise: adaptive suggestion effectiveness
#distribution of interest score across all suggestions shown
#per-participant suggestion click-through rate
#how often a suggestion was clicked

fig, axes = plt.subplots(1, 2, figsize=(12, 4))

axes[0].hist(sugg["interest_score"], bins=15, color="#4C72B0")
axes[0].set_title("Distribution of suggestion interest score")
axes[0].set_xlabel("interest_score")

axes[1].hist(adaptive_summary["suggestion_click_rate"], bins=15, color="#DD8452")
axes[1].set_title("Per-participant suggestion click-through rate")
axes[1].set_xlabel("clicked_flag mean")

fig.tight_layout()
plt.savefig("adaptive_effectiveness.png", dpi=150, bbox_inches="tight")
plt.show()

print(f"Overall suggestion click-through rate: {sugg['clicked_flag'].mean():.3f}")
print(f"Mean trigger_confidence across all adaptive_event rows: {adaptive_event['trigger_confidence'].mean():.3f}")

```

```

#Exclude ppt1 and ppt2
#took part before serendipity rate limit was set

EXCLUDE_CODES = ["ppt1", "ppt2"]

excluded_participant_ids = participant.loc[
    participant["participant_code"].isin(EXCLUDE_CODES), "participant_id"
].tolist()
print("Excluded participant_id(s):", excluded_participant_ids) #expect [52, 53]

excluded_session_ids = session_log.loc[
    session_log["participant_id"].isin(excluded_participant_ids), "session_id"
].tolist()
print("Excluded session_id(s):", excluded_session_ids) #expect [53, 54]

#adaptive_suggestion and adaptive_event only carry session_id, not participant_id
#so exclusion has to go via session_id
sugg_excl = sugg[~sugg["session_id"].isin(excluded_session_ids)]
adaptive_event_excl = adaptive_event[~adaptive_event["session_id"].isin(excluded_session_ids)]

print(f"sugg: {len(sugg)} -> {len(sugg_excl)} rows after exclusion")
print(f"adaptive_event: {len(adaptive_event)} -> {len(adaptive_event_excl)} rows after exclusion")

#adaptive_summary is per-participant
#filter on whichever identifier it actually carries
if "participant_id" in adaptive_summary.columns:
    adaptive_summary_excl = adaptive_summary[~adaptive_summary["participant_id"].isin(excluded_participant_ids)]
elif "participant_code" in adaptive_summary.columns:
    adaptive_summary_excl = adaptive_summary[~adaptive_summary["participant_code"].isin(EXCLUDE_CODES)]
elif adaptive_summary.index.name == "participant_id":
    adaptive_summary_excl = adaptive_summary[~adaptive_summary.index.isin(excluded_participant_ids)]
else:
    raise ValueError("Couldn't find participant_id/participant_code in adaptive_summary - check its columns/index")

print(f"adaptive_summary: {len(adaptive_summary)} -> {len(adaptive_summary_excl)} rows after exclusion")

fig, axes = plt.subplots(1, 2, figsize=(12, 4))

axes[0].hist(sugg_excl["interest_score"], bins=15, color="#4C72B0")
axes[0].set_title("Distribution of suggestion interest score (excl. ppt1, ppt2)")
axes[0].set_xlabel("interest_score")

axes[1].hist(adaptive_summary_excl["suggestion_click_rate"], bins=15, color="#DD8452")
axes[1].set_title("Per-participant suggestion click-through rate (excl. ppt1, ppt2)")
axes[1].set_xlabel("clicked_flag mean")

fig.tight_layout()
plt.savefig("adaptive_effectiveness_excl_ppt1_ppt2.png", dpi=150, bbox_inches="tight")
plt.show()

```

#Step 18 – Bring in thematic coding

```

CODING_XLSX = Path('/content/drive/MyDrive/DigitalMuseum/Full Study Data.xlsx')

#skiprows=4 skips the title + instruction rows above the real header row
#stops pandas silently blanking out genuine text answers like n/a
coded = pd.read_excel(
    CODING_XLSX, sheet_name="qualitative_coding", skiprows=4,
    keep_default_na=False, na_values=[""],
)

#genuinely empty cell still comes back as NaN
#every check compare response text directly
#without handling NaN
coded["response"] = coded["response"].fillna("")
coded = coded[(coded["code"] != "") & (coded["theme"] != "")] #keep only coded rows

print(f"{len(coded)} coded extracts across {coded['theme'].nunique()} themes")
coded.head()

```

#Step 19 – Data check

```

#row should have both theme and response text
#checks if a cell got cleared by accident while coding
#checks & deletes rows with no response placeholders

missing_text = coded[coded["response"] == ""]
if len(missing_text):
    print("Rows with a code/theme but no genuinely empty response cell – double check these")
    print("weren't cleared by accident when filling in the coding columns:")
    display(missing_text[["participant_code", "item", "code", "theme"]])
else:
    print("No genuinely empty response cells.")

#no response placeholders excluded from coded extracts
NO_RESPONSE_MARKERS = ["n/a"]
is_no_response = coded["response"].str.lower().str.contains(
    "".join(NO_RESPONSE_MARKERS), regex=True, na=False
)
removed = coded[is_no_response]
if len(removed):
    print(f"\nRemoving {len(removed)} row(s) – participant indicated they had no response:")
    display(removed[["participant_code", "item", "response", "code", "theme"]])
    coded = coded[~is_no_response]
else:
    print("\nNo 'no response' placeholders found.")

print(f"\n{len(coded)} coded extracts remain across {coded['theme'].nunique()} themes")

```

```

#Step 20 – Theme frequency
#how many coded extracts fall under each theme

theme_counts = coded["theme"].value_counts().rename("n_extracts")
display(theme_counts.to_frame())

fig, ax = plt.subplots(figsize=(8, 5))
theme_counts.sort_values().plot(kind="barh", ax=ax, color="#4C72B0")
ax.set_xlabel("Number of coded extracts")
ax.set_title("Theme frequency across all coded free-text responses")
plt.savefig("theme_frequency.png", dpi=150, bbox_inches="tight")
plt.show()

```

```

#Step 21 – Triangulation against quantitative data
#which themes show up under which questionnaire item
#which themes cluster around which preferred version

theme_by_item = pd.crosstab(coded["theme"], coded["item"])
print("Theme * questionnaire item:")
display(theme_by_item)

theme_by_preference = pd.crosstab(coded["theme"], coded["preferred_version"])
print("\nTheme * stated overall preference:")
display(theme_by_preference)

```

```

#Concentration check
#works out the cross-tab above
#what share of its extract sits within a preferred_version group

pref_share = (
    theme_by_preference.div(theme_by_preference.sum(axis=1), axis=0)
    .max(axis=1)
    .sort_values(ascending=False)
)
pref_share.name = "share_in_largest_preference_group"
display(pref_share.to_frame())

most_concentrated = pref_share.idxmax()
print(
    f"Most concentrated theme: '{most_concentrated}' – "
    f"{pref_share.loc[most_concentrated]:.0%} of its extracts come from a single "
    "preferred_version group."
)

```

#Step 22 – Save cleaned tables & results

```
OUT_DIR = DATA_DIR.parent / "analysis_outputs"
OUT_DIR.mkdir(exist_ok=True)

sessions.to_csv(OUT_DIR / "sessions.csv", index=False)
behaviour_summary.to_csv(OUT_DIR / "behaviour_summary.csv", index=False)
adaptive_summary.to_csv(OUT_DIR / "adaptive_summary.csv", index=False)
wide_likert.to_csv(OUT_DIR / "questionnaire_wide_likert.csv")
wide_cat.to_csv(OUT_DIR / "questionnaire_wide_categorical.csv")
qualitative.to_csv(OUT_DIR / "questionnaire_qualitative.csv", index=False)
questionnaire_results.to_csv(OUT_DIR / "stats_questionnaire_dimensions.csv", index=False)
behaviour_results.to_csv(OUT_DIR / "stats_behavioural_metrics.csv", index=False)
theme_counts.to_csv(OUT_DIR / "theme_counts.csv")
theme_by_preference.to_csv(OUT_DIR / "theme_preference_share.csv")
theme_by_item.to_csv(OUT_DIR / "theme_by_item.csv")

print("Saved to:", OUT_DIR)
print(sorted(p.name for p in OUT_DIR.glob("*.csv")))
```

G.2 Setup Output

204.0/204.0 kB 3.2 MB/s eta 0:00:00

G.3 Get Data Output

```
Drive already mounted at /content/drive; to attempt to forcibly remount,
call drive.mount("/content/drive", force_remount=True).
looking in: /content/drive/MyDrive/DigitalMuseum/csv
['adaptive_event.csv', 'adaptive_rule.csv', 'adaptive_suggestion.csv',
'artefact.csv', 'event_log.csv', 'participant.csv',
'questionnaire_item.csv', 'questionnaire_response.csv',
'session_completeness_summary.csv', 'session_log.csv',
'study_condition.csv', 'subtheme.csv', 'theme.csv', 'timeout.csv']
```

G.4 Load All Tables Output

```
Rows loaded:
session_log: 30
event_log: 8928
timeout: 62
adaptive_event: 3590
adaptive_suggestion: 2748
questionnaire_response: 362
```

G.5 Validate & Clean Raw Data Output

```
duplicate full rows: OK - none found
duplicate keys: OK - none found
event_log nulls: OK - none found
implausible values: {'dwell duration == 0 (kept, not dropped)':
np.int64(5)}
referential integrity: OK - none found
```

G.6 Build Readable Sessions Table Output

Sessions: 30 total -> 30 pass the completeness filter

session_id	participant_id	condition_id	session_start	session_end	timeout_triggered_static	timeout_triggered_adaptive	error_flag	artefact_order	session_notes	participant_code	condition_name
53	52	1	2026-08-03 09:01:07	2026-08-03 09:22:37	1	1	0	[13, 54, 9, 46, 2, 59, 23...	n/a	ppt1	static_first
54	53	2	2026-08-03 09:53:30	2026-08-03 10:21:29	1	1	0	[43, 32, 5, 40, 20, 21, 9...	n/a	ppt2	adaptive_first
58	57	1	2026-08-06 08:10:33	2026-08-06 08:35:24	1	1	0	[22, 49, 13, 8, 39, 51, 44...	n/a	ppt3	static_first
59	58	2	2026-08-06 15:49:27	2026-08-06 16:13:10	1	1	0	[5, 53, 7, 15, 32, 45, 16...	n/a	ppt4	adaptive_first
60	59	1	2026-08-06 16:14:43	2026-08-06 16:35:38	1	1	0	[43, 38, 32, 52, 13, 5, 7...	n/a	ppt5	static_first

G.7 Split Each Session into Static & Adaptive Halves Output

[data quality] 2 session/version pairs logged duplicate timeout rows - keeping the latest:

```
session_id  version_name
76          static      2
84          static      2
dtype: int64
```

session_id	version	phase_start	phase_end	duration_s
53	static	2026-08-03 09:05:29	2026-08-03 09:12:29	682.0
53	adaptive	2026-08-03 09:12:29	2026-08-03 09:19:29	420.0
54	static	2026-08-03 09:57:16	2026-08-03 10:04:16	646.0
54	adaptive	2026-08-03 10:04:21	2026-08-03 10:11:21	425.0
58	static	2026-08-06 08:17:44	2026-08-06 08:24:44	851.0

Event counts by type * version:

	adaptive	static	NaN
event_type			
adaptive_trigger	687	0	0
click	635	665	0
dwell_end	618	648	0
dwell_start	635	664	0
navigation_depth	635	664	0
panel_click	155	0	0
panel_impression	2748	0	0
revisit	36	16	0
session_end	0	0	30
session_start	0	0	30
timeout_redirect	30	32	0

adaptive-only events mistagged as non-adaptive: 0 (should be 0)

timeout_redirect events left untagged: 0 (should be 0)

G.8 Behavioural Metrics per Participant * Version Output

time_on_task_s is fixed at 420s for every participant * version by design - kept as sanity check

behaviour_summary: (60, 10) (expect 60 rows = 30 participants * 2 versions)

participant_code	version	n_clicks	mean_dwell_s	total_dwell_s	n_dwells	max_navigation_depth	n_revisits	n_unique_artefacts_viewed	time_on_task_s
ppt1	adaptive	17	12.81	205	16	17	1	16	420
ppt1	static	15	13.21	185	14	15	0	15	420
ppt10	adaptive	21	12.00	240	20	21	0	21	420
ppt10	static	20	10.05	202	20	20	0	20	420
ppt11	adaptive	18	7.65	130	17	18	0	18	420

G.9 Adaptive-Only Engagement Metrics Output

participant_code	n_adaptive_triggers	mean_trigger_confidence	n_suggestions_shown	suggestion_n_click_rate	mean_interest_score	n_panel_impressions	panel_clicks
ppt1	164	1.857097	124	0.072581	1.857	124	9
ppt10	92	2.113333	72	0.027778	2.113	72	2
ppt11	105	2.149048	84	0.000000	2.149	844	0
ppt12	121	1.672174	92	0.065217	1.672	92	6
ppt13	85	0.856471	68	0.000000	0.856	68	0

G.10 Questionnaire Responses Output

item_id	item_type	formal_name
1	multiple_choice	order_completed_first
2	likert	static_relevance
3	likert	static_surprise
4	likert	static_freedom
5	likert	static_experience
6	free_text	static_comment
7	likert	adaptive_relevance
8	likert	adaptive_surprise
9	likert	adaptive_freedom
10	likert	adaptive_experience
11	free_text	adaptive_comment
12	multiple_choice	preferred_version
13	free_text	preferred_reason
14	free_text	final_comments

participant_code	adaptive_experience	adaptive_freedom	adaptive_relevance	adaptive_surprise	static_experience	static_freedom	static_relevance	static_surprise
ppt1	5	5	5	4	3	4	4	2
ppt10	5	5	5	5	5	4	4	4
ppt11	4	4	3	4	4	4	3	4
ppt12	4	3	5	4	3	4	2	3
ppt13	4	4	4	3	4	4	4	3

G.11 Descriptive Statistics Output

=== Questionnaire dimensions (1-5 Likert) ===

dimension	version	mean	sd	n
relevance	static	3.07	0.78	30
relevance	adaptive	4.00	0.79	30
surprise	static	3.40	1.13	30
surprise	adaptive	3.63	1.07	30
freedom	static	4.00	1.02	30
freedom	adaptive	4.20	0.85	30
experience	static	4.07	0.64	30
experience	adaptive	4.47	0.51	30

=== Behavioural metrics ===

	n_clicks		mean_dwells		total_dwells		n_dwells		max_navigation_depth		n_revisits		n_unique_artefacts_viewed		time_on_task_s	
	m	sd	m	sd	m	sd	m	sd	m	sd	m	sd	m	sd	m	sd
adaptive	21.17	7.37	9.85	3.85	180.80	40.68	20.6	7.52	21.07	7.01	1.20	1.56	19.93	6.53	420.0	0.0
static	22.17	8.43	10.85	3.49	212.27	37.39	21.6	8.46	22.13	8.46	0.53	1.33	21.63	7.51	420.0	0.0

=== Preference (item 12) ===

preferred_version	count
Adaptive version	20
No preference	5
Static Version	5

=== Order completed first (item 1) vs stated condition ===

order_completed_first	count
Adaptive version	16
Static version	14

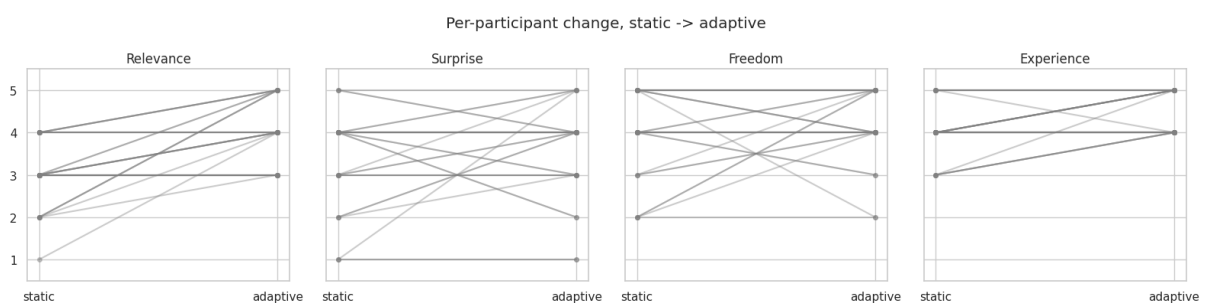
=== Actual (logged) vs self-reported order completed first ===

participant_code	actual_first	order_completed_first	match
ppt1	static	Static version	True
ppt2	adaptive	Adaptive version	True
ppt3	static	Adaptive version	False
ppt4	adaptive	Adaptive version	True
ppt5	static	Static version	True
ppt6	adaptive	Adaptive version	True
ppt7	static	Static version	True

ppt8	adaptive	Adaptive version	True
ppt9	static	Static version	True
ppt10	adaptive	Adaptive version	True
ppt11	static	Static version	True
ppt12	adaptive	Adaptive version	True
ppt13	static	Static version	True
ppt14	adaptive	Adaptive version	True
ppt15	static	Static version	True
ppt16	adaptive	Adaptive version	True
ppt17	static	Static version	True
ppt18	adaptive	Adaptive version	True
ppt19	static	Static version	True
ppt20	adaptive	Adaptive version	True
ppt22	adaptive	Adaptive version	True
ppt21	static	Static version	True
ppt23	static	Static version	True
ppt24	adaptive	Adaptive version	True
ppt25	static	Static version	True
ppt26	adaptive	Adaptive version	True
ppt27	static	Static version	True
ppt28	adaptive	Adaptive version	True
ppt29	static	Static version	True
ppt30	adaptive	Adaptive version	True

1 mismatch(es) out of 30: ['ppt3']

G.12 Within-Subject Change per Participant Slope Plot



G.13 Adaptive Suggestion Effectiveness Output

Overall suggestion click-through rate: 0.056

Mean trigger_confidence across all adaptive_event rows: 1.887

G.14 Sensitivity Check Excluding Participants Output

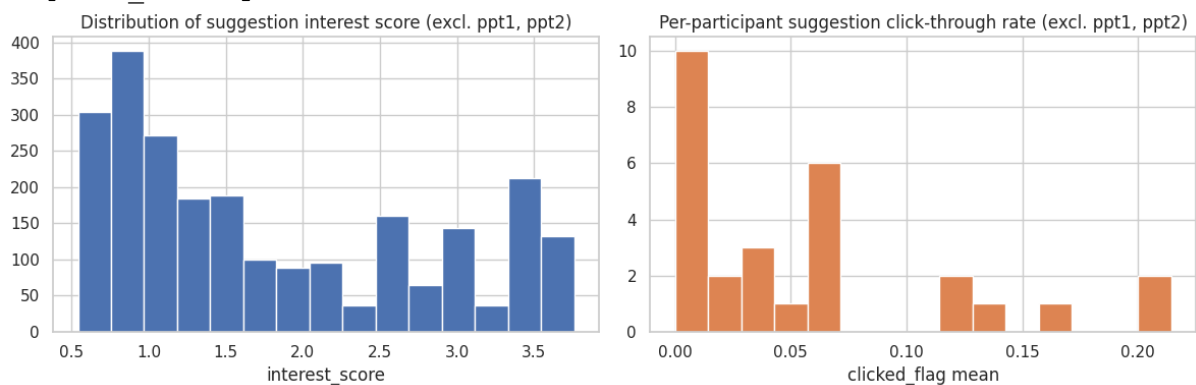
Excluded participant_id(s): [52, 53]

Excluded session_id(s): [53, 54]

sugg: 2748 -> 2404 rows after exclusion

adaptive_event: 3590 -> 3130 rows after exclusion

adaptive_summary: 30 -> 28 rows after exclusion



Overall suggestion click-through rate (excl. ppt1, ppt2): 0.052

Mean trigger_confidence across all adaptive_event rows (excl. ppt1, ppt2): 1.810

G.15 Load Thematic Coding Output

62 coded extracts across 6 themes

participant_code	item	item_text	preferred_version	response	code	theme	notes
ppt2	static_comment	Is there anything you would like to share about...	Adaptive version	The static version seems to be well suited to ...	systematic browsing; manual discovery beyond s...	Self-Directed Exploration & Autonomy	Highlights that manual scrolling revealed arte...
ppt3	static_comment	Is there anything you would like to share about...	Adaptive version	I explored based on mine and my son's interests.	family-driven exploration; interest-led browsing	Self-Directed Exploration & Autonomy	Shows multi-user influence (parent + child), r...
ppt6	static_comment	Is there anything you would like to share about...	Adaptive version	In comparison too the adaptive version navigat...	navigation friction	Exploration Constraints & Repetition	Brief but clear navigation complaint.
ppt8	static_comment	Is there anything you would like to share about...	Adaptive version	it felt like I had to view all the artefacts i...	pressure to view everything; mismatch with per...	Exploration Constraints & Repetition	Strong sense of obligation; useful for discuss...
ppt10	static_comment	Is there anything you would like to share about...	Adaptive version	layout was very clear	clear layout	Navigation Clarity, Ease & Efficiency	Good quote for illustrating static layout stre...

G.16 Thematic Coding Data Check

No genuinely empty response cells.

Removing 1 row(s) - participant indicated they had no response:

participant_code	item	response	code	theme
ppt6	final_comments	n/a	no additional feedback	Comparative Evaluation & Balanced Views

61 coded extracts remain across 6 themes

G.17 Theme Frequency Output

	n_extracts
Relevance, Personalisation & Interest Alignment	15
Navigation Clarity, Ease & Efficiency	11
Discovery, Learning & Serendipity	10
Self-Directed Exploration & Autonomy	9
Comparative Evaluation & Balanced Views	9
Exploration Constraints & Repetition	7

G.18 Triangulation Against Quantitative Data Output

Theme * questionnaire item:

	adaptive_comment	final_comments	preference_reason	static_comment
Comparative Evaluation & Balanced Views	1	4	4	0
Discovery, Learning & Serendipity	3	4	1	2
Exploration Constraints & Repetition	3	0	0	4
Navigation Clarity, Ease & Efficiency	2	1	5	3
Relevance, Personalisation & Interest Alignment	4	0	11	0
Self-Directed Exploration & Autonomy	0	0	3	6

G.19 Concentration Check Output

	share_in_largest_preference_group
Relevance, Personalisation & Interest Alignment	1.000000
Navigation Clarity, Ease & Efficiency	0.909091
Exploration Constraints & Repetition	0.714286
Discovery, Learning & Serendipity	0.600000
Comparative Evaluation & Balanced Views	0.555556
Self-Directed Exploration & Autonomy	0.444444

Most concentrated theme: 'Relevance, Personalisation & Interest Alignment'
 - 100% of its extracts come from a single preferred_version group.

G.20 Save Cleaned Tables & Results Output

Saved to: /content/drive/MyDrive/DigitalMuseum/analysis_outputs
 ['adaptive_summary.csv', 'behaviour_summary.csv',
 'questionnaire_qualitative.csv', 'questionnaire_wide_categorical.csv',
 'questionnaire_wide_likert.csv', 'sessions.csv',
 'stats_behavioural_metrics.csv', 'stats_questionnaire_dimensions.csv',
 'theme_by_item.csv', 'theme_counts.csv', 'theme_preference_share.csv']

G.21 Qualitative Coding Sheet (First 11 rows)

participant_code	item	item_text	preferred_version	response
ppt2	static_comment	Is there anything you would like to share about your experience with the static version?	Adaptive version	The static version seems to be well suited to someone who is interested in most/all artefacts as they can work systematically through the list of artefacts. Manually scrolling also allowed me to see artefacts that had not been suggested in the adaptive version.
ppt3	static_comment	Is there anything you would like to share about your experience with the static version?	Adaptive version	I explored based on mine and my son's interests.
ppt6	static_comment	Is there anything you would like to share about your experience with the static version?	Adaptive version	In comparison too the adaptive version navigation was slightly hindered
ppt8	static_comment	Is there anything you would like to share about your experience with the static version?	Adaptive version	it felt like I had to view all the artefacts in order to get the full experience even if they didn't match my interests.
ppt10	static_comment	Is there anything you would like to share about your experience with the static version?	Adaptive version	layout was very clear
ppt22	static_comment	Is there anything you would like to share about your experience with the static version?	Adaptive version	It was very repetitive.
ppt24	static_comment	Is there anything you would like to share about your experience with the static version?	Adaptive version	an improvement would be categorise items
ppt26	static_comment	Is there anything you would like to share about your experience with the static version?	Adaptive version	I was more likely to explore different categories, I was more attracted to the colourful "exhibits"
ppt28	static_comment	Is there anything you would like to share about your experience with the static version?	Adaptive version	Very unrestricting
ppt29	static_comment	Is there anything you would like to share about your experience with the static version?	Adaptive version	I enjoyed my experience with the static version, however I feel as though the absolute freedom to view anything and everything on the page did actually limit what I viewed. I noticed myself falling into the pattern of just clicking on items that came under the genres of marine life, insects, and writing since paleography, marine zoology, and the history of writing are aspects of museums that I already know that I have a deep interest in.
ppt4	static_comment	Is there anything you would like to share about your experience with the static version?	No preference	I like to make my own mind up on what to view

code	theme	notes
systematic browsing; manual discovery beyond suggestions	Self-Directed Exploration & Autonomy	Highlights that manual scrolling revealed artefacts the adaptive version hid.
family-driven exploration; interest-led browsing	Self-Directed Exploration & Autonomy	Shows multi-user influence (parent + child), relevant for identity-based exploration.
navigation friction	Exploration Constraints & Repetition	Brief but clear navigation complaint.
pressure to view everything; mismatch with personal interests	Exploration Constraints & Repetition	Strong sense of obligation; useful for discussing cognitive load.
clear layout	Navigation Clarity, Ease & Efficiency	Good quote for illustrating static layout strengths.
repetitive experience	Exploration Constraints & Repetition	"Very repetitive" is one of the clearest criticisms.
desire for categorisation	Navigation Clarity, Ease & Efficiency	Categorisation request aligns with museum UX literature.
colour-driven exploration; category-based browsing	Discovery, Learning & Serendipity	Interesting perceptual factor; relevant for visual design discussion
high browsing freedom	Self-Directed Exploration & Autonomy	"Unrestricting" is a strong autonomy key word.
freedom causing narrowing; self-reinforcing interest loop	Exploration Constraints & Repetition	Very analytically rich; shows self-reinforcing interest loops.
preference for autonomy	Self-Directed Exploration & Autonomy	Short but strong statement of user agency.

Appendix H: Ethics Approval & Risk Assessments

H.1 Signed Ethical Check List

Ethical considerations

Your research project is unlikely to result in any major ethical issues, but please read what follows in order to acquaint yourself with some of the current thinking about the ethics of doing research. Not all of what follows may be applicable to your research.

This document **MUST** be read in conjunction with the Universities [Research Guidance](#)

Ethical considerations are paramount. They need to be fully discussed with your tutor before setting out on your project. Any concerns which are relevant to your investigation should be noted and will constitute an important aspect of your final discussion. For example, particular issues concerning the involvement of children or vulnerable adults need to be thought through carefully; likewise, issues around discussion of sensitive topics; issues around intrusion, if observation strategies are employed, and so on. All these matters need to be addressed before you set about collecting data. Evidence of in-depth reflection on ethical issues should be clear in your final report.

There are certain generally accepted guidelines governing research practice of which you must be aware:

1. **Consent**
As much information as possible should be provided to participants so that they can give - or withhold - their agreement to participate. Establishing consent is not always a straightforward business and requires careful and perceptive handling.
2. **Deception**
Intentional deception of participants about the purpose and general nature of the investigation should normally be avoided. If your project involves withholding any information from your participants, you must discuss this in detail with your tutor before going ahead.
3. **Debriefing**
At the end of the study you should give participants any further information needed to complete their understanding of the nature of the research, what you hope to do with it, and how it might affect them personally at any later date.
4. **Rights to withdraw from the investigation**
Participants have the right to drop out of the study at any time and this must be made clear to them from the outset. Remember that participants also have the right to withdraw consent retrospectively and require that their data be destroyed. They should be informed about this and enabled to take appropriate action.
5. **Confidentiality and/or anonymity.**
These issues should be fully discussed with prospective participants. You need to be very clear about how 'confidentiality' and 'anonymity' are different, and what guarantees you will - and conversely will not - be able to give to participants in terms of respecting either, or both, of these.
6. **Protection of participants from physical and mental harm during the investigation.**
This is essential. It may appear to be a rather extreme consideration in relation to the small-scale research project, but you must think carefully about any levels of stress or distress which participation might cause for your participants either during or after the research.

Sensitivity to Multicultural Issues

When constructing interview schedules, questionnaires or other data-gathering instruments, it is important to be sensitive to different cultural perspectives. Depending upon the nature and location of your research, the following issues should be considered:

1. Your own prejudices and biases.
2. Though we may strive to be objective, we all have our own, often unacknowledged, prejudices. They may relate to a person's age, ethnicity, gender, sexual orientation, religion, disability, or marital, parental or socio-economic status.
3. Sensitivity to the language used in the research process to describe the different groups involved: e.g. Are explicitly derogatory terms used in describing children or adults from 'other' groups? Are subtly derogatory terms used, such as 'these people' when describing participants in a project?
4. Is the language used in data-collection instruments accessible and understandable to all participants?
5. Are members/representatives of all groups of participants involved in planning, implementing, and reviewing results from the research?
6. Have multicultural issues been addressed openly at all stages of the research?
7. Does the group of participants represent the cultural diversity of the institution/area? What implications may this have for the research findings?
8. Could the results of the research be viewed differently by different cultural groups? What has been done to ensure that their perspectives have been included?

Stage 1 Self-Assessment

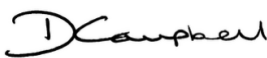
Part A

If your research involves human participants, are any of the following concerns relevant?	
Yes /No	1. The involvement of vulnerable participants or groups, such as children (under the age of 16), people with a learning disability or cognitive impairment, or persons in a dependent relationship?
Yes /No	2. The sensitivity of the research topic, e.g. the participants' sexual, political, or legal behaviour, or their experience of violence, abuse or exploitation?
Yes /No	3. The gender, ethnicity, language, or cultural status of participants?
Yes /No	4. The use of deception, trickery, or other procedures that may contravene participants' full or informed consent, without timely and appropriate debriefing, or activities that cause stress, humiliation, or anxiety, or the infliction of more than minimal pain?
Yes /No	5. Access to records of personal or other confidential information, including genetic or other biological information, concerning identifiable individuals without their knowledge or consent?
Yes /No	6. The use of intrusive interventions, such as the administration of drugs or other treatments, excessive physical exertion, or techniques such as hypnotherapy without the participants' knowledge or consent?
Yes/No	6. Research related to the NHS is strongly advised to seek advice from their supervisor before commencing the project

If you have answered 'Yes' to any of the questions, then the project is considered to be of **high ethical risk** and may need to be approved by the Departments Ethics Committee.
Please discuss your project with your supervisor.

Otherwise, your project may be considered **low ethical risk**. Please sign below and submit your self-assessment document to your supervisor and upload it to BB.

Part A

If your research involves human participants, are any of the following concerns relevant?	
Approval for Low Risk Research Projects	
I can confirm that : (confirm you have read these) - I have read the Edge Hill University Framework for Research Ethics https://www.edgehill.ac.uk/document/research-ethics-policy/ YES / NO - I have read the Computing Departments Ethics Policy Document (see BB). YES / NO - I agree to abide by their principles	
Your Signature ⁱ	
Your name	Kate Forshaw
Date:	22/06/2026
Supervisor's Signature ⁱⁱ	
Supervisor's name	Dr Dan Campbell
Date	22/06/26

H.2 Risk Profile & Key

Impact	5		R5, R13, R18, R23	R7, R24		
	4		R3, R8, R14	R1, R6, R11, R16, R20, R21		
	3		R4, R10, R15, R17, R25	R2, R9, R12, R19, R22		
	2					
	1					
		1	2	3	4	5
	Likelihood					

Impact	5	5	10	15	20	25
	4	4	8	12	16	20
	3	3	6	9	12	15
	2	2	4	6	8	10
	1	1	2	3	4	5
		1	2	3	4	5
	Likelihood					

H.3 Risk Registers

Participant & Data Risks								
Risk ID	Risk Type	Risk Description	Risk Score			Mitigation		Contingency
			I	L	R	Action	Status	
R1	Participant recruitment	Difficulty recruiting enough participants for in-person sessions	4	3	12	Clear recruitment poster; multiple channels; flexible scheduling	Completed	Extend recruitment window or widen participant pool
R2	Participant dropout	Participants withdraw before completing both prototype versions	3	3	9	Keep session short (10-15 mins); clear instructions	Ongoing monitoring	Replace with next available participant
R3	Data quality	Behavioural logs incomplete due to user error or confusion	4	2	8	Provide simple onboarding and practice click	Completed	Remove corrupted logs and recruit replacement
R4	Ethical compliance	Participant misunderstands data collection (behavioural logging)	3	2	6	Clear PIS; verbal explanation; transparency	Completed	Re-explain and offer withdrawal option
R5	Accessibility	Prototype difficult to use for participants with accessibility needs	5	2	10	Ensure readable fonts; keyboard navigation; WCAG-aligned design	Completed	Provide assisted participation or alternative task

Technical & System Risks								
Risk ID	Risk Type	Risk Description	Risk Score			Mitigation		Contingency
			I	L	R	Action	Status	
R6	System failure	Prototype crashes during participant session	4	3	12	Pre-test system; local hosting; backup laptop	Complete	Restart system; reschedule participant
R7	Logging failure	Behavioural data not recorded due to backend error	5	3	15	Test logging thoroughly; monitor console	Completed	Run session again or exclude data
R8	Database corruption	MySQL database becomes unreadable	4	2	8	Daily backups; version controls	Completed	Restore from backup
R9	Adaptive logic malfunction	Adaptive panel fails to update or updates incorrectly	3	3	9	Unit testing; simulate behaviours	Completed	Switch participant to static-only condition
R10	Device compatibility	Prototype behaves differently on different browsers	3	2	6	Standardise to one device (my laptop)	Ongoing monitoring	Provide backup browser or device

Project Management Risks								
Risk ID	Risk Type	Risk Description	Risk Score			Mitigation		Contingency
			I	L	R	Action	Status	
R11	Time constraints	Prototype takes longer to build than planned	4	3	12	Prioritise core features; iterative build	Completed	Reduce non-essential features
R12	Scope creep	Adding extra features beyond MVP delays progress	3	3	9	Stick to objectives; supervisor check-ins	Ongoing monitoring	Remove added features
R13	Evaluation delays	Difficulty scheduling in-person sessions	5	2	10	Offer flexible times; book rooms early	Completed	Extend evaluation period
R14	Analysis workload	Behavioural data takes longer to analyse than expected	4	2	8	Use Python templates; plan analysis early	Completed	Reduce dataset size
R15	Supervisor availability	Limited supervisor availability during key phases	3	2	6	Book meetings early; send updates	Ongoing monitoring	Seek second marker guidance

Legal, Ethical & Data Protection Risks								
Risk ID	Risk Type	Risk Description	Risk Score			Mitigation		Contingency
			I	L	R	Action	Status	
R16	GDPR compliance	Failure to anonymise behavioural data properly	4	3	12	No personal data collected; anonymised IDs	Completed	Delete affected dataset
R17	Consent misunderstanding	Participant signs consent but misunderstands rights	3	2	6	Clear verbal explanation; PIS clarity	Completed	Re-explain and offer withdrawal
R18	Data security	Data stored insecurely or lost	5	2	10	Encrypted storage; backups	Ongoing monitoring	Restore from backup; notify supervisor
R19	Copyright / licensing	Artefact images used without correct licensing	3	3	9	Use open-licensed museum images	Completed	Replace images immediately
R20	Bias in adaptivity	Adaptive system unintentionally reinforces narrow pathways	4	3	12	Lightweight adaptivity; diverse artefact pool	Completed	Switch to static version for affected participant

Personal Risks								
Risk ID	Risk Type	Risk Description	Risk Score			Mitigation		Contingency
			I	L	R	Action	Status	
R21	Workload management	Project workload becomes overwhelming alongside other commitments, reducing productivity	4	3	12	Create weekly micro-milestones; use structured work blocks; avoid overcommitting	Ongoing monitoring	Reduce non-essential features; request supervisor guidance on prioritisation
R22	Illness	Illness reduces ability to work during key development or evaluation phases	3	3	9	Build buffer time into schedule; complete core prototype before evaluation	Ongoing monitoring	Delay evaluation window; reduce dataset size
R23	Stress & burnout	High pressure leads to reduced focus, motivation, or quality of work	5	2	10	Use Pomodoro sessions; take planned breaks; maintain realistic expectations	Ongoing monitoring	Pause work temporarily; adjust timeline with supervisor
R24	Travel	Cruise (17-31 July) limits stable internet access and reduces available working hours	5	3	15	Complete baseline prototype build before travel; plan offline tasks	Completed	Shift evaluation to before/after travel; use local backups to continue offline
R25	Loss of motivation / boredom	Long project duration leads to dips in motivation or engagement	3	2	6	Break tasks down; track progress visually; vary tasks to stay engaged	Ongoing monitoring	Switch to lighter tasks temporarily; seek supervisor encouragement

H.4 Participant Information Sheet

Participant Information Sheet

Study Title: Behaviour-Driven Presentation in Digital Museums

Researcher: Kate Forshaw, MSc Data Analytics

Supervisor: Dan Campbell, Edge Hill University

What is this study about?

This study explores how people interact with digital museum collections and investigates whether interfaces that adapt to real-time user behaviour can enhance exploratory engagement, serendipitous discovery, and personalised browsing pathways.

What will I be asked to do?

If you agree to take part, you will stay for a short in-person session. You will:

1. Use two versions of a digital museum prototype
 - a. A static interface – where content remains the same.
 - b. An adaptive interface – where content updates based on your interaction patterns.
2. Explore each version freely for 5-7 minutes.
3. Complete a short questionnaire about your experience.

The order of the two versions will vary between participants to ensure fair comparison.

What data will be collected?

The system will automatically log behavioural interaction data, including:

- Dwell time on artefacts
- Click sequences
- Navigation patterns
- Revisits and scroll depth

These behavioural signals help the researcher understand how adaptive presentation influences exploration. No personal data (e.g. name, email, IP address) will be collected. You will not be identifiable in any dataset.

Are there any risks or disadvantages?

There are no anticipated risks. You may stop at any time without giving a reason.

Are there any benefits?

While there is no direct personal benefit, your participation will help improve understanding of how digital museum systems can better support exploratory and engaging cultural experiences.

Will my data be confidential?

Yes. All data will be anonymised, stored securely, and used only for academic research. No individual will be identifiable in the final report.

Do I have the right to withdraw?

Yes. You may withdraw at any point during the session. After the session, you may request your data to be removed within 14 days, using the anonymous participant code provided to you.

Who has reviewed this study?

This study has been reviewed and approved through Edge Hill University's ethics process.

Contact Information

If you have questions, please contact:

Researcher: Kate Forshaw – 25259628@edgehill.ac.uk

Supervisor: Dan Campbell – dan.campbell@edgehill.ac.uk

H.5 Participant Consent Form

Participant Consent Form

Study Title: Behaviour-Driven Presentation in Digital Museums

Researcher: Kate Forshaw

Participant Code: _____

Version Order: _____

Please read each statement and tick to confirm your agreement.

I confirm that:

- ☐ I have read and understood the Participant Information Sheet for this study.
- ☐ I understand what participation involves and have had the opportunity to ask questions.
- ☐ I understand that my participation is voluntary and that I may withdraw at any time without giving a reason.
- ☐ I understand that I may request my data to be withdrawn within 14 days using my participant code.
- ☐ I understand that the study will collect behavioural interaction data, to investigate how adaptive interfaces respond to user exploration, but not personal or identifying information.
- ☐ I understand that my data will be anonymised, stored securely, and used only for academic research.
- ☐ I understand that the study involves no deception and poses minimal risk.
- ☐ I consent to take part in this study.

Participant Signature: _____

Date: __ / __ / ____

Researcher Signature: _____

Date: __ / __ / ____

H.6 Participant Debrief Sheet

Participant Debrief Sheet

Study Title: Behaviour-Driven Presentation in Digital Museums

Researcher: Kate Forshaw, MSc Data Analytics

Supervisor: Dan Campbell, Edge Hill University

Participant Code: _____

Version Order: _____

Thank you for taking part!

You have just completed a study exploring how people interact with different types of digital museum interfaces. The purpose of this research is to understand whether systems that adapt to real-time user behaviour can improve exploratory engagement, serendipitous discovery, and personalised pathways through digital collections.

What was the study investigating?

You used two versions of a digital museum prototype:

- A static interface – content stays the same
- An adaptive interface – content updates based on your interaction patterns

The study examines how people explore each version and whether adaptive systems can support more engaging or meaningful browsing experiences.

What data was collected?

The system recorded behavioural interaction data, such as:

- Dwell time on artefacts
- Click sequences
- Navigation patterns
- Revisits and scroll depth

No personal or identifying information was collected. All data is anonymised and stored securely.

Why was the order of interfaces different for each participant?

To ensure fair comparison, the order was counterbalanced – some participants used the static version first, others the adaptive version. This helps reduce bias and strengthens the validity of the findings.

Your rights

- You may withdraw your data within 14 days using the anonymous participant code provided to you.
- Withdrawing will not affect you in any way.
- Your data will only be used for academic research and will never be shared in identifiable form.

Are there any concerns or questions?

If you have questions about the study or your data, please contact:

Researcher: Kate Forshaw – 25259628@edgehill.ac.uk

Supervisor: Dan Campbell – dan.campbell@edgehill.ac.uk

If you have concerns about the ethical conduct of the study, you may contact the Edge Hill University ethics team.

Thank you again for your participation!

Your contribution helps advance research into behaviour-driven interfaces, personalisation, and digital cultural experiences.

Appendix I: Supplementary Materials

I.1 Recruitment Flyer

Sign-up
here
now!

Participants





WANTED!

Help test a digital museum prototype in just 10 minutes on campus.

Who can take part?



- ✓ Anyone aged 18+
- ✓ No experience needed

- ✓ No interest in museums required
- ✓ If you use websites & have curiosity

What you'll do:



- ✓ Try two versions of a web prototype
- ✓ Explore freely - like scrolling on TikTok
- ✓ Fill in a very short questionnaire
- ✓ Help research at Edge Hill University

Running
August
3rd-23rd

INTERESTED?

Email me: 25259628@edgehill.ac.uk

I.2 Sign-Up Form – Word Copy

Study Sign-Up Form

Thank you for your interest in taking part in this study exploring how people interact with digital museum interfaces. This form is only for expressing interest, choosing a preferred time, and confirming you have read the Participant Information Sheet. Full consent will be completed in person before the study begins.

Participant Information Sheet

Please read the information sheet before completing the form:

https://edgehill-my.sharepoint.com/:w:/g/personal/25259628_edgehill_ac_uk/IQAzmfeByOaVSoVVddOlWzxnAYRj_T1ZlqY_ywUa_4HeQ1E?e=CULsz9

If you have any issues accessing it, please let me know at the end of the form.

About You

1. Full Name
2. Email Address (Used only to contact you about your session time)

Availability

3. Preferred Week (Please select all that apply to you)
 - ☐ W/C 03/08/2026
 - ☐ W/C 10/08/2026
 - ☐ W/C 17/08/2026
4. Preferred Day (Please select all that apply to you)
 - ☐ Monday
 - ☐ Tuesday
 - ☐ Wednesday
 - ☐ Thursday
 - ☐ Friday
5. Preferred Time Frame (Please select all that apply to you)
 - ☐ 9-10
 - ☐ 10-11
 - ☐ 11-12
 - ☐ 12-13
 - ☐ 13-14
 - ☐ 14-15
 - ☐ 15-16
 - ☐ 16-17

- 17-18

Pre-Consent Confirmation

6. I have read the Participant Information Sheet.
 - I agree
7. I understand that this form does not commit me to taking part and that full consent will be completed in person.
 - I agree
8. I agree to be contacted by the researcher to arrange a session time.
 - I agree

Additional Queries

9. Do you have any accessibility needs you would like me to be aware of?
10. Anything else you'd like to tell me that could affect scheduling?

Final Message

Thank you for signing up!

I will contact you shortly to confirm your session time.

If you have any questions, feel free to email me at 25259628@edgehill.ac.uk

I.3 Verbal Study Instructions

Verbal Study Instructions

MacBook Basics

- Scroll – swipe lightly using two fingers, no need to press
- Click – physically press anywhere on keypad, simply tapping won't work
- Ask if they want to practice on different page

Big Picture

- Browse one version for 7 minutes
- Redirect to transition page, can take a break for as long as they wish
- Browse the other version for 7 minutes
- Redirect to other transition page
- Answer short questionnaire

Artefact Interactions

- Click to see more details
- Source is just there for reference, no need to click
- Can close in several ways – X, “Close” button, clicking outside, escape

Freedom & Pacing

- No requirement to view everything
- No visible clock to watch
- It will move on by itself when time is up

Adaptive Panel

- Updates on its own
- No need to click it specially

Consequential Rules

- Do not refresh or use browser's back button
- Only use “Return to Home” button if they want to stop taking part

Breaks or Freezes

- Do not try to click anything if the page appears to freeze or break
- I will be here to help and handle it

I.4 Post-Study Questionnaire – Word Copy

Post-Study Questionnaire

About the Study Session

1. Which version did you complete first?
 - a. Static version
 - b. Adaptive version

Static Prototype Feedback

Perceived Relevance

2. The artefacts shown in static version felt relevant to my interests.
 - a. Strongly disagree
 - b. Disagree
 - c. Neutral
 - d. Agree
 - e. Strongly agree

Perceived Surprise

3. I encountered artefacts in the static version that were unexpected or surprising.
 - a. Strongly disagree
 - b. Disagree
 - c. Neutral
 - d. Agree
 - e. Strongly agree

Exploratory Freedom

4. The static version allowed me to explore freely without feeling restricted.
 - a. Strongly disagree
 - b. Disagree
 - c. Neutral
 - d. Agree
 - e. Strongly agree

Overall Experience

5. Overall, my experience using the static version was positive.
 - a. Strongly disagree
 - b. Disagree
 - c. Neutral
 - d. Agree

- e. Strongly agree

Optional Comment

6. Is there anything you would like to share about your experience with the static version?

Adaptive Prototype Feedback

Perceived Relevance

7. The artefacts shown in adaptive version felt relevant to my interests.
- a. Strongly disagree
 - b. Disagree
 - c. Neutral
 - d. Agree
 - e. Strongly agree

Perceived Surprise

8. I encountered artefacts in the adaptive version that were unexpected or surprising.
- a. Strongly disagree
 - b. Disagree
 - c. Neutral
 - d. Agree
 - e. Strongly agree

Exploratory Freedom

9. The adaptive version allowed me to explore freely without feeling restricted.
- a. Strongly disagree
 - b. Disagree
 - c. Neutral
 - d. Agree
 - e. Strongly agree

Overall Experience

10. Overall, my experience using the adaptive version was positive.
- a. Strongly disagree
 - b. Disagree
 - c. Neutral
 - d. Agree
 - e. Strongly agree

Optional Comment

11. Is there anything you would like to share about your experience with the adaptive version?

Comparative Reflection

12. Which version did you prefer overall?
- a. Static version
 - b. Adaptive version
 - c. No preference
13. Why did you prefer this version?

Final Thoughts

14. Any final comments about your experience today?
